

CS610 MID-Term BY Attiq Kundi

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CS610 Computer Networks

1. Asynchronous Transfer Mode uses small fixed size cell. Each cell consist of ----- octets.

55

[53](#)

52

51

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2. What will be the bandwidth rate for future (possibly 10 Gigabyte) Ethernet?

500 MHz

550 MHz

[600 MHz](#)

650 MHz

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3. Asynchronous Transfer Mode uses the concept of

Connection-less network

Connection-oriented network

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4. The fiber-modem converts data into _____ then transmits cover the optional fiber

[Pulses of light](#)

Electrons

Electromagnetic signals

None of these

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5. How many methods exist for computing routing table?

5

4

3

2

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6. To span long distance or many computers, networks must replace shared medium with

Segment transformation

Fiber optics
[Packet switches](#)
None of the above

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7. Which of the following statement is true regarding Distance-vector routing?

It is very simple to implement.
Packet switch updates own routing table first
It is used in RIP
[All of the above](#)

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8. What will be the bandwidth rate for 1000Base-T Gigabyte or ATM?

100 MHz
150 MHz
200 MHz
[250 MHz](#)

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1. Which of the following statement is true regarding Jitter?

Jitter is used for variance in transmission delays.
Jitter can occur when a packet is delayed.
Jitter is significance for voice, video and data.
All of the above

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2. The maximum cable length of a LAN is _____

Fixed
Unlimited
Fixed and Unlimited in different conditions
None of these

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3. Jitter is significance for -----

Voice
Video
Data
All of above

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4. The routing table contains information about theimmediately around it.
topology of the network

destination
both of the above
none of the above

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5. Which of the following is called virtual channel?

Connection in VCI
Connection in ATM
Connection in virtual circuit
None of the above

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6. If a bridge builds up address list for each segment that consists of physical addresses of the computer attached to that segment. So, when the bridge boots, the size of list will be__ for each segment.

- 3
- 2
- 1
- 0

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7. Frame forwarding of a bridge is based on _____.

Layer 3 source address

Layer 3 destination address

Layer 2 destination address

Layer 2 Source address

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8. The process of forwarding the packets of information is called.....

Routing

Switching

Communication

None of the above

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9. Which one of the following is a property of static routing?

It is inflexible

It is done at boot time

It has low network overload

All of the above

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10. According to a bridges Performance, if both the sources and destination are on the same segment

It forward the frame to the other segment

It does not forward the frame to the other segment

It does not forward the frame to the same segment

It forwards the frame to every other segment

The protocol address of the next hop must be _____ to an equivalent hardware address before a packet can be sent.

Encrypted

Decrypted

Translated

Non of these

A _____ address-binding table is used for each physical network.

New

Similar

[Separate](#)

Old

UDP provides connection-oriented service.

True

False

TCP provides connection oriented reliable data streaming service, whereas UDP provides connection-less unreliable messaging service.

The UDP stands for_____.

Universal Datagram Protocol

User Datagram Protocol

United Datagram Protocol

None of the given

In 3-way handshake TCP requires to generate a random _____ sequence number.

30 bit

32 bit

34 bit

none of the given

TCP stands for _____

Transport control protocol

Transmission control protocol

Terminal control protocol

None of the given

1. Which protocol working at the Transport layer provides a connectionless service between hosts?

- A. IP
- B. ARP
- C. TCP
- D. UDP**

User Datagram Protocol is used at the Transport layer to provide a connectionless service.

Chapter: 3

2. Which protocol works at the Transport layer and provides virtual circuits between hosts?

- A. IP
- B. ARP
- C. TCP**
- D. UDP

Transmission Control Protocol sets up a virtual circuit before transmitting any data. This creates a reliable session and is known as a connection-oriented session.

Chapter: 3

3. Which protocol works at the Internet layer and provides a connection service between hosts?

- A. IP**
- B. ARP
- C. TCP
- D. UDP

Internet Protocol is used to address hosts and route packets through the internetwork. The question does not refer to a connection-oriented service, which is different from a plain connection service.

Chapter: 3

4. If a host broadcasts a frame that includes a source and destination hardware address, and its purpose is to assign IP addresses to itself, which protocol at the Network layer does the host use?

- A. RARP**
- B. ARPA
- C. ICMP
- D. TCP
- E. IPX

Reverse ARP is used to find an IP address from a known hardware address.

Chapter: 3

5. If a router interface is congested, which protocol in the IP suite is used to tell neighbor routers?

- A. RARP

B. ARP

C. ICMP

D. IP

E. TCP

Internet Control Message Protocol (ICMP) is used to send redirects back to an originating router.

Chapter: 3

6. What is the valid host range the IP address 172.16.10.22 255.255.255.240 is a part of?

A. 172.16.10.20 through 172.16.10.22

B. 172.16.10.1 through 172.16.10.255

C. 172.16.10.16 through 172.16.10.23

D. 172.16.10.17 through 172.16.10.31

E. 172.16.10.17 through 172.16.10.30

First start by using the 256 mask, which in this case is $256 - 240 = 16$. The first subnet is 16; the second subnet is 32. This host must be in the 16 subnet; the broadcast address is 31 and the valid host range is 17–30.

Chapter: 3

7. What range of addresses can be used in the first octet of a Class B network address?

A. 1–126

B. 1–127

C. 128–190

D. 128–191

E. 129–192

F. 192–220

A Class B network is defined in the first octet with the numbers 128–191.

Chapter: 3

8. What range of addresses can be used in the first octet of a Class C address?

A. 1–127

B. 129–192

C. 203–234

D. 192–223

A Class C network is defined in the first octet with the numbers 192–223.

Chapter: 3

9. How many bytes is an Ethernet address?

A. 3

B. 4

C. 5

D. 6

E. 7

F. 8

G. 16

An Ethernet (MAC) address is 6 bytes long (48 bits).

Chapter: 3

10. What protocol is used to find the hardware address of a local device?

A. RARP

- B. ARP
- C. IP
- D. ICMP
- E. BootP

Address Resolution Protocol (ARP) is used to find the hardware address from a known IP address.

Chapter: 3

11. Which of the following is the broadcast address for a Class B network ID using the default subnet mask?

- A. 172.16.10.255
- B. 172.16.255.255
- C. 172.255.255.255
- D. 255.255.255.255

A Class B network address is two bytes long, which means the host bits are two bytes long. The network address must be 172.16.0.0, which is all host bits off. The broadcast address is all bits on, or 172.16.255.255.

Chapter: 3

12. Which class of IP address provides a maximum of only 254 host addresses per network ID?

- A. A
- B. B
- C. C
- D. D
- E. E

A Class C network address only has 8 bits for defining hosts. $2^8 - 2 = 254$.

Chapter: 3

13. What is the broadcast address of the subnet address 10.254.255.19 255.255.255.248?

- A. 10.254.255.23
- B. 10.254.255.24
- C. 10.254.255.255
- D. 10.255.255.255

First start with 256 mask or in this case, $256 - 248 = 8$. The first subnet is 8. The second subnet is 16, then 24. This host is in the 16 subnet, the broadcast address is 23, and the valid host range is 17–22.

Chapter: 3

14. What is the broadcast address of the subnet address 172.16.99.99 255.255.192.0?

- A. 172.16.99.255
- B. 172.16.127.255
- C. 172.16.255.255
- D. 172.16.64.127

First start with 256 mask or in this case, $256 - 192 = 64$. 64 is the first subnet; 128 is the second subnet. This host is in the 64-subnet range, the broadcast address is 127, and the valid host range is 65–126.

Chapter: 3

15. If you wanted to have 12 subnets with a Class C network ID, which subnet mask would you use?

- A. 255.255.255.252
- B. 255.255.255.248
- C. 255.255.255.240

D. 255.255.255.255

Take a look at the answers and see which subnet mask will give you what you need for subnetting. 252 gives you 62 subnets, 248 gives you 30 subnets, 240 gives you 14 subnets, and 255 is invalid. Only answer C (240) gives you what you need.

Chapter: 3

16. What is the port number range that a transmitting host can use to set up a session with another host?

A. 1–1023

B. 1024 and above

C. 1–256

D. 1–65534

Source hosts can use any port number starting at 1024.

Chapter: 3

17. Which of the following ranges are considered well-known port numbers?

A. 1–1023

B. 1024 and above

C. 1–256

D. 1–65534

The port numbers 1–1023 are defined as and considered well-known port numbers.

Chapter: 3

18. What is the broadcast address of the host subnet address 10.10.10.10 255.255.254.0?

A. 10.10.10.255

B. 10.10.11.255

C. 10.10.255.255

D. 10.255.255.255

First start with $256-254=2$. The first subnet is 2, the second subnet is 4, then 6, 8, 10, and 12. Remember that the fourth octet is host addresses. This host is a part of the subnet 10.0, the broadcast address is 11.255, and the valid host range is 10.1 through 11.254.

Chapter: 3

19. What broadcast address will the host 192.168.210.5 255.255.255.252 use?

A. 192.168.210.255

B. 192.168.210.254

C. 192.168.210.7

D. 192.168.210.15

Start with the 256 mask or in this case, $256-252=4$. This first subnet is 4. The second subnet is 8. This falls in the 4-subnet range. The broadcast address is 7, and the valid hosts are 5 and 6.

Chapter: 3

20. If you need to have a Class B network address subnetted into exactly 510 subnets, what subnet mask would you assign?

A. 255.255.255.252

B. 255.255.255.128

C. 255.255.0.0

D. 255.255.255.192

If you use the mask 255.255.255.0, that only gives you eight subnet bits, or 254 subnets. You are going

to have to use one subnet bit from the fourth octet, or 255.255.255.128. This is 9 subnet bits ($2^9 - 2 = 510$).

Chapter: 3

1. Which protocol is used to look up an IP address from a known Ethernet address?

- A. IP
- B. ARP
- C. RARP
- D. TCP

The protocol at the Network layer that finds an IP address from a known Ethernet address is Reverse ARP (RARP). See Chapter 3 for more information on IP protocols.

Chapter: 101

2. What is the subnet address of the IP address 192.168.100.30 255.255.255.248?

- A. 192.168.100.32
- B. 192.168.100.24
- C. 192.168.100.0
- D. 192.168.100.16

Start by using 256, the subnet mask, which is $256 - 248 = 8$. The first subnet is 8. The next subnet would be 16, then 24, and then 32. This host is in the 24 subnet, the broadcast address is 31, and the valid host range is 25 through 31. See Chapter 3 for more information on IP addressing.

Chapter: 101

3. Which of the following is the valid host range for the IP address 192.168.168.188 255.255.255.192?

- A. 192.168.168.129-190
- B. 192.168.168.129-191
- C. 192.168.168.128-190
- D. 192.168.168.128-192

Start by using 256, the subnet mask, which is $256 - 192 = 64$. The first subnet is 64. The next subnet would be 128. This host is in the 128 subnet, the broadcast address is 191, and the valid host range is 129 through 190. See Chapter 3 for more information on IP addressing.

Chapter: 101

4. What is the broadcast address of the subnet address 192.168.99.20 255.255.255.252?

- A. 192.168.99.127
- B. 192.168.99.63
- C. 192.168.99.23
- D. 192.168.99.31

Start by using 256, the subnet mask, which is $256 - 25 = 4$. The first subnet is 4. The next subnet would be 8, then 12, 16, 20, and 24. The broadcast address is 23, and the valid host range is 21 and 22. See Chapter 3 for more IP addressing information.

Chapter: 101

5. What is the valid host range that the host ID 192.168.10.33 255.255.255.224 is a part of?

- A. 192.168.10.32-63
- B. 192.168.10.33-63
- C. 192.168.10.33-62
- D. 192.168.10.33-61

Start by using 256, the subnet mask, which is $256 - 224 = 32$. The first subnet is 10.32. The next subnet

would be 10.64. This host is in the 10.32 subnet, the broadcast address is 10.63, and the valid host range is 10.33 through 10.62. See Chapter 3 for more IP addressing information.

Chapter: 101

1. Which protocol does DHCP use at the Transport layer?

- A. IP
- B. TCP
- C. UDP
- D. ARP

User Datagram Protocol is a connection network service at the Transport layer, and DHCP uses this connectionless service. See Chapter 3 for more information.

Chapter: 102

2. Which of the following is the valid host range for the IP address 192.168.168.188 255.255.255.192?

- A. 192.168.168.129-190
- B. 192.168.168.129-191
- C. 192.168.168.128-190
- D. 192.168.168.128-192

· $256-192=64$. $64+64=128$. $128+64=192$. The subnet is 128, the broadcast address is 191, and the valid host range is the numbers in between, or 129-190. See Chapter 3 for more information.

Chapter: 102

3. Which protocol is used to find an Ethernet address from a known IP address?

- A. IP
- B. ARP
- C. RARP
- D. BootP

If a device knows the IP address of where it wants to send a packet, but doesn't know the hardware address, it will send an ARP broadcast looking for the hardware or, in this case, Ethernet address. See Chapter 3 for more information.

Chapter: 102

4. Which class of IP address has the most host addresses available by default?

- A. A
- B. B
- C. C
- D. A and B

Class A addressing provides 24 bits for hosts addressing. See Chapter 3 for more information.

Chapter: 102

5. Which protocol does Ping use?

- A. TCP
- B. ARP
- C. ICMP
- D. BootP

ICMP is the protocol at the Network layer that is used to send echo requests and replies. See Chapter 3 for more information.

Chapter: 102

6. Which protocol is used to send a Destination Network Unknown message back to originating hosts?

- A. TCP
- B. ARP
- C. ICMP
- D. BootP

ICMP is the protocol at the Network layer that is used to send messages back to an originating router. See Chapter 3 for more information.

Chapter: 102

1. What is the subnet broadcast address the host 192.168.10.17 with four bits of subnetting will use?

- A. 192.168.10.16
- B. 192.168.10.19
- C. 192.168.10.23
- D. 192.168.10.31

When referring to only subnet bits, add the bits to the default mask. In this case, it is a Class C address, and the default mask is 255.255.255.0. Four bits would make the mask 255.255.255.240. $256-240=16$. $16+16=32$. Therefore, the subnet broadcast must be 192.168.10.31.

Chapter: 103

2. What is the subnet broadcast address the host 172.16.10.12 with nine bits of subnetting will use?

- A. 172.16.10.127
- B. 172.16.10.255
- C. 172.16.255.255
- D. 172.16.10.128

The default mask for Class B network IDs is 255.255.0.0. By adding nine subnet bits, the subnet mask is 255.255.255.128. The subnet is 172.16.10.0, and the broadcast address is 172.16.10.127. This is determined by looking at the host ID, which is 12 in this case. Since it is less than 128, the subnet must be zero in the fourth octet.

Chapter: 103

3. What is the valid host range of the IP subnet address 172.16.10.61 255.255.255.224?

- A. 172.16.10.48-63
- B. 172.16.10.33-62
- C. 172.16.10.0-254
- D. 172.16.10.60-94

· $256-224=32$. $32+32=64$. The subnet is 172.16.10.32, and the broadcast address is the number right before the next subnet (63 in this question). The valid hosts are the numbers in between the network ID and the broadcast address: 33-62.

Chapter: 103

4. What protocol is used to resolve an Ethernet address to an IP address?

- A. IP
- B. ARP
- C. RARP
- D. BootP

The RARP protocol is used to find an IP address from a known hardware address.

Chapter: 103

5. What protocol is used to resolve an IP address to an Ethernet address?

- A. IP
- B. ARP
- C. RARP
- D. BootP

The ARP protocol is used to find the hardware address from a known IP address.

Chapter: 103

Which one is NOT the function of ping program?

- Traceability
- Reach ability
- Both a and b
- None of the given

A datagram cannot be larger than _____ of a network over which it is sent.

- MTU
- Size
- IP header
- None of the given

MTU Stands for _____

- Minimum transmission unit
- Maximum transmission unit
- Multicast transmission unit
- None of the given

Fragmentation when using ICMP for path MTU should be avoided.

- **True**
- False

HEADER LEN field gives size of extension header.

- False
- **True**

The process of learning the path MTU is known as path MTU discovery.

- **True**
- False

_____ is less complex and easy to understand.

- TCP
- **UDP**
- IP
- None of the given

IPV6 128 bits address includes network prefix and _____

- **Host suffix**
- Host prefix
- source Prefix
- None of the given

The Universal Datagram is not an end-to-end protocol.

- True
- **False**

Ethernet uses a _____ static addressing scheme in which each device is assigned a unique address by the manufacturer.

- 64
- **48**
- 32
- 8

The ----- bit preamble in the Ethernet frame format that precedes the frame contains alternating 1s and 0s that allow the receiver's hardware to synchronize with the incoming signal.

- 64
- 32
- 16
- **8**

The third field of the header consists of ----- bit Ethernet frame type.

- 48
- 32
- **16**
- 8

An interface for twisted pair Ethernet must have a _____ connector, and must generate signals according to the _____ specification.

- BNC- t base 10
- **RJ 45- t base 10**
- BNC-t base 5

- RJ 45-t base 2

A bridge uses the _____ to determine which computers connect to which segment, and uses the _____ to determine whether to forward a copy of frame.

- **Source , destination**
- Source , source
- Destination, source
- None from above

Formally named _____ informally known as the thick wire Ethernet or Thick net.

- 10 Base 2
- **10 Base 5**
- 10 Base T
- None of the given

Formally named ----- informally known as thin wire Ethernet or thin net.

- **10 Base 2**
- 10 Base 5
- 10 Base T
- None of the given

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- 10 Base 2
- 10 Base 5

- **10 Base T**
- None of the given

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

- **DMA(Direct Memory Access)**
- Multiplexer
- Transceiver
- None of the given

A system with redundant bridges might have a problem with _____ in the system.

- **Loop**
- Filters
- Spamming
- All above

A Bridge can _____

- Filter a frame
- Forward a frame
- Extend a LAN
- **Do all the above**

_____ has a jitter zero

- None of the given
- Virtual Private Network
- **Isochronous Network**
- Asynchronous Network

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

- **Connectionless service paradigm**
- Connection oriented service paradigm
- Both Connectionless and Connection-oriented service paradigm
- None of the given

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

- 21-bit
- 22-bit
- 23-bit
- **24-bit**

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

- Hierarchal address
- **Default route**
- Shortest path
- None of the given

The next hop to which a packet is sent depends only on

- **Packet's destination**
- Packet's original source
- Path the packet has taken
- Non of the given

When an application----- data, it makes a copy of the data available to all other computers on the network?

- **Broadcasting**
- Multicasting
- Unicasting
- None of the given

A ----- provide a mechanism that a customer can use to set a physical address.

- Static addressing scheme
- **Configurable addressing scheme**
- Dynamic addressing scheme
- None of the given

_____ sense a signal on one cable and then transmits an amplified copy on the other cable.

- **Repeater**
- Bridge
- Hub
- None of the given

The maximum size of an Ethernet segment is _____

- 250 meters
- **500 meters**
- 700 meters
- None of the given

FDDI can transmits data at a rate of -----

- 1000 million bits per second

- **100 million bits per second**
- 10 million bits per second
- None of the given

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

- **CSMA/CD**
- CSMA/CA
- TOKEN PASSING
- None of the given

----- have advantages arisen from the size and ease of computation.

- **CRC**
- Parity
- Checksums
- None of given

The term ----- is used to denote the definition of a packet used with a specific type of network.

- Packet
- **Frame**
- Data
- None of the given

----- has no way to determine the cause of the problem.

- **Ping**
- Trace route

- ICMP
- Non of the given

In -----, network occupies the smaller area like a room a floor or a building

- **LAN**
- WAN
- MAN
- None of the given

In -----, network occupies larger areas like cities & countries.

- LAN
- **WAN**
- MAN
- None of the given

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

- False
- **True**

A typical port on an ATM switch operates at _____ or higher.

- OC-2 speed (155Mbps)
- OC-3 speed (100Mbps)
- **OC-3 speed (155Mbps)**
- OC-3 speed (155Gbps)

The product of delay and throughput measures the _____ of data that can be present on the network.

- Area
- **Volume**
- Length
- None of the given

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

- Constant Bit Rate (CBR) service
- **Variable Bit Rate (VBR) service**
- Available Bit Rate (ABR) service
- None of the given

Which of the following is a connecting device?

- Bridge
- Repeater
- Hub
- **All the given**

A bridges function in the _____ layers(s).

- Physical (MAC)
- Data link
- Network
- **Physical (MAC) and Data link**

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- Layer 2 source address

- Source node's physical address
- **Layer 2 destination address**
- Layer 3 destination address

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

- **8 octets**
- 8 bytes
- 8 bits
- None of the given

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

- **Explicit frame type**
- Ideal frame type
- Implicit frame type
- None of the given

Local Talk is a LAN technology that employs -----

- Star topology
- **Bus topology**
- Ring topology
- None of the given

The Fast Ethernet hardware operates at a rate of -----

- 10 Mbps
- **100 Mbps**

- 1000 Mbps
- None of the given

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

- **Parity**
- Checksums
- CRC
- None of given

Computer networks are often called ----- because they use packet technology.

- Ethernet
- Switch networks
- **Packet networks**
- None of the given

----- Program sends a message to a remote computer and reports whether the computer responds.

- **Ping**
- Traceroute
- ICMP
- Non of the given

In Point-to-Point topology there are two topologies.

- Tree and Ring
- Star and Ring

- **Star and Tree**
- None of the given

A network uses a -----arranges for computers to be connected in a closed loop.

- Star Topology
- **Ring Topology**
- Bus Topology
- None of the given

An interface for thin Ethernet must have an _____ connector, and must generate signals according to the _____ specification.

- RJ-45, 10 Base T
- RJ-45, 10 Base 5
- **BNC, 10 Base 2**
- BNC, 10 Base T
 - Find the class of the address.
 - 10100111 11011011 10001011 01101111

A
B
E
C
 -
 - On of the design goals for unicast route propagation is _____.

consistency
inconsistency
stability
dynamic addressing
 -
 - Propagation multicast routing information differs dramatically from unicast route propagation?

True
False

The IP multicast abstraction allows an application running on an arbitrary computer to leave a multicast group at any time. While _____ application on a computer remain a member of a group.

no

many

In IPv6 the type of address used for collection of computers with same prefix. Are known as _____.

Anycast

Unicast

Multicast

Non of the given

Special types of addresses in IPv6 used for multiple destinations; possibly not at same site. Are known as _____.

Unicast

Anycast

Multicast

Non of the given

UDP offers application programs a Message-Oriented Interface, applications can depend on protocol to preserve data boundaries.

True

False

The time for acknowledgement to arrival of packet depends on.

Distance to destination and Current traffic conditions

Current traffic conditions

Distance to destination

non of these

Cost, effort, risks, and resources are the factors included in-----

Estimation

Testing

Development

Maintenance

There are _____ possibilities to detect the destination using Trace-route

1

2

3

None of the given

_____ is used for single destination computer.

Multicast

Broadcast

unicast

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a ____ Ethernet address.

16-bit

48-bit

64-bit
128-bit

Which is not the type of error messages defined by ICMP.

Source quench
Time exceeded
Destination unreachable
none of the given

End to End delivery Service of IP datagram is _____

Connection oriented

Connectionless

both a and b

none of the given _____ is a type of address used for collection of computers with same prefix.

Cluster

unicast

Multicast

none of the given

IPv6 _____ is responsible for fragmentation. Routers simply drop datagram's larger than network

Destination

Intermediate routers

Source

Medium

_____ Source is responsible for fragmentation.

IPV4

IPV6

_____ message is sent in response to incoming datagrams with problems.

TCP/IP

IGMP

ICMP

none of the given

_____ field is used to identify a specific path through the network

FLOW LABEL

TRAFFIC CLASS

Both a and b

none of the given Connectionless service, Message-Oriented protocol, best-effort delivery service, arbitrary interaction & operating system independent are the characteristics of

_____ TCP

UDP

IP

None of the given

The process of using a routing table to select a next hop for a given datagram is called _____.

Encapsulation

Reassembling

Routing or forwarding

None of the given

A multicast routing scheme in which the protocol software builds a delivery tree from a central point is called _____

Distance Vector Multicast Routing Protocol (DVMRP)

Core Based Tree (CBT)

Protocol Independent Multicast_Sparse Mode (PIM-SM)

Protocol Independent Multicast_Dense Mode (PIM-DM)

Whenever it handles a packet, IP software needs to separate the destination address into a _____ and _____.

postfix, Infix

non of these

Infix, prefix

prefix, suffix Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by _____

None of the given

TCP

UDP

IP

_____ Protocol provides error reporting mechanism.

IGMP

SNMP

ICMP

none of the given

_____ encapsulates IP datagram as data area in hardware frame.

Network Interface Layer

Datalink Layer

Network Layer

None of the given

TTL stands for _____

Time to Learn

Time to Leave

Time to Live

none of the given _____ layer Provides reliable delivery of datagram.
Network

Transport

Datalink
none of the given

Which protocol is used to test different tools.

ICMP

IGMP
TCP/IP
none of the given

Routers use _____ to forward datagrams along prearranged path.
Traffic class

Flow label

Destination address
none of the given

NEXT HEADER field in the base header defines type of header it appears at the end of fixed-size base header.

TRUE

FALSE Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit.

Least

Maximum

Fragment
Frame

Postfix defines how much of address used to identify network.

TRUE

FALSE

_____ contains all information needed to deliver datagram to the destination.

Header

The Source can configure outgoing datagram's to avoid _____

Segmentation
Defragmentation

Fragmentation

None of the given

The Current version of IP-Version 4 is _____ old

18 years

20 years

22 years

none of given_The Header format of IPv6 is entirely different.

TRUE

FALSE

_____ shows senders preference for low latency, high Reliability.

TYPE

SERVICE TYPE

SERVICE PRIORITY

None of the given

The Network Layer Protocol ICMP stands for _____

Instant Control Message Protocol

Internet Control Message Protocol

Initial Control Message Protocol

None of the given

IPv6 address consists of _____

32 Bits

64 Bits

128 Bits

none of the given

_____ is a technique used to Limit datagram size to small MTU of any network

Segmentation

Fragmentation

Encapsulation

none of the given_ICMP message transport is acted upon by getting ICMP _____ in IP.

De-encapsulated

Encapsulated

Segmented

none of the given

IETF stands for _____

Internet Engineering Task Force

Internal Efficient Task Force

Internet Engineering Technical Force

none of the given

Which of the following protocols provide the routing information at the autonomous system level?

BGP

OSPF

RIP

OSPF and RIP

A one-to-many communication between a source and a specific group of hosts is classified as a _____ communication.

Unicast

Multicast

Broadcast

Unicast & Multicast _____ includes a 32-bits address mask with each address, which allows the address to be classful, classless, or subnetted.

RIP

OSPF

BGP

None of the given

In TCP when a computer sends a segment, the _____ and _____ fields refer to incoming data.

ACKNOWLEDGE NUMBER, WINDOW

SEQUENCE NUMBER, WINDOW

ACKNOWLEDGE NUMBER, SEQUENCE NUMBER

None of the given

_____ is used to attach two autonomous systems.

BGP

IGP

EGP

none of the given

Routing inserts or changes values in _____

MAC address

routing table

both (a) and (b)

None of the given_NAT software does not allow a PC to connect with the Internet and act as a NAT device at the same time.

True

False

Each autonomous system used to communicate among autonomous systems by chooses an IGP.

True

False

Interior Gateway Protocols (IGPs) and Exterior Gateway Protocols (EGPs) two broad classes of Internet Routing Protocol.

True

False

The computer uses _____ to inform Local router about the last application when it leaves.

ICMP

IGMP

SNMP

None of the given

IPV6 address with _____ leading zeros is interpreted to hold an IPV4 address.

96

100

120

none of the given. **For _____, information about forwarding is stored in a routing table, which is initialized at system initialization and must be updated as network topology changes.**

Efficiency

Security

Accuracy

Anomalies

Class A mask is 255.0.0.0 which is used for _____

Unicasting

Multicasting

Subnetting

All of the given

When one computer sends an ARP message to another the message travels inside the hardware frame. Technically, placing a message inside a frame for transport is not called encapsulation.

True

False

Which one of these is not a main feature of connectionless service:

It includes extension of LAN abstraction.

It has universal addressing and the data is delivered in packets frames), each with a header.

It combines collection of physical networks into a single virtual network.

It has universal addressing and the data is delivered in packets frames), without a header.

_____ of TCP/IP layering model, corresponds to basic network hardware.

- • **Physical Layer**
- • Network Interface Layer
- • Internet Layer
- • Transport Layer

_____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

- • Physical Layer
- • Network Interface Layer
- • Internet Layer
- • **Transport Layer**

_____ is called an end-to-end protocol because it provide a connection directly from an application on one computer to an application on a remote computer.

- • IP
- • **UDP**
- • TCP
- • None of the given

_____ uses distance vector approach to define routing

- • BGP
- • OSPF
- • **RIP**
- • None of the given

_____ is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

- • **Flood-and -Prune**
- • Configuration-and -Tunneling

- • Core-Based Discovery
- • None of the given

Router that decrements TTL to ___ sends ICMP time exceeded message, with router s address as source address

- • 3
- • 2
- • 1
- • 0

Protocol addresses are abstractions provided by _____.

- • hardware
- • software
- •
- • internet

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

- • True
- • False

ARP is almost always used to bind a ___-bit IP address to a ___-bit Ethernet address.

- • 32, 48
- • 24, 32
- • 32, 64
- • 32, 128

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

- • False
- • True

In which method of Address Resolution Protocol the protocol address is determined by hardware address? Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- • T
- • D
- • C
- • T, C

Which method of Address Resolution Protocol is useful with any hardware? Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- • T
- • C
- • D
- • C, D

In which method of Address Resolution Protocol the implimentation is more difficult? Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- • T, C
- • T
- • C
- • D

Which method of Address Resolution Protocol resolution with minimum delay?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

- • T, D
- • C
- • T
- • **T, C**

The general form of an IP datagram is with a header followed by data. The header contains information that controls where and how the datagram is to be sent.

- • **True**
- • False

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system.

- • True
- • **False**

----- was especially concerned about the lack of high powered computers.

- • IEEE
- • APRA (wrong spelling)
- • EIA
- • **None**

Missing eot indicates sending computer crashed in frame format.

- • **True**
- • False

The -----term refers to the general concept of a small block of data.

- • **Packets**

- • Data
- • Frame
- • None of given

CRC can detect more errors than a simple checksum.

- • **True**
- • False

The network that uses a -----, usually consist of a single long cable to which computer attach.

- • Star topology
- • **Bus topology**
- • Ring topology
- • None of the given

LAN that use ATM technology have a -----

- • Star topology
- • Bus topology
- • Ring topology
- • **None of the given**

A bridge uses ----- to determine which computer is connected to which segment, and uses the ----- to determine whether to forward a copy of frame.

- •
- •
- •
- •

----- has a jitter zero

- • Virtual Private Network
- • **Isochronous Network**
- • Asynchronous Network
- • None of the given

The network with Throughput T and Delay D has a total ----- bits in transit at a time.

- • $D + T$
- • $D - T$
- • **$D \times T$**
- • D / T

One repeater -----, two repeaters ----- the maximum cable length limitation.

- • **Double, triple**
- • Double, 4 time
- • half, triple
- • Double, half

End-to-end delivery service is connection oriented.

- • True
- • **False**

A single networking technology is best for all needs.

- • True
- • **False**

Twice NAT allows a site to run servers.

- • **True**

- • False

_____ device is used for multicasting.

- • Hub
- • Switch
- • **Router**
- • none of the given

_____ does not depend on any particular unicast routing protocol.

- • PIM-DM
- • **PIM-SM**
- • PIM-AM
- • none of the given

A routing table contains_____

- • The destination network ID
- • The hop count to reach the network
- • **The router ID of the next hop**
- • All of the given

_____ can be used to propagate information about remote networks.

- • **Dynamic routing**
- • Static routing
- • Address resolution
- • None of the given

_____ protocol is designed to use within an organization.

- • OSPF

- • MEOSPF
- • **MOSPF**
- • none of the given

NAPT stands for _____

- • Network Address and Protocol Translation
- • **Network Address and Port Translation**
- • Network Address and Packet Translation
- • None of the given

In dynamic routing, the routing table is initialized when system boots.

- • **True**
- • False

OSPF includes _____ address mask with each address.

- • 30Bit
- • **32Bit**
- • 34Bit
- • none of the given

Twice NAT fails if an application uses the IP addresses instead of Domain Name.

- • **True**
- • False

_____ uses window mechanism to control the flow of data.

- • IP
- • UDP
- • **TCP**

- • none of the given

TCP uses _____ mechanism to control the flow of data.

- • door
- • **window**
- • acknowledgment
- • retransmission

IGPs stand for _____

- • Internal Gateway Protocols
- • **Interior Gateway Protocols**
- • Intermediate Gateway Protocols
- • None of the give

_____ protocol uses distance vector algorithm.

- • IGP
- • BGP
- • **RIP**
- • none of the given

_____ measures distance in network hops, where each network between the source and destination counts as single hop.

- • BGP
- • OSPF
- • **RIP**
- • Non of these

Network Address Translation (NAT) requires device to perform packet translation.

- • **True**
- • False

We use the term _____ to refer to a measure of the path that routing software use when choosing a route.

- • routing path
- • **routing metric**
- • routing
- • switching

Part of the 3-way handshake used to create a connection, requires each end to generate a random _____ sequence number.

- • **32 bit**
- • 16 bit
- • 8 bit
- • 64 bit

Reconstruction of original datagram is called reassembly.

- • **True**
- • False

Reliability is the responsibility of the _____ layer.

- • **Transport**
- • Network
- • Physical
- • Internet

_____ is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

- • **Flood-and -Prune**
- • Configuration-and -Tunneling
- • Core-Based Discovery
- • None of the given

In Direct point to point communication adding the Nth computer requires----- new connections.

- • None of the given
- • N^2
- • **N-1**
- • $(N^2 - N)/2$

The number of connections needed for N computer in direct point to point communication is equal to:

- • **$(N^2 - N)/2$**
- • $N(N-1)$
- • N^2
- • None of the given

Hardware that calculates a CRC uses two simple components.

- • AND unit and XOR unit
- • **Shift register and XOR unit**
- • Shift register and AND unit
- • None of the given

In Direct point to point communication adding the Nth computer requires----- new connections.

- • None of the given

- • N^2
- • **$N-1$**
- • $(N^2 - N)/2$

The number of connections needed for N computer in direct point to point communication is equal to:

- • **$(N^2 - N)/2$**
- • $N(N-1)$
- • N^2
- • None of the given

Hardware that calculates a CRC uses two simple components.

- • AND unit and XOR unit
- • **Shift register and XOR unit**
- • Shift register and AND unit
- • None of the given

The Gigabit Ethernet hardware operates at a rate of -----

- • 10 Mbps
- • 100 Mbps
- • **1000 Mbps**
- • None of the given

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- • Greedy algorithm
- • Distance vector algorithm
- • **Dijkstra's algorithm**

- • Non of the given

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

- • **Connectionless service paradigm**
- • Connection-oriented service paradigm
- • Both Connectionless and Connection-oriented service paradigm
- • None of the given

An Internet Address (IP address) is a unique _____ binary number assigned to a host and used for all communication with host

- • 48-bit
- • **32-bit**
- • 24-bit
- • None of the given

The address _____ identifies the physical network to which the computer is attached, while the _____ identifies an individual computer on that network.

- • **prefix , suffix**
- • suffix , prefix
- • suffix , suffix
- • None of the given

_____ places the boundary between the first and second octets

- • **Class A**
- • Class B
- • Class C
- • Class D

_____ places the boundary between the second and third octets.

- • Class A
- • **Class B**
- • Class C
- • Class D

_____ places the boundary between the third and fourth octets.

- • Class A
- • Class B
- • **Class C**
- • Class D

_____ field of header indicates whether a datagram is a fragment or a complete datagram.

- • **FLAGS**
- • FLAGMENT OFFSET
- • IDENTIFICATION
- • None of the given

_____ provides connectionless service.

- • TCP
- • **UDP**
- • IP
- • None of the given

UDP and TCP are both _____ layer protocols

- • Physical
- • Data link

- • Network
- • **Transport**

_____ identifies which application program on receiving computer should receive the data
Logical address

- • Source port
- • **Source port**
- • Destination Port
- • None of the given

_____ identifies the application program that sent the data.

- • Destination Port
- • **Source port**
- • Logical address
- • None of the given

The Border Gateway Protocol (BGP) uses _____ for all communication

- • UDP
- • **TCP**
- • Both UDP and TCP
- • None of the given

Which of the following protocols allows the sender and receiver to enforce policies.

- • RIP
- • OSPF
- • **BGP**
- • RIP and OSPF

ICMP message transport is acted upon by getting ICMP encrypted in IP.

- • True
- • **False**

These ICMP message transport is acted upon _____ as frames on _____

- • Intranet, LAN
- • Internet, WAN
- • Intranet, WAN
- • **Internet, LAN**

Address mask defines how many bits of address are in suffix?

- • True
- • **False**

A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?

- • **True**
- • False

The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address_____.

- • **itself**
- • prefix
- • suffix
- • mask

Find the class of the address:

11110011 10011011 11111011 00001111

- • A

• • C

• • **E**

• • B

As the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of three possible sizes.

Select correct option:

True

False

Question # 4 of 20 (Start time:06:44:07 PM)

Total M - 1

Twice NAT allows a site to run servers.

Select correct option:

True

False

Preliminary version of IP was called _____.

Select correct option:

IP - New Generation (IPng)

IP - Next Generation (IPng)

IP - Net Generation (IPng)

None of the given

As the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of two possible sizes.

Select correct option:

True

False

_____ uses window mechanism to control the flow of data.

Select correct option:

IP

UDP

TCP

none of the given

TCP uses window mechanism to control the flow of data.

The Internet service providers coordinate with the Internet assigned number authority to obtain their network numbers.

Select correct option:

True

False

Which protocol is used to test different tools.

Select correct option:

ICMP

IGMP

TCP/IP

none of the given

USING ICMP TO TEST REACHABILITY:

ICMP can also be used to test different tools. An Internet host A, is reachable from another host B, if datagrams can be delivered from A to B. Ping program tests reachability. It sends datagram from B to A, that echoes back to B. it uses ICMP echo request and echo reply messages. Internet layer includes code to reply to incoming ICMP

echo request messages. _____ identifies the application program that sent the data.

Select correct option:

DestinationPort

Source port

Logical address

None of the given

The Source can configure outgoing datagram's to avoid _____

Select correct option:

Segmentation

Defragmentation

Fragmentation

None of the given

Fragmentation should be avoided. Source can configure outgoing datagrams to avoid fragmentation. Source determines path MTU- smallest network MTU on path from source to destination.

Question # 18 of 20 (Start time:06:57:12 PM)

Total M - 1

The time for acknowledgement to arrive depends on _____

Select correct option:

Distance to destination

Current traffic conditions

Both a and b

none of the given

The time for acknowledgement to arrive depends on:

- Distance to destination
- Current traffic conditions

which is not the type of error messages defined by ICMP.

Select correct option:

Source quench

Time exceeded

Destination unreachable

none of the given

ERROR MESSAGES:

These are as follows:

- Source quench
- Time exceeded
- Destination unreachable
- Redirect
- Fragmentation required

Question # 1 of 20

_____ is used to attach two autonomous systems.

Select correct option:

BGP

IGP

EGP

none of the given

Question # 2 of 20

Due to revolutionalization of IP-V6 the speed has increased from _____

Select correct option:

56kbps to 512kbps

512kbps to 1gbps

56kbps to 1gbps

none of the given

Scale is also dramatically changed. Size from a few tens to a few tens of millions of computers has been revolutionized. Speed has increased from 56Kbps to 1Gbps. Also there is an increased frame size in hardware

Question # 3 of 20

Whenever it handles a packet, IP software needs to separate the destination address into a prefix.

Select correct option:

True

False

COMPUTING THE CLASS OF AN ADDRESS:

Whenever it handles a packet, IP software needs to separate the destination address into a prefix and suffix.

Question # 4 of 20

TTL stands for _____

Select correct option:

Time to Learn

Time to Leave

Time to Live

none of the given

Question # 5 of 20

IGPs stand for _____

Select correct option:

Internal Gateway Protocols

Interior Gateway Protocols

Intermediate Gateway Protocols

None of the given

Question # 6 of 20

_____ identifies which application program on receiving computer should receive the data

Select correct option:

Logical address

Source port

Destination Port

None of the given

Field Destination Port identifies which application program on receiving computer should receive the data. While field source port identifies the application programs that sent the data.

Question # 7 of 20

_____ encapsulates IP datagram as data area in hardware frame.

Select correct option:

Network Interface Layer

Datalink Layer

Network Layer

None of the given

The network interface layer encapsulates an entire datagram in the data area of a hardware frame

vuzs.info

Question # 8 of 20

Class A mask is 255.0.0.0 which is used for _____

Select correct option:

Unicasting

Multicasting

Subnetting

All of the given

class A mask is 255.0.0.0 which is used for subnetting.

Question # 9 of 20

NAT is not useful at a residence with Cable Modem or DSL connectivity.

Select correct option:

True

False

Question # 10 of 20

Autonomous System chooses a routing protocol to exchange routing information which is passed before being summarized to another group.

Select correct option:

True

False

Question # 11 of 20

_____ layer Provides reliable delivery of datagram.

Select correct option:

Network

Transport

Datalink

none of the given

Transport layer provides reliable delivery. Network layer – IP – can detect and report errors without actually fixing them. It focuses on datagram delivery. Application layer is not interested in differentiating among delivery problems at intermediate routers

Question # 12 of 20

The process of using a routing table to select a next hop for a given datagram is called_____.

Select correct option:

Encapsulation

Reassembling

Routing or forwarding

None of the given

The process of using a routing table to select a next hop for a given datagram is called routing or forwarding.

Question # 13 of 20

The process of learning the path MTU is known as path MTU discovery.

Select correct option:

True

False

The process of learning the path MTU is known as path MTU discovery.

Question # 14 of 20

As the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of _____ possible sizes.

Select correct option:

three

two

four

five

Three

As the Internet grew, the original Classful addressing scheme became a limitation.

The IP address space was being exhausted because all networks had to choose one of three possible sizes. Many addresses were unused.

Two new mechanisms were invented to overcome the limitations, which are as follows:

- Subnet addressing
- Classless addressing

Question # 15 of 20

End to End delivery Service of IP datagram is _____

Select correct option:

Connection oriented

Connectionless

both a and b

none of the given

Question # 16 of 20

EGP stands for _____

Select correct option:

Extension Gateway Protocol

Exterior Gateway Protocol

Explicit Gateway Protocol
none of the given

Question # 17 of 20

Hashing is the transformation of a string of characters into a usually shorter fixed-length value or a key that represents the original string.

Select correct option:

True

False

Hashing is the transformation of a string of characters into a usually shorter fixed-length value or a key that represents the original string. Hashing is used to index and retrieve items in a database because it is faster to find the item using the shorter hashed key than to find it using the original value. It is also used in many encryption algorithms.

Question # 18 of 20

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a ____ Ethernet address.

Select correct option:

16-bit

48-bit

64-bit

128-bit

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a 48-bit Ethernet address

Question # 19 of 20

Reliability is not the responsibility of the Transport layer.

Select correct option:

True

False

Reliability is the responsibility of the Transport layer. In TCP/IP, TCP provides reliable transport service. Most Internet applications use TCP as no other protocol has proved to work better.

Question # 20 of 20

IPv6 address with _____ leading zeros is interpreted to hold an IPv4 address.

Select correct option:

<http://www.vuzs.info/>

96

100

120

none of the given

IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address **Question # 1 of 20**

NAT software does not allow a PC to connect with the Internet and act as a NAT device at the same time.
Select correct option:

True

False

Question # 2 of 20

TCP uses the term segment to refer to a _____.

Select correct option:

packet

message

both (a) and (b)

None of the given

both a and b?

TCP uses single format for all messages. TCP uses the term segment to refer to a message. Each message sent from TCP on one machine to TCP on another machine uses this format including data and acknowledgement.

Question # 3 of 20

The Universal Datagram Protocol is a message-oriented protocol.

Select correct option:

True

False

UDP has the following characteristics.

- It is an end-to-end protocol. It provides application-to-application communication.
- It provides connectionless service.
- It is a Message-Oriented protocol.
- It uses best-effort delivery service.
- It follows arbitrary interaction.
- It is operating system independent

Question # 4 of 20

_____ field tells the receiver how to order fragments within a given datagram.

Select correct option:

FLAGS

FLAGMENT OFFSET

IDENTIFICATION

None of the given

The FRAGMENT OFFSET field tells a receiver how to order fragments within a given datagram

Question # 5 of 20

IP datagram can contains _____ octets.

Select correct option:

0-65,535

1-65,535

1-65,536

none of the given

Datagrams can have different sizes i.e.

Header area is usually fixed (20 octets) but can have options. Data area can contain between 1 octet and 65.535 octets (216-1).

Usually, data area is much larger than header.

Question # 6 of 20

The amount of buffer space available at any time is called the window.

Select correct option:

True

False

Question # 7 of 20

_____ Source is responsible for fragmentation.

Select correct option:

IPV4

IPV6

IPv6 source (not intermediate routers) is responsible for fragmentation. Routers simply drop datagrams larger than network MTU (Maximum Transmission Unit). So source must fragment datagram to reach destination.

Question # 8 of 20

Routers use _____ to forward datagrams along prearranged path.

Select correct option:

Traffic class

Flow label

Destination address

none of the given

Routers use flow label to forward datagrams along prearranged path

Question # 9 of 20

_____ protocol uses three way handshake to begin a connection.

Select correct option:

UDP

TCP

IP

none of the given

Question # 10 of 20

Typical internet routing uses uses a combination of two metrics_____.

Select correct option:

routing metrics
administrative cost and routing metrics
hop cost

administrative cost and hop count

Question # 14 of 20

Network Address and Port Translation (NAPT) is by far the most popular form of _____

Select correct option:

Network Address Transmission

Network Address Translation

Network Address Transformation

None of the given

Question # 15 of 20

The routers within an autonomous system use a _____ to exchange routing information.

Select correct option:

Interior Gateway protocols(IGPs)

Exterior Gateway protocol (EGPs)

Both Interior Gateway protocols(IGPs) and Exterior Gateway protocol (EGPs)

None of the given

It is used among routers within autonomous system. The destinations lie within IGP.

Question # 17 of 20

Interior Gateway Protocols (IGPs) and Exterior Gateway Protocols (EGPs) two broad classes of Internet Routing Protocol.

Select correct option:

True

False

Question # 18 of 20

The Network Layer Protocol ICMP stands for _____

Select correct option:

Instant Control Message Protocol

Internet Control Message Protocol

Initial Control Message Protocol

None of the given

Question # 19 of 20

The Current version of IP-Version 4 is _____ old

Select correct option:

18 years

20 years

22 years
none of given

Question # 20 of 20 (Start time: 11:05:43 AM) Total M - 1

TCP is a connectionless and reliable transport protocol.

Select correct option:

True

False

Question # 1 of CBR stands for.....

Constant Byte Rate

Constant Bit Rate

Connection break Recovery

None of the above

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

Question # 2 of

The routing table in interior and exterior switches has which one of the following?

It should have next hop for each possible destination

The next hop in table must be on shortest path to destination

Both (a) and (b)

None of the above

Question # 3 of 10

Which one of the following is a method for computing routing table information?

Manual entry

Static routing

Boot time routing

None of the above

How to compute routing table information:

Static routing - at boot time

Dynamic routing - allow automatic updates by a program

Question # 4 of 10

Which one of the following is the responsibility of Dijkstra's algorithm?

To compute the shortest path between two nodes

To extract next hop information from path information

To insert next hop information into routing table

All of the above

While computing shortest path, first we assume graph representation of network at each node then we use Dijkstra's algorithm to compute shortest path from each node to every other node. Then extract next hop information from resulting path information and

insert next hop information into routing tables.

Question # 5 of 10

Routing table is used to keep the information about.....

Destination

Medium of transmission

Routers used in network

All of the above

The routing table contains list of destination networks and next hop for each destination.

Question # 6 of 10 (Start time: 04:48:43 PM) Total M - 1

Frame relay is.....

Connection oriented service

Connectionless service

It is typically ranges from 5Mbps to 1000Mbps.

None of the above

It is used for Telco service for delivering blocks of data. It is connection based service and must contract with Telco for circuit between two endpoints. It is typically 56kbps or 1.5Mbps and can run to 100Mbps.

Question # 7 of 10 (Start time: 04:49:43 PM) Total M - 1

Which of the following is not a guided medium?

Twisted-pair cable

Fiber-optic cable

Atmosphere

Coaxial cable

Question # 8 of 10

Which of the following statement is wrong regarding PVC?

PVC last as long as the customer pay the periodic fee for its use.
Forwarding tables are automatically restored after power of equipment failure.

Forwarding table entries for PVC's are dynamically configured.

All of the above are wrong

ATM can provide customers with virtual circuits that look like traditional leased digital circuits. Such permanent virtual circuits (PVC) last as long as the customer pay the periodic fee for its use. The forwarding tables are automatically restored after power of equipment failure. The forwarding table entries for such permanent VC's are statically configured, the terms used by Telco's for this is provisioning

Question # 9 of 10

Which of the following statement is true regarding ATM?

Select correct option:

ATM is a single technology that is designed to meet the goals of both LANs and WANs.

ATM uses the concept of connection-oriented networking.

Both (a) and (b)

None of the above

ATM is a single technology that is designed

to meet the goals of both LANs and WANs.

ATM uses the concept of connection-oriented networking

Question # 10 of 10

Dijkstra's algorithm is used to compute the.....

shortest path between two nodes

number of nodes used in the network

distance between different nodes

none of the above

While computing shortest path, first we assume graph representation of network

at each node then we use Dijkstra's algorithm to compute shortest path from each node to

every other node. Then extract next hop information from resulting path information and

insert next hop information into routing tables

1. If a computer on the network shares resources for others to use, it is called ____

a. **Server**

b. Client

c. Mainframe

2. Terminators are used in _____ topology.

a. **Bus**

b. Star

3. In _____ topology, if a computer's network cable is broken, whole network goes down.

- a. Bus
- b. Star

Answer : a

4. For large networks, _____ topology is used.

- a. Bus
- b. Star
- c. Ring

5. ISO stands for

- a. International Standard Organization
- b. International Student Organization
- c. Integrated Services Organization

6. ISO OSI model is used in

- a. Stand alone PC
- b. Network environment

7. Network cable lies on _____ layer

- a. Application
- b. Network
- c. Physical

8. _____ layer decides which physical pathway the data should take.

- a. Application
- b. Network
- c. Physical

9. ISDN is an example of _____ network

- a. Circuit switched
- b. Packet switched

10. X.25 is an example of _____ network

- a. Circuit switched
- b. Packet switched

11. _____ allows LAN users to share computer programs and data.

- a. Communication server
- b. Print server
- c. File server

12. Print server uses _____ which is a buffer that holds data before it is send to the printer.

- a. Queue
- b. Spool
- c. Node

13. A standalone program that has been modified to work on a LAN by including concurrency controls such as file and record locking is an example of _____

- a. LAN intrinsic software
- b. LAN aware software
- c. Groupware
- d. LAN ignorant software

14. The _____ portion of LAN management software restricts access, records user activities and audit data etc.

- a. Configuration management
- b. Security management
- c. Performance management

15. What is the max cable length of STP?

- a. 100 ft
- b. 200 ft
- c. 100 m
- d. 200 m

16. What is the max data capacity of STP?

- a. 10 mbps
- b. 100 mbps
- c. 1000 mbps
- d. 10000 mbps

17. Which connector STP uses?

- a. BNC
- b. RJ-11
- c. RJ-45
- d. RJ-69

18. What is the central device in star topology?

- a. STP server
- b. Hub/switch
- c. PDC
- d. Router

19. What is max data capacity for optical fiber cable?

- a. 10 mbps
- b. 100 mbps
- c. 1000 mbps
- d. 10000 mbps

20. Which of the following architecture uses CSMA/CD access method?

- a. ARCnet
- b. Ethernet

Question 1

NAT is not useful at a residence with Cable Modem or DSL connectivity.

True

False

NAT is useful at a residence with Cable Modem or DSL connectivity

Question 2

_____ includes a 32-bits address mask with each address, which allows the address to be classful, classless, or subnetted.

RIP

OSPF

BGP

None of the given

OSPF includes a 32-bit address mask with each address, which allows the address to be classful, classless, or subnetted.

Question 3

NAPT stands for _____

Network address and protocol translation

Network address and port translation

Network address and packet translation

None of the given

But Network Address and Port Translation (NAPT) (which is another modified form of NAT)

Question 4

Which of the following are interior routing protocols?

RIP

OSPF

BGP

RIP and OSPF

The Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) are the two most well-known interior routing protocols used in the TCP/IP suite

Question 5

NAT stands for _____

Network address translation

Network address transmission

Network address test

None of the given

Question 6

In IP routing, forwarding refers to _____ transfer.

Packet

Datagram

Message

None of the given

FORWARDING refers to datagram transfer.

Question 7

Basic NAT does not work well for communication initiated from the Internet.

True

False

Basic NAT does not work well for communication initiated from the Internet

Question 8

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system

True

False

OSPF has designed as an Interior Gateway Protocol used to pass routing information among routers within an autonomous system.

Each Autonomous System chooses a routing protocol to exchange routing information which is summarized before being passed to another group.

Question 9

OSPF is based on-----

Distance vector routing

Link state routing

Path vector routing

Distance vector and link state routing

LINK-STATE ALGORITHM:

OSPF uses link-state routing.

Question 10

The Border Gateway Protocol (BGP) uses_____ for all communication

UDP

TCP

UDP and TCP

None of the given

"It uses reliable transport (TCP).

BGP uses the TCP/IP protocol on port 179 to establish connections

Question 11

_____ identifies the application program that sent the data.

Destination

Source port

Logical address

None of the given (not sure)

(Reference not available)

Question 12

_____ can be used to propagate information about remote networks

Dynamic routing

Static routing

Address resolution

None of the given

Dynamic routing can be used to propagate information about remote networks.

Question 13

Which of the following protocol uses UDP for all message transmission?

OSPF

BGP

RIP

BGP AND RIP

RIP uses UDP for all message transmissions

Question 14

OSPF stands for _____

Open Source Path First Protocol

Open shortest path first

Open Secure Path First Protocol

open shortest path first

(OSPF) known as the Open Shortest Path First Protocol).

Question 15

Typical internet routing uses uses a combination of two metrics_____.

Mouthing metrics

Administrative cost and routing metrics

Hop cost

Administrative cost and hop count

Typical internet routing uses uses a combination of two metrics Administrative cost and hop count

Question 16

_____ is a current version of BGP.

2

3

4

None of these

The current version is four (BGP-4).

Question 17

OSPF doesn't allow a router to introduce routes learned from another means.

True

False

OSPF allows a router to introduce routes learned from another means (e.g., from BGP).

Question 18

Preliminary version of IP was called _____.
Select correct option:

- IP - New Generation (IPng)
- IP - Next Generation (IPng)
- IP - Net Generation (IPng)
- None of the given

A preliminary version of IP was called IP- Next Generation (IPng).

Question 19

Source is responsible for fragmentation.
Select correct option:

IPv4

IPv6

IPv6 source (not intermediate routers) is responsible for fragmentation

Question 20

IPv6 address consists of _____.
Select correct option:

- 32 Bits
- 64 Bits

128 Bits

none of the given

IPV6 addresses are 128 bits

Question

Header contains all information needed to deliver datagram to the destination computer. But which one of the following is not included:

Destination address

Source address

Rectifier

Other delivery information

Header contains all information needed to deliver datagram to the destination computer

Destination address

Source address

Identifier

Other delivery information

_____ **field is used to identify a specific path through the network**

Flow label

Traffic class

Both a and b

None of the given

Flow label is partitioned into a TRAFFIC CLASS field and a separate FLOW LABEL field used to identify a specific path thorough the network.

The Source can configure outgoing datagram's to avoid _____

Segmentation

Defragmentation

Fragmentation

Source can configure outgoing datagrams to avoid fragmentation

Question

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit.

Least

Maximum

Fragment

Frame

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the Maximum Transmission Unit (MTU).

Question

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a _____ Ethernet address.

16-bit

48-bit

64-bit

128-bit

ARP is almost always used to bind a 32-bit IP address to a 48-bit Ethernet address

Question

Fragmentation when using ICMP for path MTU should be avoided.

True
False

Question

End to End delivery Service of IP datagram is _____

Connection oriented
Connectionless
both a and b
none of the given

End-to-end delivery service is connection less.

Question

HEADER LEN field gives size of extension header.

False
True

Question

_____ Serve Same purpose in Internet as frames on LAN

Packets
Virtual Packets
Virtual Frames
Both b and c

VIRTUAL PACKETS:

These packets serve same purpose in Internet as frames on LAN.

Question

Header contains all information needed to deliver datagram to the destination computer. But which one of the following is not included

Destination address
Source address
Rectifier
Other delivery information

Header contains all information needed to deliver datagram to the destination computer

- *Destination address*
- *Source address*
- *Identifier*
- *Other delivery information*

Question

_____ encapsulates IP datagram as data area in hardware frame.

Select correct option:

Network Interface Layer

Data link Layer

Network Layer

None of the given

Network interface layer encapsulates IP datagram as data area in hardware frame.

Question

_____ field is used to identify a specific path through the network
Select correct option:

Flow label

Traffic class

Both a and b

None of the given

Flow label is partitioned into a TRAFFIC CLASS field and a separate FLOW LABEL field used to identify a specific path thorough the network.

Question

IPV6 address with _____ leading zeros is interpreted to hold an IPV4 address.
Select correct option:

96

100

120

none of the given *IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address*

Question

A datagram cannot be larger than _____ of a network over which it is sent.
Select correct option:

MTU

Size

IP header

None of the given

Datagram encapsulated in a hardware frame must be smaller than the MTU for that hardware

Question

The Source can configure outgoing datagram's to avoid _____

Segmentation

Defragmentation

Fragmentation

None of the given

Fragmentation should be avoided. Source can configure outgoing datagrams to avoid fragmentation

Question

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit.

Select correct option:

Least

Maximum

Fragment

Frame

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the Maximum Transmission Unit (MTU).

Question

_____ Protocol provides error reporting mechanism.

Select correct option:

IGMP

SNMP

ICMP

none of the given

Internet Control Message Protocol (ICMP) provides error reporting mechanism.

Question

_____ Source is responsible for fragmentation.

Select correct option:

IPV4

IPV6

IPv6 source (not intermediate routers) is responsible for fragmentation

MTU Stands for _____

Minimum transmission unit

Maximum transmission unit

Multicast transmission unit

None of the given

MTU (Maximum Transmission Unit)

CS-610 (Computer Networks)

Grand Quiz Dec 2020

Prepared By: MCS of Virtualians

If you found any error in the file, Please correct it accordingly

1. In Cyclic Redundancy Checking, CRC is _____.
 - a. Divisor
 - b. Quotient
 - c. Remainder page 20
 - d. Dividend
2. The satellite or radio topology in which all computers are connected to each other via satellite or radio wave is a kind of
 - a. Broadcast network page 5
 - b. Point-to-Point network
3. Which of the following two operations are performed by Ethernet CD?
 - I. It monitors outgoing signals.
 - II. Grabbled signal is interpreted as a collision.
 - III. Signals are sent back to the sender.
 - a. I and III
 - b. II and IV
 - c. II and III
 - d. I and II page 28
4. The process of forwarding a packet is called _____.
 - a. Routing page 58
 - b. Processing
 - c. Hierarchical Addressing
 - d. Source Addressing
5. Which statement is correct about Network Interface Card (NIC)?

I. NIC contains sufficient hardware to process data independent of system CPU

II. NIC looks like any other I/O device to system CPU

III. NIC also receives interrupt on incoming data.

a. I and II

b. II and III

c. I and III

d. All of the given options

page 40

6. While transmitting data from source A to destination B, 4 bits are changed during the transmission then the error is a _____.
- a. **Burst error** page 22
 - b. Single error
 - c. Double error
 - d. Logic error
7. The switch that has no attached computers is called _____.
- a. Packet Switch
 - b. Exterior Switch
 - c. **Interior Switch** page 59
 - d. External Switch
8. _____ is the most popular wiring scheme because of lowest cost.
- a. 5 Base T
 - b. **10 Base T** page 43
 - c. 100 base T
 - d. 1000 base T
9. In _____ technique, all members of the network can send data only on the specific time slot?
- a. CDMA
 - b. FDMA
 - c. **CSMA** page 43
 - d. TDMA
10. If a sender with a 100 Megabit NIC and receiver with 10 Megabit NIC wants to communicate with each other at which speed they can communicate?
- a. **100 Megabit** page 40
 - b. 110 Megabit
 - c. 1000 Megabit
 - d. 10 Megabit

11. Missing _____ in a frame format indicates receiving computer missed beginning of the message.

a. eot

b. **soh** page 16

c. sot

d. eoh

12. The _____ of errors is easier than the _____ of errors.

a. correction; detection

b. **detection; correction** cs601 handouts

c. correction; creation

d. detection; manipulation

13. The _____ of errors is difficult than the _____ of errors.

- a. **correction; detection** google
- b. detection; correction
- c. correction; creation
- d. detection; manipulation

14. Source Independence phenomenon allows fast and efficient _____.

- a. Source Addressing
- b. Routing
- c. **Packet Switching** page 58
- d. Store and forward

15. In distributed route computation process each packet switch computes its routing table locally and sends messages to the _____.

- a. Router
- b. Graph
- c. **Neighbors** page 63
- d. WAN

16. The ____ layer of the OSI model takes the data from the physical layer and perform error checking.

- a. Transport
- b. **Data link** google
- c. Application
- d. Presentation

17. CBR stands for _____.

- a. Constant Byte Rate

- b. Constant Bit Rate**
- c. Connection break Recovery
- d. Constant Borrow Rate

18. A network that uses _____ usually consist of a single long cable to which
computer attach.

- a. Star Topology
- b. Ring Topology

- c. Bus Topology** page 26
- d. Bus and Ring Topology

19. If simultaneous transmission occurs, the frames interfere with each other and this phenomenon is called _____.

- a. packet loss

- b. collision** page 28
- c. delayed transmission
- d. coordination

20. In _____ all computers are attached to a central hub

a. Ring topology

b. Star topology

page 31

c. Hub topology

d. Bus topology

21. For adding a new computer in a point to point network _____ new connection

will be required

a. N

b. N-1

page 23

c. N-2

d. 2n

22. The fastest Ethernet hardware operates at a rate of _____

a. 1000 bps

b. 100 Mbps

page 27

c. 10 Mbps

d. 1000 Mbps

23. Which of the following the benefit to point to point communication?

a. Quick to configure

b. Security implementation

page 23

c. Cost

d. Less availability

24. ATM offers _____ speed and _____ delay

a. High, High

b. High, low

page 72

- c. Low, low
- d. Low, high

25. Thick Ethernet is known as _____

- a. 10 Base 2
- b. 10 Base 5 google
- c. 10 Base T
- d. 10 Base 10

26. Both interior and exterior switches _____ packet and they also need routing

tables.

- a. Create
- b. Delay
- c. Forward page 59
- d. Start

27. _____ is the rate at which data can be sent through the network

a. **Throughput** page 79

b. Propagation delay

c. Access delay

d. Switching delay

28. In a ping command, small –sized information packets of _____ are sent.

a. TCP

b. UDP

c. **ICMP** page 11

d. IGMP

29. Many WANs use _____ addressing of efficiency

a. Sequence

b. Forward

c. **Hierarchical** page 56

d. Sequential

30. Which of the following is not a function of ping commands?

a. Local computer network issue

b. **Congestion in the network**

c. Relay nodes page 11 and google d.
Remote host availability

31. Packet switched can be linked together to form a _____

a. **LAN**

b. WAN page 56

c. Bridge

d. None of the given option

32. Fixed network is a type of network which is classified with respect to the _____ factor

a. Size

b. Connectivity

c. Medium

d. Mobility

page 6

33. Parity bit can detect _____ error(s)

a. 4

b. 3

c. 2

page 19 (even or odd)

d. 1

34. When we will try to connect to the internet which of the following protocol will be used?

a. Telnet

b. **TCP/IP**

page 84

c. ICMP

d. IPx

35. How many computers are mandatory to form a network?

a. 1

b. **2**

google

c. 4

d. 8

36. _____ is standard wireless LAN that uses radio signals at 2.4GHz.

a. IEEE 803.16

b. IEEE 802.15

c. IEEE 802.3

d. **IEEE 802.11**

page 29

37. LAN interface hardware obeys _____ access rule.

a. **CSMA/CD**

page 34

b. Round robin

c. First come first served

d. High priority

38. The most important task a bridge performs is _____.

a. Packet switching

b. Line controlling

c. Noise controlling

d. **Frame filtering**

page 50

39. ATM uses small, fixed-sized packets called cells and each cell has _____

octets.

a. 47

b. 52

c. 53 page 66

d. 55

40. _____ can't be guaranteed with variable length packets as easily as it can

with fixed length cells.

a. Errors

b. Noise

c. Quality of Service page 72

d. None of the given

41. Which multiplexing technique transmits digital signals?

a. WDM

b. FDM

c. **TDM** page 15

d. GSM

42. Connections in Asynchronous transfer mode are called _____.

a. **Virtual channels** page 67

b. Virtual Path Identifier

c. Cells

d. Cell tax

43. A _____ is a device that combines multiple signals and forwards into a single

line.

a. Modem

b. **Multiplexer** page 41

c. Repeater

d. Layer 3 Switch

44. _____ is inflexible.

a. Routing

b. Searching

c. Processing

d. **Static routing** page 62

45. If a hub is used to connect many computers in a LAN, then only _____

computers can communicate at a given time.

a. Three

b. **Two** page 54

c. Four

d. One

46. _____ is the term used for variance in transmission delays.

a. Friction

b. Segment

c. Checksum

d. **Jitter** page 66

47. _____ is also called self healing network.

a. ATM

b. FDDI

c. **Token Ring** page 46

d. Frame relay

48. _____ and _____ are the two frame types.

- a. Internal and external
- b. Single and dual
- c. Implicit and explicit
- d. **Multicast and broadcast**

page 35

49. Bluetooth uses shared medium and _____ instead of coaxial cable.

- a. **Radio waves**
- b. Infrared waves
- c. Micro waves
- d. Low frequency waves

page 29

50. Which of the following is not a guided medium?

- a. Twisted-pair cable
- b. Fiber-optic cable
- c. **Atmosphere**
- d. Coaxial cable

google

51. As a result of ARPA research, the first network was established which was named as _____.

- a. INTERNETWORKING
- b. ARPNET
- c. **ARPANET**
- d. PACKET NETWORK

page 9

52. Asynchronous transfer mode (ATM) can dynamically establish a _____.

- a. **Switched virtual circuit**
- b. Virtual connection identifier
- c. Virtual path indicator

page 70

d. None of the given options

53. A typical port on an ATM switch operates at _____ or _____ higher.

a. OC-2 speed (155Gbps)

b. OC-2 speed (1000Mbps)

c. OC-3 speed (155Mbps)

page 72

d. OC-3 speed (100Gbps)

54. 10Base-T Ethernet is often called a _____.

a. ring shaped star

b. ring shaped bus

c. star shaped bus

page 46

d. star shaped ring

55. _____ allows automatic updates.

- a. **Dynamic routing** page 62
- b. Static routing
- c. Routing
- d. Default routing

56. Handling variable length packets leads to memory _____.

- a. **Fragmentation** page 72
- b. Management
- c. Location
- d. Allocation

57. In order to forward a copy of frame, the bridge uses _____ address.

- a. Network
- b. **Destination** page 51
- c. IP
- d. Source

58. We may use _____ to extend local area network.

- a. Bridge, Router
- b. Bridge, Hub, Router
- c. **Bridge, Hub, Repeater** page 149
- d. Bridge, Hub, Repeater, Router

59. All of the given options are the types of delay except _____.

- a. Propagation
- b. Access
- c. **Throughput** page 79
- d. Queuing

60. While computing shortest path in a graph, next hop information is inserted into

_____.

a. **Routing table** page 62

b. Trailer

c. Header

d. Register

61. LAN interface uses _____ to copy frame data directly from main memory

a. **DMA** page 34

b. FDDI

c. Hard disk

d. Flash

62. The Gigabit Ethernet hardware operates at a rate of _____.

- a. 10 Mbps
- b. 100 Mbps

c. 1000 Mbps

reference book page 263

d. 1000 Gbps

63. In the early resource sharing era, which resources were shared among users?

a. Computation

page 9

b. Floppy Drives

c. Hard Disk

d. Printers

64. If two buildings are located far from each other then bridge, and _____ can

be used to connect two LAN.

a. Optical fiber

page 51

b. Ethernet modem

c. Switch

d. NIC

65. Which type of network consists of wireless connection of laptops in a university classroom?

a. WAN

b. MAN

c. Fog

d. LAN

google

66. A typical bridge has _____.

a. One NIC, a CPU a memory and a ROM.

b. Two NICs, a CPU a memory and a ROM.

- c. RAM, ROM and CPU.
- d. A cache memory, ROM and CPU.

67. _____ is used for compressed audio and video, where the data rate depends on the level of compression that can be achieved.

- a. Constant Bit Rate
- b. Stable Bit Rate
- c. **Variable Bit Rate**
- d. Available Bit Rate

page 71

68. There are _____ types of data stuffing.

- a. 1
- b. **2**
- c. 0
- d. 5

page 17

69. Dijkstra's algorithm can accommodate weights on _____ in graph.

a. Nodes

b. Edges page 62

c. Path

d. Connections

70. _____ masks your IP address.

a. Firewall

b. Antivirus

c. VPN

d. Incognito mode

71. Static routing computes routing table information at _____ time.

a. Load

b. Boot page 62

c. Execute

d. One

72. _____ was the first automated tool required to see whether the given

computer is online or not.

a. Probing program

page 11

b. PING program

c. status program

d. None of the above

73. In which type of error detection, the arithmetic sum is calculated before and after the transmission of data at both communication ends?

- a. CRC
- b. Party

c. **Checksum**

page 19

d. Hash Function

74. Computer sends a connection request to the _____ to which it is attached

while establishing a switched virtual circuit

a. Bridge

b. **Switch**

page 70

c. VPI

d. Virtual circuit

75. The bridges communicate with each other on the network and use _____

algorithm to decide which bridge will not forward frames if a cycle occurs.

a. Dijkstra

b. Shortest path first

c. Vector distance

d. **Distributed Spanning Tree**

page 53

76. Ethernet uses a _____ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

a. 64

b. 48

c. 32

d. 16

77. Weights on graph edges reflect cost of _____.

a. Nodes

b. Traversing edge page 63

c. Trending edge

d. Path

78. A _____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

a. Static addressing scheme page 34

b. Configurable addressing scheme

c. Dynamic addressing scheme

d. None of the given

79. In _____ topology, if the switch goes down all the computers will be disconnected.

a. Star

b. Ring

c. Bus

d. Mesh

80. Which type of network consists of wired computers in a university classroom?

a. MAN

- b. PAN
- c. WAN

d. LAN

81. If you as a network administrator want to know the traffic flow of your data while communicating with a remote computer which of the following will be used to know about intermediate routers?

a. Ipconfig

b. Ping

page 11

c. Traceroute

d. Arp

82. Hardware address is also called _____.

a. Physical address

b. Logical address

c. IP address

d. Port address

83. Thick Ethernet also requires _____ to avoid signal reflectance.

- a. Transceivers
- b. AUI cable
- c. Drop cable
- d. Terminators

page 41

84. Which of the following network is using the TCP/IP model globally?

- a. PAN
- b. SAN
- c. LAN
- d. The Internet

85. VPNs connections are sometimes called _____.

- a. Tunnels
- b. Open networking
- c. Routing
- d. Public network

page 75

86. To reduce _____ computers can be connected and distributed to a number of

hubs, then the hubs can be connected to a switch.

- a. Costs
- b. Quality
- c. Efficiency
- d. Noise

page 54

87. If a _____ is used the communication between two computers will not

affect the other pair of computers and they can communicate at the same time

- a. Hub
- b. Switch

page 54

- c. Repeater
- d. Amplifier

88. The length of hardware address is _____.

- a. One to ten bytes
- b. One to five bytes
- c. One to six bytes
- d. One to seven bytes

page 34

89. A network analyzer is also called _____.

- a. Network manager
- b. Network administrator
- c. Network monitor
- d. Networksupervisor

page 37

90. No any error detection scheme is perfect because transmission errors can affect

_____.

- a. Data
- b. Additional Information
- c. **Additional information & data** google
- d. Transmission medium

91. The token ring network topology is a ring but wiring topology is a _____.

- a. **Star** page 44
- b. bus
- c. tree
- d. mesh

92. _____ Addresses are used in an ethernet frame.

- a. IP
- b. Port
- c. Logical
- d. **MAC**

93. AUI cable connects from NIC to _____.

- a. BNC Connector
- b. RJ-45 Connector
- c. **Transceiver** page 41
- d. Terminator

94. Public networks are _____.

- a. More secure than LAN
- b. **Flexible** page 74

- c. Limited in boundary
- d. Intranet

95. The maximum cable length of a LAN is fixed because _____.

- a. Additional hardware are not suitable for the LAN extension.
- b. The electrical signal level gets weaker as it travels.
- c. Additional software are not suitable for the LAN extension
- d. The electrical signal level gets stronger as it travels.

Page 48

96. In _____ network topology is separated from route computation.

- a. Local area
- b. Link-state routing
- c. Static routing
- d. Dynamic routing

page 64

97. Which layer of the OSI Model directly communicates and controls the transmission medium:

- a. Transmission
- b. Physical
- c. Application
- d. Network

98. In _____ technique data is treated as a sequence of integers and their arithmetic sum is computed and the carry bits are added to the final sum.

- a. CRC
- b. Bit stuffing
- c. Checksum
- d. 2D Parity

page 19

99. A network with _____ jitter takes more time to deliver some packets than others.

- a. Zero
- b. High
- c. Low
- d. Moderate

page 80

100. A network uses _____ to arrange computers in a single closed loop.

- a. Star Topology
- b. Mesh Topology
- c. Ring Topology
- d. Bus Topology

page 25

101. Which of the following tasks is not done by the second layer of the OSI model?

- a. Flow control
- b. Framing

c. **Channel coding**

google

d. Error control

102. ATM is a single technology designed to meet the goals of both _____
and _____.

a. LAN & PAN

b. **LAN & WAN**

page 66

c. LAN & MAN

d. MAN & WAN

103. LAN and WAN are classified according to their _____.

a. **Size**

page 4

b. Connectivity

c. Medium

d. Mobility

104. The identifier that is used for data transfer in the virtual circuit network is called _____.
- a. Virtual circuit identifier
 - b. Global address
 - c. Network identifier
 - d. IP identifier
105. In Direct point to point communication adding the Nth computer requires _____ new connections
- a. N^2
 - b. $N-1$ page 23
 - c. $(N^2 - N)/2$
 - d. $N(N-1)$
106. Which of the following is the correct function of bridge?
- a. MTU path discovery
 - b. Byte stuffing
 - c. Routing information
 - d. Filtering and forwarding a frame page 50, 51
107. Point to point topology is
- a. Size based
 - b. Mobility based
 - c. Connectivity based page 5
 - d. Medium based
108. IEEE LLC / SNAP header is _____ which is used to specify the type of data.
- a. 8 octets google
 - b. 16 octets
 - c. 32 octets
 - d. 64 octets

109. IEEE LLC / SNAP header is _____ which is used to specify the type of data.

- a. CRC
- b. Bit stuffing
- c. Checksum
- d. 2D Parity

110. What is the largest network in existence?

- a. The internet google
- b. A PAN
- c. A Fog
- d. Edge Network

111. The frame format of any LAN technology must consists of ____ and ____ -

- a. Payload and CRC
- b. Frame header and CRC
- c. Payload and Frame type

d. **Frame header and payload**

page 36

112. _____ can occur a LANs when a packet is delayed because the network is busy

a. **Jitter**

page 66

- b. Loop
- c. Deadlock

d. variance

113. Which of the following cannot be used to extend a LAN?

- a. Repeaters
- b. Switches
- c. Routers

d. **Fiber modems**

114. Links-state algorithm is used in _____

- a. RIP
- b. IGRP
- c. RIPv2

d. **OSPF**

page 64

115. According to size, there are ____ classifications of networks

a. **2**

page 2

- b. 3
- c. 4
- d. 5

116. In which topology terminators are used at the end of the cable?

a. Star

b. **Bus**

c. Ring

- d. Mesh
- 117. Switch simulates a _____ LAN connecting segments
 - a. Single
 - b. Single shared
 - c. **Bridged** page 54
 - d. Complex
- 118. PVS stands for _____
 - a. **Permanent virtual circuit** page 69
 - b. Private virtual circuit
 - c. Public virtual circuit

d. Plastic virtual circuit

119. Hardware that calculates a CRC used two simple components _____

a. AND unit and XOR Unit

b. **Shift register and XOR unit** page 20

c. Shift register & AND unit

d. Shift register and Shift XCR unit

120. Thin ethernet is known as _____

a. **10 base 2** google

b. 10 base 5

c. 10 base T

d. All of the above

121. If a sender with a 1 Gigabit NIC and receiver with 100 Megabit NIC wants to communicate with each other than at which speed they can communicate

a. 11 Megabit

b. **100 Megabit**

c. 1 Gigabit

d. 1.1 Gigabit

122. _____ is used as an alternative of packet for a specific computer network technology

a. Datum

b. Token

c. Ether

d. **Frame** page 14

123. The sharing of a medium and its link by two or more devices is called _____

a. Modulation

b. Encoding

c. **Multiplexing**

d. Line discipline

124. The topology each computer is connected to a central hub is called

a. Ring topology

b. **Star topology** page 5

c. Tree topology

d. Mesh topology

125. Asynchronous transfer mode (ATM) is designed to work on fiber but can be used with _____

a. wireless

b. **Twisted pair** page 72

c. Wifi

d. Bluetooth

126. The basic responsibility of NIC is _____

- a. To access medium network
- b. To resource allocation
- c. To access memory
- d. All of the given option

127. Frame relay is _____

- a. Connection oriented service page 64
- b. Connectionless service
- c. Typically ranges for 5 Mbps to 1000Mbps
- d. Call based

128. The length of time required to send a variable length packet is _____

- a. 5 second
- b. Variable page 72
- c. Constant
- d. Fix

129. ABR stands for _____

- a. Asynchronous bit rate
- b. Asynchronous byte recovery
- c. Asynchronous bit redundancy
- d. Available bit rate page 71

130. In Token Ring, if all computers are ready to transmit it enforces _____

access

- a. First come first served
- b. Round Robin page 30
- c. Shorter delay
- d. Last come first served

131. The Ethernet standard specifies that frames are sent using the _____

scheme.

- a. Differential Manchester
- b. Not Return to Zero (NR2)
- c. Manchester Encoding
- d. Return to Zero (RZ)

page 27

132. ATM uses the concept of _____.

- a. Connection-oriented
- b. Connection-less
- c. variable packet size
- d. Fixed and variable packet size

page 66

133. Which of the following is a benefit of point to point communication?
- a. Quick to configure
 - b. Security implementation**
 - c. Cost
 - d. Less availability
134. _____ is not used to extend LAN.
- a. Hub
 - b. Fiber modem**
 - c. Repeater
 - d. Bridge
135. Which types of VPNs are used for corporate connectivity across companies residing in different geographical locations?
- a. Remote access VPNs
 - b. Peer-to-peer VPNs
 - c. Country-to-country VPNs
 - d. **Site-to-site VPNs** google
136. Star topology is the kind of a. Tree topology
- b. **Point-to-point topology** page 05
 - c. Broadcast topology
 - d. Ring topology
137. In a _____ state, a bridge allows simultaneous use of each segment.
- a. Ready
 - b. Steady**
 - c. Final
 - d. Startup
138. The network occupies larger areas like cities & countries is called
- a. LAN
 - b. WAN**
 - c. MAN

d. All of the above

139. In link state routing _____ send link-state information about local connections.

a. Bridges

b. Switches

c. Routers

d. Hubs

140. Transmission media are usually categorized as _____.
a. **Guided or unguided**
b. Fixed or unfixed
c. Determinate or indeterminate
d. Metallic or nonmetallic
141. Even parity can be used to check for _____ bit/ bits of errors.
a. **1** page 18
b. 2
c. 3
d. 0
142. Routing table entries can be _____ with a default route.
a. **Collapsed**
b. Redirected
c. Emerged
d. Guaranteed
143. The system administrator must coordinate to avoid the conflict in _____ hardware addressing scheme.
a. Static
b. Dynamic
c. **Configurable**
d. Fixed
144. _____ network does not depend on CSMA/CD.
a. **Ethernet**
b. Fast Ethernet
c. Gigabit Ethernet
d. Wireless
145. Which of the following technologies can extend the diameter of LAN medium?
a. Fiber optic
b. Repeater
c. Diameter of LAN medium is not extendable

d. Both. Fiber optic and Repeaters

146. Connections are formed in Asynchronous transfer mode by starting values in

_____ in ATM switches, as opposed to making actual electrical connections.

a. Heap

b. Stack

c. Virtual Circuit

d. **Memory locations**

page 67

147. The _____ scheme must be reliable to prevent conflicts.
- a. Static
 - b. **Dynamic** page 34
 - c. Configurable
 - d. Fixed
148. Metropolitan area network is used for _____.
- a. **City** page 55
 - b. Building
 - c. Country
 - d. Continent
149. Which addressing can be only used in the local area networks?
- a. **IP** google
 - b. Physical
 - c. Port
 - d. Protocol
150. The sharing of a _____ among the workers in an office is a goal of resource sharing.
- a. Mouse
 - b. **Printer** page 9
 - c. Keyboard
 - d. Motherboard
151. ATM uses fixed size, small cells, 48 octet's data and _____.
- a. 3 Octets header
 - b. **5 Octets header** page 65
 - c. 6 Octets header
 - d. 7 Octets header
152. Which of the followings is not an advantage of framing?
- a. Missing eot indicates sending computer crashed
 - b. Missing soh indicates receiving computer missed beginning of message

c. **Delimiters do not contain any user data**

page 16

d. Bad frame is discarded

153. The _____ field of Ethernet frame has the purpose of receiver synchronization.

a. **Preamble**

page 36

b. Frame Type

c. Data

d. CRC

154. _____ was the first internet available for the first time.

- a. ARPANAT
- b. The ARPNET
- c. The NET

d. **ARPANET** page 9

155. We need packets rather than bits because _____

- a. To compress the data
- b. Packets move faster
- c. **Sender and receiver need to coordinate to detect transmission errors. Page 14**
- d. Packets can only move between two terminals

156. When an application _____ any data, it makes a copy of the data available to all other computers on the network.

a. **Multicast** page 143

- b. Broadcast
- c. Anycast
- d. Unicast

157. VBR stands for _____

- a. Virtual Bit Rate
- b. Virtual Byte Rate
- c. Variable Byte Redundancy

d. **Variable Bit Rate** page 71

158. Odd parity can be used to check for _____ bit/ bits of errors.

a. **0** page 18

- b. 1
- c. 3
- d. 2

159. Each computer in LAN is identified by a physical address, usually imprinted on the_____.

- a. Modem
- b. Router
- c. Switch

d. Network Interface Card

160. Which is the incorrect statement about 'Repeaters'?

a. A repeater listen to all traffic and recognizes frame format. Page 49

b. One repeater doubles, two repeaters triple the maximum cable length limitation

c. Maximum cable length cannot be increased as many times by repeaters

d. Every repeater introduces a delay in the network

161. The number of connections needed for 8 computer in direct point to point communication is equal to _____.
a. 8
b. 7
c. 26
d. **28** page 23
162. The CSMA/CD mechanism does not work with _____.
a. Two Ethernet segments.
b. Less cable length
c. High throughput
d. **Long delays** page 49
163. _____ was especially most concerned about the lack of highly powered computers.
a. **ARPA** page 9
b. IEEE
c. IEFT
d. PTA
164. When there is heavy traffic on the network, the queuing time is _____.
a. Zero
b. Fixed
c. Decreased
d. **Increased** page 80
165. _____ standard includes logical link control (LLC) sub network attachment point (SNAP) header
a. **IEEE 802.2** page 37
b. IEEE 802.11
c. IEEE 803.2
d. IEEE 803.11
166. To model a WAN, we use a graph in which the nodes model switches and the edges model direct connection between _____.
a. **Switches** page 59
b. Routers
c. Hubs
d. LANs

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2. Each computer on LAN is identified by a physical address, usually imprinted on the-----

- a. Modem
- b. Router
- c. Switch

Network interface card (pg#34 +google)

2 The basic idea behind the operation of network analyzer is a computer with a network interface that receives all frames which is called ----- • Selective mode

- d. Transfer mode
- e. Dense

f. Promiscuous mode (pg#32)

3 The identifier that is used for data transfer in the virtual circuit network is called-----

d. Virtual circuit identifier (pg#78 +google)

e. Global address

- f. Network identifier
- g. IP identifier

•

4) Which of the following is a benefit of packet switching?

- Change in transmission time
- Broken data discovery**
- Dedicated paths for data sending
- Hefty data transmission is not easy
-

5) Physical addressing in a WAN is similar to LAN as each packet has format with-----

• Header (pg#56)

- a Title
- b Frame
- c Address

d

3. was first automated tool required to see whether the given computer is online or not.

- a Probing program

- PING program (pg#11)

c. Status program

d. None of the given

2 In FDDI, in case of station failure, remaining stations loop back and -----

- Data frames are discarded

- Re-route data through spare ring (pg#31)

a Collision will occur

b Re-transmit data through same ring

4. The Frame format of any LAN technology must consist of----- and ----- • Payload and

CRC

a Frame header and CRC

b Payload and frame type

- Frame header and payload (pg#36)

J. In which type of error detection, the arithmetic sum is calculated before and after the transmission of data at both communications end?

CRC

Parity

A Hash function

JJ. In a ping command result, time denotes the -----

A TLT

B TTL

C BRT

D RTT (pg#11)

•

d. Which of the following cannot be used to extend a LAN?

Repeaters

Switches

e. Fiber modems

2 A typical port on an ATM switch operates at ----- or higher.

b. OC-2 speed(155Gbps)

c. OC-2 speed(1000Mbps)

d. OC-3speed(155Mbps) (pg#73)

e. OC-3speed(100Gbps)

3 CBR stands for -----

• Constant Byte Rate

d. Connection Break Recovery

e. None of the above

6. If two buildings are located far from each other then bridge and ----- can be used to connect two LAN.

• Optical fiber (pg#51)

Ethernet modem

Switch

NIC

7. Variable length packets require ----- to accommodate the largest possible packet and thus to detect the end of the packet
- e. Software
 - f. Instruction
 - g. Network ownership
 - h.
- 2 ----- is used for compressed audio and video ,where the data rate depends on the level of compression that can be achieved
- b. Constant Bit Rate
 - c. Bit Rate
 - d. Variable Bit Rate (pg#71)
 - e. Frame Relay
- 3 The type of the error occurs due to a hardware failure e.g. the second bit of every character will/can be damage is called-----
- 8. Burst error
 - 9. Vertical error (pg#22)

- e. Parity error
 - f. Single bit error
- 2 In order to forward a copy of frame, the bridge uses-----
- d. Network
 - e. Destination
 - f. IP
 - g. Source
- 3 ATM header is about ----- of the cell.
- 10%
- 4 The system administrator must coordinate to avoid the conflict in ----- hardware addressing scheme.
- d. Static
 - e. Dynamic
 - f. Configurable (pg#34)
 - g. Fixed
 -
9. If any switch of the destination computer does not agree to setting up the ----- an error message is sent back and the switched virtual circuit is not established.
- a. Virtual path
 - b. Public virtual circuit
 - c. Circuit
 - Virtual circuit (pg#70)
- 22) To reduce ----- , computers can be connected and distributed to a number of hubs, then the hubs can be connected to a switch.
- Costs (pg#54)
 - Quality
 - Efficiency
 - Noise
- 23) Asynchronous Transfer Mode (ATM) as a single -----networking technology has only minimal provision for inter-operation with other technologies.
- Wired
 - Virtual circuit
 - LAN
 -

Universal (pg#72)

24) LAN is used for local area but ----- can extend LAN across large distances.

- Bridge
- WAN
- Hub
- Satellite Bridge (pg#55)

25) The recent executed instruction ----- in spatial aspect of the locality of reference.

- Is executing again next
- won't be executed again
- will be executed at a later time
- nearby and execution will be executed in future (pg#24+google)

2 In distributed route computation process each packet switch computes its routing table locally and sends message to the:

- c. Router
- d. Graph
- e. Neighbors (pg#63)
- f. WAN

10. Which of the following are the issues/problems with broadcasting technique.

a. For every Broadcasting frame on the network each computer see computational resources and places the contents into memory which interrupts the CPU.

If a pair of computer use broadcasting instead of sending them directly all other computers waste CPU time while discarding the frames.

I and II (pg#35)

II and III

I and III

All of the given options

e. Networks are sometime called Intranet

Private (pg#73)

Public

Virtual

Wireless

f. Handling variable length packets leads to memory-----

Fragmentation (pg#72)

Management

Location

Allocation

g. The bridges communicate with each other on the network and use ----- algorithm to decide which bridge will not forward frames if a cycle occurs.

Dijkstra

Shortest path first

Vector distance

Distributed Spanning Tree (pg#53)

h. ATM is a single technology designed to meet the goal of both-----and-----

LAN & Pan

LAN & WAN (pg#66)

LAN & MAN

All of above

i. Asynchronous transfer mode (ATM) designed to work on fibe

Coaxial cables

Twisted pair (pg#72)

LAN

Ethernet

- j. uses counter rotating rings in which data flows in opposite direction.

Token Ring

ATM

Frame delay

- e. For hefty networks ----- topology is used.

Bus

Mesh

Star (Google)

Pool

- f. A packet switch may connect to ----- and to other packet switch.

Computers (pg#56)

Router

Coaxial cable

Hub

- e. Quality of Service desires are specified all connection set up time, and are never altered for the-----

ATM

Connection duration (pg#70)

Virtual connections

LAN

- f. A network that uses ----- usually consist of a single long cable to which computer attach.

Star topology

Ring topology

Bus topology (pg#26)

Bus and Ring topology

- g. Asynchronous Transfer Mode(ATM) can dynamically establish a -----

Switched virtual circuit (pg#70)

Permanent virtual circuit

Virtual path indicator

Connection

- h. The length of time required to send a variable length packet is ----- • 5 seconds

Variable (pg#72)

Constant

Fix

- i. Which layer of the OSI model directly communicates and controls the transmission medium.

Transmission

Physical (Google)

Application

Network

- j. Which of the following is not a function of ping command?

Local computer network issue

Congestion in the network

Relay nodes (pg#11)

Remote host availability

- k. is not used to extend LAN.

- Hub (pg#149)

Fiber modem

Bridge

Repeater

- c. Frames include additional information to detect/correct errors are insert by ----- • Sender

Receiver (pg#18)

Network devices

Sender and Receiver

- d. SMDS is-----

- Connectionless service (pg#65)

Connection oriented service

Used for delivering blocks of data

None of the given

- e. The topology each computer is connected to a central hub is called ----- • Ring topology

- Star topology (pg#5)

Tree topology

Mesh topology

- e. If 0 sender with a 1Gigabit NIC and receiver with 100Megabits NIC wants to communicate with each other at which speed they can communicate?

11Megabit

100Megabit (Google)

1 Gigabit

11 Gigabit

c. The ----- field of Ethernet frame has the purpose of receiver synchronization.

- Preamble (pg#36)
- Frame type
- Data
- CRC

48) are more frequently being designed to allow specification of the quality of service required by the user.

- Switch
- VIP
- VCI

- Networks (pg#70)

49) -----and ----- are two types of frame type.

- Internal and external
- Single and dual

- Implicit and explicit (pg#35)

- Multicast and broadcast

50) According to a Bridge performance, if both the source and destination are on the same segment

- It forwards the frame to the other segment
- It does not forward the frame to the other segment (Pg#50)
- It does not forward the frame to the same segment
- It forwards the frame to every other segment

51) According to size there are ----- classification of networks.

- 2 (pg#4)
- 3 • 4
- 5

52) We need packets rather than bits because ----- • To compress the data

- Packets move faster
- Sender and receiver needs to coordinate to detect transmission error (pg#14) •
Packets only move between two terminals

53) Point –to – point topology is

- Size based
- Mobility based
- Connectivity based (pg#5)
- Medium based

54) Even parity can be used to check for ----- bit/bits of errors.

- 1 (pg#18)
- 2
- 3
- 0
-

e. is inflexible.

Routing

Searching

Processing

Static routing (pg#62)

f. The switch that has computers attached with it is called ----- switch.

21. Interior

Packet

Front

- e. cannot be extended arbitrarily far or to handle arbitrarily many computers.

WAN

LAN (pg#55)

MAN

LED

- f. In CSMA, if simultaneous transmission occurs, the ----- interfere with each other and this phenomenon is called collision.

Computers

Networks

Frames (pg#28)

Cable wires

- g. The ----- of errors is easier than ----- of errors.

Correction, detection

Detection, correction (Google)

Correction, creation

Detection, manipulation

- h. Frame relay is-----

Connection oriented service (pg#64)

Connectionless service

It is typically ranges from 5Mbps to 1000Mbps

None of the above

is a measure of the rate at which data can be sent through the network.

Throughput (pg#79)

Propagation delay

Access delay

Switching delay

- e. Which of the following technologies can extend the diameter of LAN medium?

Fiber optic

Repeaters

Diameter of LAN medium is not extended

- Both fiber optic and Repeaters (pg#48,49)

e. The information about ----- is kept in routing tables.

Entries

Destinations (pg#58)

Source

Packet

f. Bluetooth uses shared medium and ----- instead of coaxial cable.

Radio waves (pg#29)

Infrared waves

Micro waves

Low frequency waves

Service paradigms are similar to the telephone system.

Connection-oriented (pg#75)

Connectionless

Public-oriented

Private-oriented

e. Connections in Asynchronous transfer mode are called-----

- Virtual channels pg#67)

Virtual connections

Cells

Cell tax

- e. The maximum cable length of a LAN is fixed because -----
- Additional hardware are not suitable for the LAN extension

The electrical signal level gets weaker as it travels (pg#48)

Additional software are not suitable for the LAN extension

The electrical signal level gets stronger as it travels

- f. The sharing of a ----- among the workers in an office in an office is a goal of resource goal.

Mouse

Printer (pg#9)

Keyboard

Motherboard

- g. If a small set of bits changes near a specific location due to lightning or electric motor starting nearby etc.

then these types of errors are called -----

Burst error (pg#22)

Vertical error

Parity error

Single bit error

- c. The fiber-modem converts digital data into ----- then transmit over the optical fiber.

Pulses of light (pg#48)

Electronics

Electromagnetic signals

None of these

- d. The process of forwarding the packet information is called-----

Routing (pg#58)

Processing

Hierarchical addressing

Source addressing

- e. Dijkstra's algorithm can accommodate weights on ----- in graph.

Nodes

Edges (pg#62)

Paths

Connections

was the first internet available for the first time.

ARPANAT

The ARPANT

ARPANET (pg#9)

The Net

- e. The CSMA/CD mechanism does not work with ----- • Two Ethernet segments

Less cable length

High throughput

- Long delays (pg#49)

networks are expensive to install and maintain.

Public

Wide area

Metropolitan area

Private (pg#74)

- c. Weights on graph edges reflect cost of -----

Nodes

Traversing edges (pg#63)

Trending edges

Paths

- e. What is the largest network in existence?

Internet (pg#4)

A PAN

A FOG

Edge Network

- f. The value of delay in computer network measured in-----

Hours

Minutes

Seconds or fraction of seconds (pg#79)

Years

- d. Packet switches can be linked together to form-----

LAN

WAN (pg#56)

Bridge

Network

- e. A typical Bridge has-----

One NIC, a CPU, a Memory and a ROM

Two NIC, a CPU, a Memory and a ROM (pg#50)

RAM, ROM and CPU

A cache Memory, ROM and CPU

- f. Source Independent Phenomenon allow fast and efficient ----- • Source Addressing

Routing (pg#58)

Packet switching

Store and Forward

- g. ABR stands for-----

- Available Bit Rate (pg#71)

Asynchronous Bit Rate

Asynchronous Byte Rate

Asynchronous Bit Redundancy

- c. Switch stimulates a ----- LAN connecting segments.

Single

Single shared

Bridged (pg#54)

Complex

d. LAN and WAN are classified according to their-----

Size (pg#4)

Connectivity

Mobility

Medium

e. The identifier that is used for data transfer in the virtual circuit network is called-----

Virtual Circuit Identifier (Google)

Global Address

Network Identifier

IP Identifier

f. Local talk uses ----- (physical Star) to stimulate a Bus topology.

Bridges

Hubs (pg#47)

Routers

Switch

is the variation in delay for packet belonging to the same flow.

Reliability

Jitter (pg#66)

Delay

Band Width

- e. The basic idea behind the operation of network analyzer is a computer with a network interface that receive all frames, which is called-----

Selective mode

Transfer mode

Dense

Promiscuous mode (pg#38)

- f. The Token Ring Network topology is a ring but wiring topology is a-----

Star (pg#44)

Bus

Tree

Mesh

- g. Dynamic routing can work around ----- automatically.

Network Failure (pg#62)

LAN

Switch

- e. The number of connection needed for N computers in direct point to point communication is equal to:

$N(N-1)$

$(N^2-N)/2$ (Pg#23)

$N-1$

N^2

- f. The type of error occurs due to a hardware failure e.g. the second bit of every character will/can be damaged is called:

Burst error

Vertical error (pg#22)

Parity error

Single bit error

Is the term used for variance in transmission delays.

Friction

Asynchronous transfer mode

Checksum

Jitter (pg#66)

d. Which is incorrect statement about bridge?

A Bridges uses two NICs to connect two segments

A Bridge is more intelligent than Repeater

Bridge do not recognize frame formats, they just amplify and retransmit the electric signal (pg#50) • A Bridge listen to all traffic and recognize frame format

e. When there is a heavy traffic on the network, the queuing time is -----

• Increase

f. Data delivery from one computer to another computer is accomplished through-----

LAN

Fragmentation

Hierarchal addressing

Store-and-Forward (pg#56)

g. CPU can't process data at -----speed.

High

Clock

Slow

Network (pg#39)

- e. Network occupies large areas like cities and countries is called

LAN

WAN (Pg#4)

MAN

PAN

- f. A network that permits many sources to take turns accessing a shared communication is a form of-

TDM(Time division multiplexing) (Google)

PDM

FDM

WDM

- g. ASYNCHRONOUS TRANSFER MODE is a signal ----- networking technology has only minimal provision for inter-operation with other technologies.

Wired

LAN

Virtual circuit

Universal (pg#72)

- h. Which one of the following is the computing routing table information?

Manual entry

Static routing (pg#62)

Boot time routing

None of the above

- i. All of the following are types of delay except ----- • Propagation

Access

Throughput (pg#79)

Queuing

- j. There are ----- types of switch according to the attached computers.

Four

Three

Five

Two (pg#59)

k. IBM's token ring uses ----- (physically a star topology) to stimulate logic ring network.

Hubs (pg#47)

Switch

Router

Bridge

l. ATM uses small, fixed sized packets called cells. Each cell has ----- octets.

52

53 (pg#66)

50

25

m. LAN interface hardware obeys ----- access rule.

CSMA/CD (pg#34)

Round robin

First come first serve

High priority

n. The interface hardware does not include ----- type of frame.

Multicast frames

Broadcast frames

- Fewcast frames (pg#35)

d. The frames that are distend to the station itself
Packet switches can be linked together to form-----

LAN

WAN (pg#56)

Bridge

Network

e. Which two topologies are included in point to point topology?

Ring and star topology

- Star and tree topology (page#5)

c. Bus and mesh topology

Which of the following is a benefit of point to point communication? •
Quick to configure

- Security implementation (page#23)

χ. Cost

δ. Less availability

c. No any error detection scheme is perfect because transmission error can effect-----

- Data

- Additional information

- Additional information and data (Google)

ε. Transmission medium

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data. A cyclicity redundancy slightly more difficult to computer can detect more errors then a Parity or Check sum mechnasim.

e. Thick Ethernet is known as-----

- 10Base2

- 10Base5 (google)

δ. 10BaseT

ε. 10Base10

e. ATM uses fixed sized, small cells, 48 octet's data and -----

- 3 Octets header

- 5 Octets header (page#65)

δ. 6 Octets header

ε. 7 Octets header

c. ATM offers-----speed and -----
delay. • high, high

- high, low (google)

δ. low, low

ε. low, high

e. Both interior and exterior switches ----- packets and they also need routing
tables. • Create

- Delay

χ. Start

c. VPNs connections are sometimes called-----

ε. Open networking

φ. Routing

γ. Public network

c. A network analyzer is also called ----- --.

χ. Network manager

δ. Network administrator

• Network monitor (page#37)

ε. Network supervisor

e. ATM uses the concept of ----- --.

ε. Connection-less

φ. Variable packet size

γ. Fixed and variable packet size

Which of the following is not a guided medium?

η. Twisted-pair cable

ι. Fiber-optic cable

χ. Coaxial cable

c. A network uses ----- to arrange computers in a single closed loop. • Star topology

• Ring topology (page#29)

ε. Mesh topology

φ. Bus topology

c. There are ----- types of data stuffing.

• 1

• 2 (page#17)

ε. 0

φ. 5

d. A network with ----- jitter takes more time to deliver some packets to other. • Zero

ε. Low

φ. Moderate

c. masks your IP address. • Firewall

ε. ntivirus

ε. Incognito mode

e. In ----- , network topology is separated from route computation. • Local area

• Link-state routing (page#64)

χ. Static routing

δ. Dynamic routing

e. is the most popular wiring scheme because of lowest cost. • 5Base T

δ. 100 Base T

ε. 1000 Base T

Which of the following two operations are performed by Ethernet CD?

I.It monitors outgoing signals

e. Grabbled signal is interrupted as a collision

e. Signals are sent back to the sender

IV. Choose a shorter delay

ε. I and III

φ. II and IV

γ. II and III

• I and II (page#28)

e. Public networks are-----

- More source than LAN

ε. Limited in boundary

φ. Intranet

e. address is used in an Ethernet frame. • IP

- Port

- Logical

d. AUI cable connects from NIC to -----

- BNC connector
- RJ-45 connector

ε. Terminator

e. If a ----- is used, the communication between two computers will not affect the other pair of computers and they can communicate the same time.

- Hub

ε. Repeater

φ. Amplifier

e. Source Independence Phenomenon allows fast and efficient-----

- Source addressing

χ. Packet switching

δ. Store and forward

c. is also called self healing network. • ATM

χ. Token ring

δ. Frame delay

e. Which multiplexing technique transmits digital signals? • WDM

• FDM

ε. GSM

e. The length of hardware address is-----

• One to ten bytes •
One to five bytes

• One to six bytes (page#34)

ε. One to seven bytes

e. Transmission media are usually categorized as ----- --.

• Guided and unguided (google)

χ. Fixed or unfixed

δ. Determinate or indeterminate

ε. Metallic or nonmetallic

2 A ----- is a device that combines multiple signals and forward into a single line. • Modem

101. Repeater

102. Layer 3 Switch

e. Which statement is correct about Network Interface Card NIC?

I.NIC contains sufficient hardware to process data independent of system CPU.

102. NIC looks like any other I/O device to CPU.

2 NIC also receive interrupt on incoming data.

I and II

II and III

I and III

All of the given options. (page #40)

e. Which addressing can only be used in the local area networks? • IP (Google)

• Physical •
Port

• Protocol

f. Which of the following network is using the TCP/IP model globally? • PAN

• SAN

103. LAN

104. The Internet (Google)

2 The sharing of a medium and its link by two or more devices is called ----- . • Modulation

• Encoding

• Multiplexing (Google)

• Line discipline

3 The fast Ethernet hardware operates at a rate of ----- .

104. 1000bps

105. 100Mbps (page# 27)

106. 10Mbps

107. 1000Mbps

e. The number of connection needed for 8 computers in point to point communication is equal to-
.....

106. 8

107. 7

108. 26

109. 28 (page # 23)

2 uses counter rotating rings in which data flows in opposite directions. • Token Ring

• FDDI (page #31) •
ATM

107. Frame delay

108. In which topology terminators are used at the end of the cable? • Star
.....

2 Ring

3 Mesh

109. The Ethernet standard specifies that frames are sent using the ----- scheme. • Differential Manchester

- None Return To Zero

- Manchester Encoding (page#27) •

Return To Zero

110. Hardware address is also called -----.

2 Physical address (page# 33)

3 Logical address

4 Port address

5 IP address

111. When an application ----- any data, it makes a copy of the data available to all other computers on the network.

- Multicast

ε. Anycast

φ. Unicast

112. allows automatic updates.

113. Dynamic routing (page #62)

114. Static routing

115. Routing

116. Default routing

2 -----standard includes logical link control (LLC) sub network attachment point(SNAP) header.

115. IEEE 802.11

116. IEEE 803.2

117. IEEE 803.11

2 In a Token Ring, if all computers are ready to transmit it enforces ----- access. • First come first serve

118. Shortest delay

119. Last come first serve

119. Missing-----in a frame format indicates receiving computer missed beginning of the message.

- ect •
- eoh

- soh (page#16)

δ. eoh

2 Computer sends a connection request to the ----- to which it is attached while establishing a switched virtual circuit.

- Bridge

120. VPI

121. Virtual circuit

2 Distance-Vector routing is used in ----- --.

3. ISIS

4. NLSP

5. OSPF

6. Metropolitan area network is used for ----- --.

7. Building

8. Country

9. Continent

10. networks are sometimes called Intranet.

11. Public

12. Virtual

13. WAN

14. Ethernet uses a----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

- 64

- 48 (google)

15. 32

16. 16

17. Fixed network is a type of network which is classified with respect to the ----- factor.

- Size
- Connectivity
- Medium

- Mobility (page#6)

18. Which is the incorrect statement about “Repeaters”?

- A repeater listens to all traffic and recognize frame format. (page#49)

19. One repeater doubles, two repeaters triple the maximum cable length limitations.

20. Maximum cable length cannot be increased as many times by repeaters.

21. Every repeater introduces a delay in the network

22. Which of the following tasks is not done by the second layer of the OSI model?

- Flow control

- Framing

- Channel coding (google)

23. Error control

24. In ----- all computers are attached to a central hub. • Ring Topology

25. Bus Topology

26. Tree Topology

27. Connections are formed in Asynchronous transfer mode by storing values in ----- in ATM switches, as opposed to making actual electrical connections.

- Heap •
Stack

- Virtual Circuit

- Memory locations (page#67)

28. Thick Ethernet also requires ----- to avoid signal reflectance. • Transceivers

- AUI cable

29. Drop cable

30. Terminators(page#41)

31. To model a WAN, we use a graph in which the nodes model switches and the edges model direct connection between ----- --.

32. Routers

33. Hubs

34. LANs

35. Routing table entries can be ----- with a default route. • Collapsed (page #60)

- Redirected •
- Emerged •
- Guaranteed

36. IEEE LLC/SNAP header is ----- , which is used to specify the type of data.

- 8 octets (google)

37. 16 octets

38. 32 octets

39. 64 octets

CS-610 (Computer Networks)

Grand Quiz Dec 2020

Prepared By: MCS of Virtualians

If you found any error in the file, Please correct it accordingly

1. In Cyclic Redundancy Checking, CRC is _____ .

40.Divisor

41.Quotient

c. Remainder page 20

a. Dividend

42.The satellite or radio topology in which all computers are connected to each other via satellite or radio wave is a kind of

a. Broadcast network page 5

a. Point-to-Point network

43.Which of the following two operations are performed by Ethernet CD?

I. It monitors outgoing signals.

44. Grabbled signal is interpreted as a collision.

a. Signals are sent back to the sender.

IV. Choose a shorter delay.

45.I and III

46.II and IV

47.II and III

d. I and II page 28

4. The process of forwarding a packet is called _____ .

a. Routing page 58

- a. Processing
- b. Hierarchical Addressing
- c. Source Addressing

48. Which statement is correct about Network Interface Card (NIC)?

NIC contains sufficient hardware to process data independent of system CPU

- b. NIC looks like any other I/O device to system CPU
- III. NIC also receives interrupt on incoming data.

49. I and II

50. II and III

51. I and III

d. All of the given options

page 40

6. While transmitting data from source A to destination B, 4 bits are changed during the transmission then the error is a _____.

a. **Burst error** page 22

52. Single error

53. Double error

54. Logic error

7. The switch that has no attached computers is called _____.

55. Packet Switch

56. Exterior Switch

c. **Interior Switch** page 59

d. External Switch

8. _____ is the most popular wiring scheme because of lowest cost.

a. 5 Base T

b. **10 Base T** page 43

57. 100 base T

58. 1000 base T

9. In _____ technique, all members of the network can send data only on the _____ specific time slot?

59. CDMA

60. FDMA

c. **CSMA** page 43

d. TDMA

10. If a sender with a 100 Megabit NIC and receiver with 10 Megabit NIC wants to communicate with each other at which speed they can communicate?

a. **100 Megabit** page 40

61. 110 Megabit

62.1000 Megabit

63.10 Megabit

11. Missing _____ in a frame format indicates receiving computer missed

_____ beginning of the message.

a. eot

b. soh _____ **page 16**

64.sot

65.eoh

12. The _____ of errors is easier than the _____ of errors.

a. _____
correction; detection

b. detection; correction _____ **cs601 handouts**

66.correction; creation

67.detection; manipulation

13. The _____ of errors is difficult than the _____ of errors.

a. correction; detection **google**

68.detection; correction

69.correction; creation

70.detection; manipulation

14. Source Independence phenomenon allows fast and efficient _____.

71.Source Addressing

72.Routing

c. Packet Switching **page 58**

d. Store and forward

15. In distributed route computation process each packet switch computes its

routing table locally and sends messages to the _____.

73.Router

74.Graph

c. Neighbors **page 63**

d. WAN

16. The _____ layer of the OSI model takes the data from the physical layer and perform error checking.

a. Transport

b. Data link **google**

75.Application

76.Presentation

17. CBR stands for _____.

77.Constant Byte Rate

78.Constant Bit Rate

79.Connection break Recovery

80.Constant Borrow Rate

18. A network that uses _____ usually consist of a single long cable to which computer attach.

a. Star Topology

b. Ring Topology

c. Bus Topology **page 26**

d. Bus and Ring Topology

19. If simultaneous transmission occurs, the frames interfere with each other and this phenomenon is called _____.

a. packet loss

b. collision **page 28**

81.delayed transmission

82.coordination

20. In _____ all computers are attached to a central hub

a. Ring topology

b. Star topology **page 31**

83. Hub topology

84. Bus topology

21. For adding a new computer in a point to point network _____ new connection will be required

a. N

b. N-1 **page 23**

c. N-2

d. $2n$

22. The fastest Ethernet hardware operates at a rate _____ of a. 1000 bps

b. 100 Mbps **page 27**

85. 10 Mbps

86. 1000 Mbps

23. Which of the following the benefit to point to point communication? a. Quick to configure

b. Security implementation **age 23**

87. Cost

88. Less availability

24. ATM offers _____ speed and _____ delay

a. High, High

b. High, low **page 72**

c. Low, low

d. Low, high

25. Thick Ethernet is known as _____

a. 10 Base 2

b. 10 Base 5 **google**

c. 10 Base T

d. 10 Base 10

26. Both interior and exterior switches _____ packet and they also need routing tables.

a. Create

b. Delay

c. Forward **page 59**

d. Start

27. _____ is the rate at which data can be sent through the network

a. Throughput page 79

b. Propagation delay

c. Access delay

d. Switching delay

28. In a ping command, small –sized information packets of _____ are sent.

a. TCP

b. UDP

c. ICMP page 11

d. IGMP

29. Many WANs use _____ addressing of efficiency

a. Sequence

b. Forward

c. Hierarchical page 56

d. Sequential

30. Which of the following is not a function of ping commands?

89. Local computer network issue

90. Congestion in the network

c. Relay nodes page 11 and google

d. Remote host availability

31. Packet switched can be linked together to form a _____

a. LAN

b. WAN page 56

91. Bridge

92. None of the given option

32. Fixed network is a type of network which is classified with respect to the
— factor
—

93. Size

94. Connectivity

95. Medium

d. Mobility page 6

33. Parity bit can detect _____ error(s)

a. 4

b. 3

c. 2 page 19 (even or odd)

d. 1

34. When we will try to connect to the internet which of the following protocol will be used?

a. Telnet

b. TCP/IP **page 84**

96.ICMP

97.IPx

35. How many computers are mandatory to form a network?

a. 1

b. 2 **google**

98.4

99.8

36. _____ is standard wireless LAN that uses radio signals at 2.4GHz.

a. IEEE 803.16

b. IEEE 802.15

c. IEEE 802.3

d. IEEE 802.11 **page 29**

37. LAN interface hardware obeys _____ access rule.

a. CSMA/CD **page 34**

b. Round robin

c. First come first served

d. High priority

38. The most important task a bridge performs is _____.

a. Packet switching

b. Line controlling

c. Noise controlling

d. Frame filtering

page 50

39. ATM uses small, fixed-sized packets called cells and each cell _____
has octets.

100. 47

101. 52

c. 53

page 66

d. 55

40. _____ can't be guaranteed with variable length packets as easily as it can
_____ with fixed length cells.

102. Errors

103. Noise

c. Quality of Service

page 72

d. None of the given

41. Which multiplexing technique transmits digital signals?

a. WDM

b. FDM

c. TDM page 15

d. GSM

42. Connections in Asynchronous transfer mode are called ____.

a. Virtual channels page 67

104. Virtual Path Identifier

105. Cells

106. Cell tax

43. A ____ is a device that combines multiple signals and forwards into a single line.

a. Modem

b. Multiplexer page 41

107. Repeater

108. Layer 3 Switch

44. ____ is inflexible.

a. Routing

b. Searching

c. Processing

d. Static routing page 62

45. If a hub is used to connect many computers in a LAN, then only ____ computers can communicate at a given time.

a. Three

b. Two page 54

c. Four

d. One

46. _____ is the term used for variance in transmission delays.

109. Friction

110. Segment

111. Checksum

d. Jitter **page 66**

47. _____ is also called self healing network.

a. ATM

b. FDDI

c. Token Ring **page 46**

d. Frame relay

48. _____ and _____ are the two frame types.

112. Internal and external

113. Single and dual

114. Implicit and explicit

d. Multicast and broadcast **page 35**

49. Bluetooth uses shared medium and _____ instead of coaxial cable.

a. Radio waves **page 29**

a. Infrared waves

b. Micro waves

c. Low frequency waves

115. Which of the following is not a guided medium?

a. Twisted-pair cable

b. Fiber-optic cable

c. Atmosphere **google**

d. Coaxial cable

116. As a result of ARPA research, the first network was established which was named as _____.

a. INTERNETWORKING

b. ARPNET

c. ARPANET **page 9**

d. PACKET NETWORK

52. Asynchronous transfer mode (ATM) can dynamically establish a _____.

a. Switched virtual circuit **page 70**

117. Virtual connection identifier

118. Virtual path indicator

119. None of the given options

53. A typical port on an ATM switch operates at _____ or _____ higher.

- a. OC-2 speed (155Gbps)
- b. OC-2 speed (1000Mbps)

c. OC-3 speed (155Mbps) **page 72**

- d. OC-3 speed (100Gbps)

54.10Base-T Ethernet is often called a _____ .

- a. ring shaped star
- b. ring shaped bus

c. star shaped bus **page 46**

- d. star shaped ring

55. _____ allows automatic updates.

a. Dynamic routing **page 62**

- b. Static routing
- c. Routing
- d. Default routing

56. Handling variable length packets leads to memory _____.

a. Fragmentation **page 72**

- b. Management
- c. Location
- d. Allocation

57. In order to forward a copy of frame, the bridge uses _____ address.

- a. Network

b. Destination **page 51**

- c. IP
- d. Source

58. We may use _____ to extend local area network.

120. Bridge, Router

121. Bridge, Hub, Router

c. Bridge, Hub, Repeater **page 149**

- d. Bridge, Hub, Repeater, Router

59. All of the given options are the types of delay except _____.

- a. Propagation
- b. Access

c. **Throughput** page 79

a. Queuing

122. While computing shortest path in a graph, next hop information is inserted into

_____.

a. **Routing table** page 62

b. Trailer

c. Header

d. Register

61. LAN interface uses _____ to copy frame data directly from main memory

a. **DMA** page 34

b. FDDI

c. Hard disk

d. Flash

62. The Gigabit Ethernet hardware operates at a rate of _____ .

123. 10 Mbps

124. 100 Mbps

c. 1000 Mbps reference book page 263

a. 1000 Gbps

125. In the early resource sharing era, which resources were shared among users?

a. Computation page 9

b. Floppy Drives

c. Hard Disk

d. Printers

64. If two buildings are located far from each other then bridge, and _____ can be used to connect two LAN.

a. Optical fiber page 51

a. Ethernet modem

b. Switch

c. NIC

126. Which type of network consists of wireless connection of laptops in a university classroom?

a. WAN

b. MAN

c. Fog

d. LAN google

66.A typical bridge has _ .

a. One NIC, a CPU a memory and a ROM.

b. Two NICs, a CPU a memory and a ROM. page 50

127. RAM, ROM and CPU.

128. A cache memory, ROM and CPU.
67. _ is used for compressed audio and video, where the data rate

depends on the level of compression that can be achieved.

129. Constant Bit Rate

130. Stable Bit Rate

c. Variable Bit Rate

page 71

d. Available Bit Rate

68. There are _ types of data stuffing.

a. 1

b. 2 **page 17**

c. 0

d. 5

69. Dijkstra's algorithm can accommodate weights on _____ in graph.

a. Nodes

b. Edges page 62

131. Path

132. Connections

70. _____ masks your IP address.

133. Firewall

134. Antivirus

135. VPN

136. Incognito mode

71. Static routing computes routing table information at _____ time.

a. Load

b. Boot page 62

c. Execute

d. One

72. _____ was the first automated tool required to see whether the given computer is online or not.

a. Probing program

b. PING program page 11

a. status program

b. None of the above

137. In which type of error detection, the arithmetic sum is calculated before and after the transmission of data at both communication ends?

a. CRC

b. Parity

c. Checksum page 19

d. Hash Function

74. Computer sends a connection request to the _ _ to which it is attached
while establishing a switched virtual circuit _____

a. Bridge

b. Switch **page 70**

c. VPI

d. Virtual circuit

138. The bridges communicate with each other on the network and use
algorithm to decide which bridge will not forward frames if a cycle occurs. _____

a. Djikstra

b. Shortest path first

c. Vector distance

d. Distributed Spanning Tree **page 53**

76. Ethernet uses a _____ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

- a. 64
- b. 48**
- c. 32
- d. 16

77. Weights on graph edges reflect cost of _____.

a. Nodes

b. Traversing edge **page 63**

c. Trending edge

d. Path

78. A _____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

a. Static addressing scheme **page 34**

139. Configurable addressing scheme

140. Dynamic addressing scheme

141. None of the given

79. In _____ topology, if the switch goes down all the computers will be disconnected.

- a. Star**
- b. Ring
- c. Bus
- d. Mesh

142. Which type of network consists of wired computers in a university classroom?

- a. MAN
- b. PAN
- c. WAN

d. LAN

143. If you as a network administrator want to know the traffic flow of your data while communicating with a remote computer which of the following will be used to know about intermediate routers?

a. Ipconfig

b. Ping **page 11**

144. Traceroute

145. Arp

82. Hardware address is also called _____.

146. Physical address

147. Logical address

148. IP address

149. Port address

83. Thick Ethernet also requires _____ to avoid signal reflectance.

150. Transceivers

151. AUI cable

152. Drop cable

d. Terminators page 41

153. Which of the following network is using the TCP/IP model globally?

a. PAN

b. SAN

c. LAN

d. The Internet

85. VPNs connections are sometimes called _____.

a. Tunnels page 75

154. Open networking

155. Routing

156. Public network

86. To reduce _____ computers can be connected and distributed to a number of hubs, then the hubs can be connected to a switch.

a. Costs page 54

157. Quality

158. Efficiency

159. Noise

87. If a _____ is used the communication between two computers will not

affect the other pair of computers and they can communicate at the same time

a. Hub

b. Switch page 54

c. Repeater

d. Amplifier

88. The length of hardware address is _____ .

a. One to ten bytes

b. One to five bytes

c. One to six bytes

page 34

d. One to seven bytes

89. A network analyzer is also called _____ .

a. Network manager

b. Network administrator

c. Network monitor

page 37

d. Network supervisor

160. No any error detection scheme is perfect because transmission errors can affect

— _____ .

161. Data

162. Additional Information

c. Additional information & data **google**

d. Transmission medium

91. The token ring network topology is a ring but wiring topology is a ____.

a. Star **page 44**

163. bus

164. tree

165. mesh

92. _____ Addresses are used in an ethernet frame.

166. IP

167. Port

168. Logical

169. MAC

93. AUI cable connects from NIC to _____ .

a. BNC Connector

b. RJ-45 Connector

c. Transceiver **page 41**

d. Terminator

94. Public networks are ____.

a. More secure than LAN

b. Flexible **page 74**

c. Limited in boundary

d. Intranet

95. The maximum cable length of a LAN is fixed because .

a. Additional hardware are not suitable for the LAN extension.

b. The electrical signal level gets weaker as it travels.

Page 48

c. Additional software are not suitable for the LAN extension

d. The electrical signal level gets stronger as it travels.

96. In ____ network topology is separated from route computation.

a. Local area

b. Link-state routing

page 64

c. Static routing

d. Dynamic routing

170. Which layer of the OSI Model directly communicates and controls the transmission medium:

- a. Transmission
- b. Physical**
- c. Application
- d. Network

98. In _____ technique data is treated as a sequence of integers and their arithmetic sum is computed and the carry bits are added to the final sum.

- a. CRC
- b. Bit stuffing
- c. Checksum**
- d. 2D Parity

page 19

99. A network with _____ jitter takes more time to deliver some packets than others.

- a. Zero
- b. High**
- c. Low
- d. Moderate

page 80

100. A network uses _____ to arrange computers in a single closed loop.

- 171. Star Topology**
- 172. Mesh Topology**

c. Ring Topology

page 25

- d. Bus Topology

173. Which of the following tasks is not done by the second layer of the OSI model?

- a. Flow control**
- b. Framing**

c. Channel coding

google

d. Error control

102. ATM is a single technology designed to meet the goals of both _____
and _____.

174. LAN & PAN

b. LAN & WAN **page 66**

175. LAN & MAN

176. MAN & WAN

103. LAN and WAN are classified according to their ____.

a. Size **page 4**

177. Connectivity

178. Medium

179. Mobility

104. The identifier that is used for data transfer in the virtual circuit network is called _____.

- a. **Virtual circuit identifier**
- b. Global address
- c. Network identifier
- d. IP identifier

180. In Direct point to point communication adding the Nth computer requires _____ new connections

- a. N^2
- b. **N-1** **page 23**
- c. $(N^2 - N)/2$
- d. $N(N-1)$

181. Which of the following is the correct function of bridge?

- a. MTU path discovery
- b. Byte stuffing
- c. Routing information
- d. **Filtering and forwarding a frame** **page 50, 51**

182. Point to point topology is

- a. Size based
- b. Mobility based
- c. **Connectivity based** **page 5**
- d. Medium based

108. IEEE LLC / SNAP header is _____ which is used to specify the type of date.

- a. **8 octets** **google**
- b. 16 octets
- c. 32 octets

d. 64octets

109. IEEE LLC / SNAP header is _____ which is used to specify the type of date.

c. CRC

d. Bit stuffing

e. **Checksum**

f. 2D Parity

183. What is the largest network in existence?

a. **The internet** **google**

184. A PAN

185. A Fog

186. Edge Network

111. The frame format of any LAN technology must consists of _____ and _____-

187. Payload and CRC

188. Frame header and CRC

189. Payload and Frame type

d. Frame header and payload

page 36

112. _____ can occur a LANs when a packet is delayed because the network is busy

a. Jitter

page 66

a. Loop

b. Deadlock

c. variance

190. Which of the following cannot be used to extend a LAN?

a. Repeaters

b. Switches

c. **Routers**

d. Fiber modems

191. Links-state algorithm is used in _____

a. RIP

b. IGRP

c. RIPv2

d. OSPF

page 64

115. According to size, there are _____ classifications of networks

a. 2

page 2

d. 3

e. 4

f. 5

192. In which topology terminators are used at the end of the cable?

a. Star

b. Bus

c. Ring

d. Mesh

117. Switch simulates a _ LAN connecting segments

193. Single

194. Single shared

c. Bridged page 54

d. Complex

195. PVS stands for _____

a. Permanent virtual circuit page 69

196. Private virtual circuit

197. Public virtual circuit

- a. Plastic virtual circuit
198. Hardware that calculates a CRC used two simple components _____
- a. AND unit and XOR Unit

b. Shift register and XOR unit **page 20**

199. Shift register & AND unit
200. Shift register and Shift XCR unit
120. Thin ethernet is known as _ _

a. **10 base 2** **google**

a. 10 base 5

b. 10 base T

c. All of the above

201. If a sender with a 1 Gigabit NIC and receiver with 100 Megabit NIC wants to communicate with each other than at which speed they can communicate

a. 11 Megabit

b. **100 Megabit**

c. 1 Gigabit

d. 1.1 Gigabit

122. _ _ is used as an alternative of packet for a specific computer

network technology

202. Datum

203. Token

204. Ether

d. Frame **page 14**

205. The sharing of a medium and its link by two or more devices is called _

a. Modulation

b. Encoding

c. **Multiplexing**

d. Line discipline

206. The topology each computer is connected to a central hub is called

a. Ring topology

b. Star topology page 5

b. Tree topology

c. Mesh topology

207. Asynchronous transfer mode (ATM) is designed to work on fiber but can be used with

a. wireless

b. Twisted pair page 72

208. Wifi

209. Bluetooth

126. The basic responsibility of NIC is _____
- a. **To access medium network**
 - b. To resource allocation
 - c. To access memory
 - d. All of the given option
210. Frame relay is _____
- a. **Connection oriented service** **page 64**
 - a. Connectionless service
 - b. Typically ranges for 5 Mbps to 1000Mbps
 - c. Call based
211. The length of time required to send a variable length packet is _____
- a. 5 second
 - b. **Variable** **page 72**
 - c. Constant
 - d. Fix
129. ABR stands for _____
- a. Asynchronous bit rate
 - b. Asynchronous byte recovery
 - c. Asynchronous bit redundancy
 - d. **Available bit rate** **page 71**
130. In Token Ring, if all computers are ready to transmit it enforces _____
access
212. First come first served
- b. **Round Robin** **page 30**
 - a. Shorter delay
 - b. Last come first served

213. The Ethernet standard specifies that frames are sent using the scheme.

a. Differential Manchester

b. Not Return to Zero (NR2)

c. Manchester Encoding **page 27**

d. Return to Zero (RZ)

132. ATM uses the concept of _ _.

a. **Connection-oriented** **page 66**

214. Connection-less

215. variable packet size

216. Fixed and variable packet size

217. Which of the following is a benefit of point to point communication?
- a. Quick to configure
 - b. **Security implementation**
 - c. Cost
 - d. Less availability
134. _____ is not used to extend LAN.
- e. Hub
 - f. **Fiber modem**
 - g. Repeater
 - h. Bridge
218. Which types of VPNs are used for corporate connectivity across companies residing in different geographical locations?
- a. Remote access VPNs
 - b. Peer-to-peer VPNs
 - c. Country-to-country VPNs
 - d. **Site-to-site VPNs** google
219. Star topology is the kind of
- a. Tree topology
 - b. **Point-to-point topology** page 05
220. Broadcast topology
221. Ring topology
137. In a _____ state, a bridge allows simultaneous use of each segment.
- a. Ready
 - b. **Steady**
 - c. Final
 - d. Startup
222. The network occupies larger areas like cities & countries is called
- a. LAN

b. **WAN**

c. MAN

d. All of the above

139. In link state routing _____ send link-state information about local connections.

a. Bridges

b. **Switches**

c. Routers

d. Hubs

140. Transmission media are usually categorized as _____.

223. **Guided or unguided**
224. Fixed or unfixed
225. Determinate or indeterminate
226. Metallic or nonmetallic
141. Even parity can be used to check for _____ bit/ bits of errors.

- a. **1** **page 18**
b. 2
c. 3
d. 0
142. Routing table entries can be _____ with a default route.

227. **Collapsed**
228. Redirected
229. Emerged
230. Guaranteed
143. The system administrator must coordinate to avoid the conflict in
_____ hardware addressing scheme.

231. Static
232. Dynamic
233. **Configurable**
234. Fixed
144. _____ network does not depend on CSMA/CD.

- a. **Ethernet**
b. Fast Ethernet
c. Gigabit Ethernet
d. Wireless

235. Which of the following technologies can extend the diameter of LAN medium?

- a. Fiber optic
- b. Repeater
- c. Diameter of LAN medium is not extendable
- d. **Both. Fiber optic and Repeaters**

236. Connections are formed in Asynchronous transfer mode by starting values in _ _ _ in ATM switches, as opposed to making actual electrical connections.

237. Heap

238. Stack

239. Virtual Circuit

d. Memory locations

page 67

147. The _____ scheme must be reliable to prevent conflicts.

240. Static

b. Dynamic page 34

241. Configurable

242. Fixed

148. Metropolitan area network is used for _____.

a. City page 55

a. Building

b. Country

c. Continent

243. Which addressing can be only used in the local area networks?

a. IP google

b. Physical

c. Port

d. Protocol

150. The sharing of a _____ among the workers in an office is a goal of resource sharing.

a. Mouse

b. Printer page 9

244. Keyboard

245. Motherboard

151. ATM uses fixed size, small cells, 48 octet's data and _____.

246. 3 Octets header

b. 5 Octets header page 65

a. 6 Octets header

b. 7 Octets header

247. Which of the followings is not an advantage of framing?

a. Missing eot indicates sending computer crashed

b. Missing soh indicates receiving computer missed beginning of message

c. Delimiters do not contain any user data **page 16**

d. Bad frame is discarded

153. The _____ field of Ethernet frame has the purpose of receiver synchronization.

a. Preamble **page 36**

248. Frame Type

249. Data

250. CRC

154. _ _ was the first internet available for the first time.

251. ARPANAT

252. The ARPNET

253. The NET

d. ARPANET page 9

254. We need packets rather than bits because

a. To compress the data

b. Packets move faster

c. Sender and receiver need to coordinate to detect transmission errors. Page 14

d. Packets can only move between two terminals

156. When an application _ any data, it makes a copy of the data
_____ available to all other computers on the network.

a. Multicast page 143

b. Broadcast

c. Anycast

d. Unicast

157. VBR stands for _ _

a. Virtual Bit Rate

b. Virtual Byte Rate

c. Variable Byte Redundancy

d. Variable Bit Rate page 71

158. Odd parity can be used to check for _ bit/ bits of errors.

a. 0 page 18

e. 1

f. 3

g. 2

255. Each computer in LAN is identified by a physical address, usually imprinted on the _____.

a. Modem

b. Router

c. Switch

d. **Network Interface Card**

256. Which is the incorrect statement about 'Repeaters'?

a. **A repeater listen to all traffic and recognizes frame format.**

Page 49

b. **One** repeater doubles, two repeaters triple the maximum cable length limitation

c. Maximum cable length cannot be increased as many times by repeaters

d. Every repeater introduces a delay in the network

161. The number of connections needed for 8 computer in direct point to point communication is equal to _____.

257. 8

258. 7

259. 26

d. 28 page 23

260. The CSMA/CD mechanism does not work with _____.

a. Two Ethernet segments.

b. Less cable length

c. High throughput

d. Long delays page 49

163. _____ was especially most concerned about the lack of highly powered computers.

a. ARPA page 9

b. IEEE

c. IEFT

d. PTA

164. When there is heavy traffic on the network, the queuing time is _____.

261. Zero

262. Fixed

263. Decreased

d. Increased page 80

165. _____ standard includes logical link control (LLC) sub network attachment point (SNAP) header

a. IEEE 802.2 page 37

- a. IEEE 802.11
- b. IEEE 803.2
- c. IEEE 803.11

264. To model a WAN, we use a graph in which the nodes model switches and the edges model direct connection between ____.

- a. **Switches** **page 59**
- b. Routers
- c. Hubs
- d. LANs

167. 167.



CS610 Computer Networks

Grand Quiz Fall 2020

Department of MCS

Compose By

MUNIR HUSSAIN &

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Semester 3rd

OBJECTIVE

1. _____ is inflexible.
 - a. Routing
 - b. Searching
 - c. Processing
 - d. **Static Routing**
2. The fiber-modem converts digital data into _____ the transmit over the optical fiber.
 - a. **Pulses of Light**
 - b. Electronics
 - c. Electromagnetic signal
 - d. None of these
265. Which of the following remain same in the header of the packet in a datagram during the entire journey of the packet?
 - a. Source Address
 - b. **Destination Address**

c. Next node Address

d. checksum

266. The process of forwarding the packet of information is called

a. **Routing**

b. Processing

c. Hierarchical Addressing

d. Source Addressing

267. _____ is a tool that send a message to remote computer and report whether the computer responds.

—a. TCP

b. **Ping**

c. Fragment Offset

d. Flag

6. Dijkstra's algorithm can accommodate weights on _____ in graph.

e. Nodes

f. **Edges**

g. Path

h. Connections

268. _____ was the first internet available for the first time.

- 269. ARPANAT
- 270. The ARPNET
- 271. The Net
- 272. **ARPANET**

8. The _____ field of Ethernet frame has the purpose receive

synchronization.

- a. **Preamble**
- b. Frame type
- c. Data CRC
- d. None of these

273. In which type of error detection, the arithmetic sum is calculated before and after the transmission of data both communication ends.

- a. CRC
- b. Parity
- c. **Checksum**
- d. Hash function

274. Frame Relay is

- a. **Connection oriented Address**
- b. Connectionless service
- c. It is typically range from the 5Mbps to 1000Mbps
- d. None of these

11. The CSMA/CD mechanism does not work with _____.

- e. Two Ethernet
- f. Less Cable length
- g. High Through Put
- h. **Long delays**

275. Networks are expensive to install and maintain.

- a. Public
- b. Wide Area
- c. Metropolitan Area

d. Private

13. Weights on graph edges reflect cost of _____ .

e. Nodes

f. Traversing edge

g. Trending edge

h. Path

276. What is the largest network in existence?

a. The Internet

b. A PAN

c. A Fog

d. Edge Network

15. The value of delay in computer network measured in _____ .

277. Hours

278. Minutes

279. Seconds of fraction of seconds

280. Years

16. Packet switches can be linked together to form _____ .

281. LAN

282. WAN

283. Bridge

284. Network

17. A typical bridge has _____.

- a. One NIC, a CPU, a Memory and a ROM
- b. **Two NIC, a CPU, a Memory and a ROM**
- c. RAM, ROM and CPU
- d. None of these

285. Source Independence phenomenon allow fast and efficient

- a. Source Addressing
- b. **Routing**
- c. Packet Switching
- d. Store and Forward

286. ABR stand for _____

- a. **Available Bit rate**
- b. Asynchronous bit Rate
- c. Asynchronous byte Recovery
- d. None of these

287. Switch simulates a LAN connecting segment.

- a. Single
- b. Single shared _____
- c. **Bridged**
- d. Complex

21. LAN and WAN are classified according to their _____.

288. **Size**

289. Medium

290. Mobility

291. Connectivity

292. The identifier that is used for data transfer in the virtual circuit networks is called _____.

293. **Virtual circuit identifier**

294. Global address

295. Network identifier

296. IP identifier

23. Local Talk uses _____ (physical star) to simulate a bus topology.

- a. Bridges
- b. Hubs**
- c. Switch
- d. Router

297. _____ is the variation in delay for packet belonging to the same flow.

- a. Reliability
- b. Jitter**
- c. Delay
- d. Band width

25. SMDs is _____.

- e. Connection less service**
- f. Connection oriented service
- g. Use for delivering block of data
- h. None of these

298. The basic idea behind the operation of network analyzer is a computer with a network interface that receive all frames, which is called ____.

299. Selective Mode

300. Transfer Mode

301. Dense

d. Promiscuous Mode

27. In _____ technique data is treated as a sequence of integers and their

arithmetic sum is computed and the carry bits are added to the final sum.

- a. CRC
- b. Bit stuffing
- c. **Checksum**
- d. 2D parity

302. The Token ring network topology is ring but wiring topology is a _____.

- a. **Star**
- b. Bus
- c. Tree
- d. Mesh

29. Dynamic routing can network can work around _____ automatically.

303. Network Failure

304. LAN

305. Switch

306. Network overhead

307. The type of error occurs due to a hardware failure for example the second bit of every character will/can be damaged is called _____.

- a. **Burst error**
- b. Vertical error
- c. Parity error
- d. Single bit error

308. The number of connection needed for N computer in direct point to point communication is equal to.

- a. $N(N-1)$
- b. **$(N^2 - N)/2$**
- c. $N-1$
- d. $N-2$

309. is the term used for variance in transmission delay.

- a. Friction
- b. Asynchronous transfer mode
- c. Checksum
- d. **Jitter**

310. When there is heavy traffic on the network, the queuing time is

- a. Zero
- b. Remains the same
- c. Decrease
- d. **Increase**

311. In distributed route computation process each packet switch
computes its routing table locally and sends message to the

- a. Router
- b. Graph
- c. **Neighbors**
- d. WAN

35. AUI cable carries digital signal from NIC to .

312. Modem

313. CPU

314. **Transceiver**

- a. Network medium
- 315.** The is a standard wireless LAN that uses radio signals at 2.4 GHz
- a. IEEE 802.14
 - b. IEEE 802.15
 - c. IEEE 803.3
 - d. **IEEE 802.11**
- 316.** Each computer on LAN is identified by a physical address, usually imprinted on the-----
- a. Modem
 - b. Router
 - c. Switch
 - d. **Network interface card**
- 317.** Which of the following is a benefit of packet switching?
- a. Change in transmission time
 - b. **Broken data discovery**
 - c. Dedicated paths for data sending
 - d. Hefty data transmission is not easy
- 318.** Physical addressing in a WAN is similar to LAN as each packet has format with-----
- a. **Header**
 - b. Title
 - c. Frame
 - d. Address
- 319.** was first automated tool required to see whether the given computer is online or not.
- a. Probing program
 - b. **PING program**
 - c. Status program
 - d. None of the given
- 320.** In FDDI, in case of station failure, remaining stations loop back and----
- a. Data frames are discarded
 - b. **Re-route data through spare ring**

- c. Collision will occur
- d. Re-transmit data through same ring

321. The Frame format of any LAN technology must consist of-----
and

- a. Payload and CRC
- b. Frame header and CRC
- c. Payload and frame type
- d. **Frame header and payload**

322. In a ping command result, time denotes the -----

- a. TLT
- b. TTL
- c. BRT
- d. **RTT**

323. Which of the following cannot be used to extend a LAN?

- a. Repeaters
- b. **Switches Routers**
- c. Fiber modems

324. A typical port on an ATM switch operates at ----- or higher.

- a. OC-2 speed(155Gbps)
- b. OC-2 speed(1000Mbps)
- c. **OC-3speed(155Mbps)**
- d. OC-3speed(100Gbps)

325. CBR stands for ----- • Constant Byte Rate

- a. **Constant Bit Rate**
- b. Connection Break Recovery
- c. None of the above

326. If two buildings are located far from each other then bridge and-----
-can be used to connect two LAN.

- a. **Optical fiber**
- b. Ethernet modem
- c. Switch
- d. NIC

327. Variable length packets require ----- to accommodate the largest possible packet and thus to detect the end of the packet

- a. **Hardware**
- b. Software
- c. Instruction
- d. Network ownership

49-----is used for compressed audio and video ,where the data rate depends on the level of compression that can be achieved

- e. Constant Bit Rate
- f. Bit Rate
- g. **Variable Bit Rate**
- h. Frame Relay

328. In order to forward a copy of frame, the bridge uses-----

- a. Network
- b. **Destination**
- c. IP

329. ATM header is about ----- of the cell.

- a. 100%

b. **10%**

c. 20%

d. 50%

330. The system administrator must coordinate to avoid the conflict in-----
----- hardware addressing scheme.

a. Static

b. Dynamic

c. **Configurable**

d. Fixed

331. Each computer on LAN is identified by a physical address,
usually imprinted on the-----

Modem

Router

Switch

Network interface card

332. Which of the following is a benefit of packet switching?

a. Change in transmission time

b. **Broken data discovery**

- c. Dedicated paths for data sending
- d. Hefty data transmission is not easy

333. Physical addressing in a WAN is similar to LAN as each packet has format with---

- a. **Header**
- b. Title
- c. Frame
- d. Address

56 ----- was first automated tool required to see whether the given computer is online or not.

Probing program

PING program

Status program

None of the given

e. In FDDI, in case of station failure, remaining stations loop back and-----

Data frames are discarded

Re-route data through spare ring

Collision will occur

Re-transmit data through same ring

f. The Frame format of any LAN technology must consist of----- and -----

Payload and CRC

Frame header and CRC

Payload and frame type

Frame header and payload

g. In which type of error detection, the arithmetic sum is calculated before and after the transmission of data at both communications end?

CRC

Parity

Checksum

Hash function

334. In a ping command result, time denotes the -----

TLT

TTL

BRT

RTT

335. Which of the following cannot be used to extend a LAN?

Repeaters

Switches

Routers

Fiber modems

62.A typical port on an ATM switch operates at- or higher.

b. OC-2 speed(155Gbps)

c. OC-2 speed(1000Mbps)

d. OC-3speed(155Mbps)

e. OC-3speed(100Gbps)

336. CBR stands for-----

a. Constant Byte Rate

b. Constant Bit Rate

c. Connection Break Recovery

d. None of the above

337. If two buildings are located far from each other then bridge and can be used to connect two LAN.

a. Optical fiber

b. Ethernet modem

c. Switch

d. NIC

65. Variable length packets require to accommodate the largest possible packet and

thus to detect the

end of the packet

Hardware

Software

Instruction

Network ownership

338. is used for compressed audio and video ,where the

data rate depends on the level of compression that can be achieved

a. Constant Bit Rate

b. Bit Rate

c. Variable Bit Rate

d. Frame Relay

e.

e. The type of the error occurs due to a hardware failure e.g. the second bit of every character will/can be damage is called-----

Burst error

Vertical error

Parity error

Single bit error

339. In order to forward a copy of frame, the bridge uses-----

Network

Destination

IP

Source

340. The system administrator must coordinate to avoid the conflict in hardware addressing scheme.

- 341. Static
- 342. Dynamic
- 343. Configurable**
- 344. Fixed

70 ----- If any switch of the destination computer does not agree to setting up the ----- an error message is sent back and the switched virtual circuit is not established.

- a. Virtual path
- b. Public virtual circuit
- c. Circuit
- d. Virtual circuit**

345. To reduce ----- , computers can be connected and distributed to a number of hubs, then the hubs can be connected to a switch.

- a. **Costs**
- b. Quality
- c. Efficiency
- d. Noise

346. Asynchronous Transfer Mode (ATM) as a single networking technology has only minimal

provision for inter-operation with other technologies.

Wired

Virtual circuit

LAN

Universal

347. LAN is used for local area but -- can extend LAN across large distances.

Bridge

WAN

Hub

Satellite Bridge

348. The recent executed instruction in spatial aspect of the locality of reference.

Is executing again next

won't be executed again

will be executed at a later time

nearby and execution will be executed in future

- b. In distributed route computation process each packet switch computes its routing table locally and sends message to the:

Router

Graph

Neighbors

WAN

- c. Which of the following are the issues/problems with broadcasting technique.

For every Broadcasting frame on the network each computer see computational resources and places the contents into memory which interrupts the CPU.

349. If a pair of computer use broadcasting instead of sending them directly all other computers waste CPU time while discarding the frames.

- a. **I and II**
- b. II and III
- c. I and III
- d. All of the given options

77 ----- Networks are sometime called Intranet

Private

Public

Virtual

Wireless

350. Handling variable length packets leads to memory-----

- a. **Fragmentation**
- b. Management
- c. Location
- d. Allocation

79. The bridges communicate with each other on the network and use

decide which bridge

will not forward frames if a cycle occurs.

Dijkstra

Shortest path first

Vector distance

Distributed Spanning Tree

351. ATM is a single technology designed to meet the goal of both-----and---

a. LAN & Pan

b. LAN & WAN

c. LAN & MAN

d. All of above

352. Asynchronous transfer mode (ATM) designed to work on fibe

Coaxial cables

Twisted pair (pg#72)

LAN

Ethernet

82 ----- uses counter rotating rings in which data flows in opposite direction.

b. Token Ring

c. **FDDI** (pg#31)

d. ATM

e. Frame delay

353. For hefty networks ----- topology is used.

a. Bus

b. Mesh

c. **Star** (**Google**)

d. Pool

84.A packet switch may connect to and to other packet switch.

Computers (pg#56)

Router

Coaxial cable

Hub

354. Quality of Service desires are specified all connection set up time, and are never altered for the-----

a. ATM

b. Connection duration (pg#70)

c. Virtual connections

d. LAN

355. A network that uses ---- usually consist of a single long cable to which computer attach.

Star topology

Ring topology

Bus topology (pg#26)

Bus and Ring topology

356. Asynchronous Transfer Mode(ATM) can dynamically establish a-----

Switched virtual circuit (pg#70)

Permanent virtual circuit

Virtual path indicator

Connection

357. The length of time required to send a variable length packet is-----

5 seconds

Variable (pg#72)

Constant

Fix

b. Which layer of the OSI model directly communicates and controls the transmission medium.

Transmission

Physical (Google)

Application

Network

358. Which of the following is not a function of ping command?

Local computer network issue

Congestion in the network

Relay nodes (pg#11)

Remote host availability

359. is not used to extend LAN.

Hub

Fiber modem

Bridge

Repeater (pg#149)

360. Frames include additional information to detect/correct errors are insert by-----

--

-

Sender

Receiver (pg#18)

Network devices

Sender and Receiver

b. SMDS is-----

Connectionless service (pg#65)

Connection oriented service

Used for delivering blocks of data

None of the given

c. The topology each computer is connected to a central hub is called-----

Ring topology

b. **Star topology** (pg#5)

d. Tree topology

e. Mesh topology

361. If 0 sender with a 1Gigabit NIC and receiver with 100Megabits NIC wants to communicate with each other at which speed they can communicate?

a. 11Megabit

b. **100Megabit (Google)**

c. 1 Gigabit

d. 11 Gigabit

96. The ----- field of Ethernet frame has the purpose of receiver synchronization.

- a. **Preamble** (pg#36)
- b. Frame type
- c. Data
- d. CRC

97 ----- are more frequently being designed to allow specification of the quality of service required by the user.

- a. Switch
- b. VIP
- c. VCI
- d. **Networks** (pg#70)

98. -----and ----- are two types of frame type.

- d. Internal and external
- e. Single and dual
- f. **Implicit and explicit** (pg#35)
- g. Multicast and broadcast

362. According to a Bridge performance, if both the source and destination are on the same segment

- a. It forwards the frame to the other segment
- b. **It does not forward the frame to the other segment (Pg#50)**
- c. It does not forward the frame to the same segment
- d. It forwards the frame to every other segment

100. ----- According to size there are ----- classification of networks.

2(pg#4)

3

4

e. We need packets rather than bits because-----

To compress the data

Packets move faster

Sender and receiver needs to coordinate to detect transmission error (pg#14)

Packets only move between two terminals

363. Point-to-point topology is

Size based

Mobility based

by

c. **Connectivity based (pg#5)**

d. Medium based

103.

Even parity can be used to check for bit/bits of

errors.

a. **1** (pg#18)

364. 2

365. 3

366. 0

104 ----- is inflexible.

a. Routing

- a. Searching
- b. Processing
- c. **Static routing** (pg#62)

367. The switch that has computers attached with it is called switch.

- a. **Exterior** (pg#59)
- b. Interior
- c. Packet
- d. Front

106 ----- cannot be extended arbitrarily far or to handle arbitrarily many computers.

- 368. WAN
- 369. **LAN** (pg#55)
- 370. MAN
- 371. LED

107 In CSMA, if simultaneous transmission occurs, they interfere with each other and this

Phenomenon is called collision.

- 372. Computers
- 373. Networks
- 374. **Frames** (pg#28)
- 375. Cable wires

108. The _____ of errors is easier than of errors.

Correction, detection

Detection, correction (Google)

Correction, creation

Detection, manipulation

376. Frame relay is-----

- a. **Connection oriented service** (pg#64)
- b. Connectionless service
- c. It typically ranges from 5Mbps to 1000Mbps
- d. None of the above

110 is a measure of the rate at which data can be sent through the network.

- e. **Throughput**(pg#79)
- f. Propagation delay
- g. Access delay
- h. Switching delay

377. Which of the following technologies can extend the diameter of LAN medium?

- a. Fiber optic
- b. Repeaters
- c. Diameter of LAN medium is not extended
- d. **Both fiber optic and Repeaters** (pg#48,49)

378. The information about is kept in routing tables.

- a. Entries
- b. **Destinations** (pg#58)

379. Source

380. Packets

113 ----- Service paradigms is similar to the telephone system.

- a. **Connection-oriented** (pg#75)

381. Connectionless

- a. Public-oriented
- b. Private-oriented

382. Bluetooth uses shared medium and instead of coaxial cable.

- a. **Radio waves** (pg#29)
- b. Infrared waves
- c. Micro waves
- d. Low frequency waves

383. Connections in Asynchronous transfer mode are called-----

- a. **Virtual channels** pg#67)
- b. Virtual connections
- c. Cells
- d. Cell tax

384. The maximum cable length of a LAN is fixed because-----

- a. Additional hardware are not suitable for the LAN extension
- b. **The electrical signal level gets weaker as it travels** (pg#48)
- c. Additional software are not suitable for the LAN extension
- d. The electrical signal level gets stronger as it travels

385. The sharing of a --- among the workers in an office in an office is a goal of resource goal.

- a. Mouse
- b. **Printer** (pg#9)
- c. Keyboard
- d. Motherboard

386. If a small set of bits changes near a specific location due to lightning or electric motor starting nearby etc. then these types of errors are called-----

- a. **Burst error** (pg#22)
- b. Vertical error
- c. Parity error
- d. Single bit error

119 The fiber-modem converts digital data into then transmit over the optical fiber.

387. **Pulses of light** (pg#48)

388. Electronics
389. Electromagnetic signals
390. None of these
391. The process of forwarding the packet information is called-----
--

- a. **Routing** (pg#58)
- b. Processing
- c. Hierarchal addressing
- d. Source addressing
- f. Dijkstra's algorithm can accommodate weights on in graph.

Nodes

Edges (pg#62)

Paths

Connections

392.Source Independent Phenomenon allow fast and efficient-----

- a. Source Addressing
- b. **Routing** (pg#58)

Packet switching

Store and Forward

c. ABR stands for-----

Available Bit Rate (pg#71)

Asynchronous Bit Rate

Asynchronous Byte Rate

Asynchronous Bit Redundancy

d. Switch stimulates a LAN connecting segments.

Single

Single shared

Bridged (pg#54)

Complex**125.** The Token Ring Network topology is a ring but wiring topology is a-----

e. **Star** (pg#44)

f. Bus

g. Tree

h. Mesh

393. Dynamic routing can work around automatically.

a. **Network Failure** (pg#62)

b. LAN

c. Switch

d.

Network Overhead

394. The number of connection needed for N computers in direct point to point communication is equal to:

$N(N-1)$

$(N_2-N)/2$ (Pg#23)

$N-1$

N_2

395. The type of error occurs due to a hardware failure e.g. the second bit of every character will/can be damaged is called:

Burst error

Vertical error (pg#22)

Parity error

Single bit error

396. Which is incorrect statement about bridge?

A Bridge uses two NICs to connect two segments

A Bridge is more intelligent than Repeater

Bridge do not recognize frame formats, they just amplify and retransmit the electric signal (pg#50)

397. A Bridge listens to all traffic and recognizes frame format. The satellite or radio topology in which all computers are connected to each

other via satellite radio wave is a kind of-----

a. **Broadcast network** (pg#5)

b. Point to point network

131. In direct point to point communication adding the nth computer required new connection.

a. $N(N-1)$

- c. $(N_2 - N)/2$
- d. **N-1** (pg#23)
- e. N_2

398. Which of the following statement is correct about connectors of wiring technologies?

- a. **AUI connector is used for Thicknet** (pg#44)
- b. Rj-45 connector is used for Thinnet
- c. BNC connector is used for Thicknet

399. All of the above statement are correct Data delivery from one computer to another computer is accomplished through-----

- a. LAN
- b. Fragmentation
- c. Hierarchical addressing
- d. **Store-and-Forward**(pg#56)

134. The bridges communicate with each other on the network and use algorithm to decide which bridge will not forward frame if a cycle occurs.

- e. Dijkstras
- f. Shortest path first
- g. Vector distance
- h. **Distributed spanning tree** (pg#53)

400. CPU can't process data at - speed.

- a. High
- b. Clock
- c. Slow
- d. **Network** (pg#39)

401. Network occupies large areas like cities and countries is called

- a. LAN
- b. **WAN** (Pg#4)
- c. MAN
- d. PAN

402. A network that permits many sources to take turns accessing a shared communication is a form of-----

- a. **TDM(Time division multiplexing)** (**Google**)
- b. PDM
- c. FDM
- d. WDM
- e.
- f.

by

403. ASYNCHRONOUS TRANSFER MODE is a signal-- networking technology has only minimal provision for inter-operation with other technologies.

- a. Wired
- b. LAN
- c. Virtual circuit
- d. **Universal** (pg#72)

404. Which one of the following is the computing routing table information?

- a. Manual entry
- b. **Static routing** (pg#62)

Boot time routing

405. None of the above All of the following are types of delay except-----

Propagation

Access

Throughput (pg#79)

Queuing

406. There are ----- types of switch according to the attached computers.

Four

Three

Five

Two (pg#59)

b. IBM's token ring uses -- (physically a star topology) to stimulate logic ring network.

Hubs (pg#47)

Switch

Router

Bridge

143. ATM uses small, fixed sized packets called cells. Each cell has octets.

c. 52

d. **53** (pg#66)

e. 50

f. 25

407. The interface hardware does not include type of frame.

a. Multicast frames

b. Broadcast frames

c. **Few cast frames** (pg#35)

d. The frames that are destined to the station itself

408. Which two topologies are included in point to point topology?

a. Ring and star topology

b. **Star and tree topology** (page#5)

c. Bus and mesh topology

THE END



CS610 Solved Grand Quiz 2020

Question No 1

_____ is designed to detect transmission errors, send one extra bit of information with each character.

Select Correct Option



Checksum



CRC



VRC

Question No 2

The _____ of errors is difficult than the _____ of errors.



Select Correct Option



Correction, detection (For More Visit VUStudents.pk)



Detection, correction



Correction, creation



Detection, manipulation

Question No 3

The recent executed instruction _____ in temporal aspect of the locality of reference.

Select Correct Option



Wont be executed soon



Is temporarily not referenced



Will be executed soon again (For More Visit VUStudents.pk)



Will be showing the final results

Question No 4

ATM uses the concept of _____ networking.

Select Correct Option



Connection-oriented

(For More Visit VUStudents.pk)



Connection-less

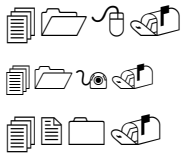


Connected

Wired Question No 5

As a result of ARPA research, the first network was established which was named as _____.

Select Correct Option



INTERNETWORKING

ARPNET

PACKET NETWORK

Question No 6

The _____ of errors is easier than the _____ of errors.

Select Correct Option



Correction, detection



Detection, correction(For More Visit VUStudents.pk)



Correction, creation



Detection, manipulation

Question No 7

_____ addresses are used in an ethernet frame.

Select Correct Option



IP(For More Visit VUStudents.pk)



Port



Logical



MAC

Question No 8

Even Parity can be used to check for _____ bit/bits of errors.

Select Correct Option

✓ **1**(For More Visit VUStudents.pk)

✓ 2

✓ 3

✓ 0

Question No 9

_____ is the most popular wiring scheme because of lowest cost.

Select Correct Option



5 Base T

✓

10 Base T(For More Visit VUStudents.pk)



100 Base T



1000 Base T

Question No 10

In which topology bi-directional communication links among each link exists.

Select Correct Option



Star



Bus



Mesh

Question No 11

A network with _____ jitter takes much longer to deliver some packets than others.

Select Correct Option



Zero



Low



Moderate

Question No 12

What is the largest network in existence?

Select Correct Option



The Internet (For More Visit VUStudents.pk)



A PAN



A Fog



Edge Network

\\

Question No 13

Missing _____ in a frame indicates receiving computer missed beginning of the message.

Select Correct Option



eot



soh

(For More Visit VUStudents.pk)



sot



eoh

Question No 14

In a ping command result time denotes the _____.

Select Correct Option



TLT



BRT



RTT

Question No 15

VPNs connection are sometimes called _____.

Select Correct Option



Tunnels

(For More Visit VUStudents.pk)



Open networking



Routing



Public Network

Question No 16

Source independence phenomenon allows fast and efficient _____.

Select Correct Option



Source addressing



Packet switching



Store and forward

Question No 17

10Base-T Ethernet is often called a _____.

Select Correct Option



Ring shaped star



Ring shaped bus



Star shaped bus

(For More Visit VUStudents.pk)



Star shaped ring

Question No 18

LAN interface hardware obeys _____ access rule.

Select Correct Option

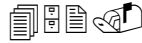
✓ **CSMA/CD** (For More Visit VUStudents.pk)



Round Robin



First come first served

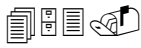


High priority

Question No 19

The number of connections needed for N computer in direct point to point communication is equal to:

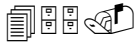
Select Correct Option



N (N-1)



N-1



N^2

Question No 20

An ATM network is built from interconnected _____.

Select Correct Option



Bridges



Virtual Circuits



ATM switches (For More Visit VUStudents.pk)



LAN

Question No 21

If a hub is used to connect many computers on a LAN, then only _____ computers can communicate at a given time.

Select Correct Option



Three



Four



One

Question No 22

Which one is a method for building routing tables?

Select Correct Option



Program



Graph Algorithm (For More Visit VUStudents.pk)



Store and forward

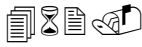


Manual entry

Question No 23

When there is heavy traffic on the network, the queuing time is _____.

Select Correct Option



Zero



Remains the same



Decreases



Increases.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))

Question No 24

_____ was especially most concerned about the lack of highly powered computers.

Select Correct Option



ARPA



IEFT



PTA

Question No 25

A _____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

Select Correct Option

- ✓ **Static addressing scheme** (For More Visit VUStudents.pk)
- ✓ Configurable addressing scheme
- ✓ Dynamic addressing scheme
- ✓ None of the given

Question No 26

In a ping command result, time denotes the _____.

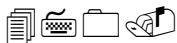
Select Correct Option



TLT



BRT



RTT

Question No 27

Which of the following task is not done by the second layer of the OSI model?

Select Correct Option



Flow control (For More Visit VUStudents.pk)



Framing



Channel coding



Error control

Question No 28

ATM header is about _____ of the cell.

Select Correct Option



5%



20%



25%

Question No 29

_____ can't be guaranteed with variable length packets as easily as it can with fixed length cells.

Select Correct Option



Error



Noise



Quality Service(For More Visit VUStudents.pk)

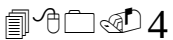


None of the given

Question No 30

Parity bit can detect _____ error(s).

Select Correct Option



4



3



2



1(For More Visit VUStudents.pk)

1)Shows sender preference for low latency, high reliability.

483. Type

484. Service type

485. Service priority

486. None of the given UDP follows four types of interactions.

a. True

b. false

487. Postfix defines how much of a address used to identify network.

True

b) False

488. The process of learning the path MTU is known as path MTU discovery.

True

False

30Protocol provides error reporting mechanism IGMP

SNMP

ICMP

None of the given

489. MTU stands for

a. Minimum transmission unit

b. Maximum transmission unit

c. Multicast transmission unit

d. None of the given

7) NEXT HEADER field in the base header defines type of header it appears at the end of fixed- size base header.

490. True

491. False

492. The maximum segment size of TCP flow control is

- a. 3000 octets
- b. 2000 octets
- c. 1000 octets
- d. None of the given

493. UDP uses best –effort delivery service.

494. True

495. False

496. Router use to forward data grams along prearranged path.

- a. Traffic class



Flow label

- b. Destination address
- c. None of the given

497. Serve same purpose in internet as frames on

LAN. a) Packets

498. Reliability is not the responsibility of the transport layer.

499. True

a. False

12) H.LEN shows the header length in unite of _____ bit.

b. 34

c. 32

500. the user datagram protocol is not end-to-end protocol. a) True

b) False

501. UDP provides connection oriented service.

a. True

b) False

17) HEADER LEN field gives size of the extension header.

502. T
True
False

503. TTL

stands for a) Time _____

to learn b) Time to

leave

a. Time to Live

504. Non of the given 19) the amount of buffer space available at any time is called the window.

505. True

506. False

20) NAT device storage State information in transition on table

507. True

a. Basic NAT does not work well for communication initiated from the internet.

a) True

508. False NAT software does not allow a PC to connect with the Internet and act as a NAT device at the same time.

509. True

b) False

510. Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the
Transmission Unit. _____

a) Least

511. Maximum

b) Fragment

512. Frame

513. NAT is not useful at a residence with Cable Modem or DSL connectivity.

a. True

b. False

514. NAT stands for

a. Network Address Translation

b. Network Address Transmission

c. Network Address Test

d. None of the given

515. Which is not the type of error messages defined by ICMP

a. Source quench

b. Time exceeded

c. Destination unreachable

d. None of the given

27) One of the parameters which motivated IP for change is address space.

The address space allows for over a million networks. But most networks

are class C and too small for many organizations.

e. 32-bit

f. 128-bit

g. 16-bit

h. 64-bit

516. Twice NAT fails if an application uses the IP address instead of Domain Name.

True

Fals

517. ☐ In IPV6 the 128 bit address unwidely in dotted decimal; requires numbers.

12

14

16

None of the given

30) TCP achieves _____ by retransmission.

518. Efficiency

519. Accuracy

Reliability

None of the given

520. Reliability is not the responsibility of the Transport layer.

- a. True
- b. False

32) TCP achieves _____ by retransmission.

- c. Efficiency
- d. Accuracy

e. Reliability

f. None of the given

521. Source is responsible for fragmentation.

IPV4
IPV6

34) A popular use of _____ is TCP Splicing. It interconnects two independent

TCP connections and performs segment rewriting.

NAT

NAPT

Twice NAT

All of these

b. Which protocol is used to test different tools. NAT stands for

522. ICMP

523. IGMP

524. TCP/IP

525. None of the given

36) TCP uses _____ format for all messages.

- a. Single
- b. Double
- c. Multiple
- d. None of the given

526. Preliminary version of IP was called _____

IP- New Generation (IPng)

527.) b) IP- Next Generation (IPng)

528. IP- Net Generation (IPng)

529. None of the given

24) UDP diagram format is _____ bits.

- a. Network Address Translation
- b. Network Address Transmission
- c. Network Address Test
- d. None of the given

530. Which is not the type of error messages defined by ICMP

- a. Source quench
- b. Time exceeded
- c. Destination unreachable
- d. None of the given

531. _____ Encapsulates IP datagram as data area in hardware frame.

532. Network Interface

533. Datalink Layer

- a. _____ protocol uses three way handshake to begin a connection.

TCP _____

534. NAPT stands for _____

Network Address and Port Translation

42) For _____ information about forwarding is stored in a routing table, which is initialized and must be updated as network topology changes.

Efficiency

535. The network layer Protocol ICMP _____ stands for Internet Control Message Protocol

44) ICMP message transport is acted upon by getting ICMP _____ in IP.

Encapsulate 45) Header contains all information needed to datagram to the destination computer, but which one of the following is not included

Rectifier

46) System rebooting is one of the major problems in the reliable delivery

a)true

536.)TCP uses the term segment to refer
to a message

47)) ----- provides computer to computer communication

537. IP

b)TCP

c)ICMP

d)IGMP

538. Network Address Translate(NAT) requires device to perform packet
translation a) true

539. false

540.) ----- is a type of address used for collection of computers with
same prefix

a. Cluster

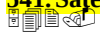
CS610 Midterm Solved McQs Papers By Waqar Sidhu (File 1)

Question No 1

In broadcast topology there are two categories _____.

Select Correct Option

541. Satellite/Radio and Ring.....(For More Visit VUStudents.pk)



Satellite/Radio and Star



Ring and Star



Satellite/Radio

Question NO 2

In ----- , network occupies the smaller area like a room a floor or a building.

Select Correct Option

✓ LAN (For More Visit VUStudents.pk)

✓ WAN

✓ MAN

✓ PAN

Question NO 3

The Ethernet standard specifies that frame are send using the ----- --.

Select Correct Option

✓ Manchester Encoding (For More Visit VUStudents.pk)

✓ Nonreturn to zero(NRX)

✓ Return to zero(RZ)

✓ Differential Manchester

Question No 4

A -----relies on the hardware manufacturer to assign a unique physical to each network interface.

Select Correct Option

545. Static addressing scheme (For More Visit VUStudents.pk)



Configurable addressing scheme



Dynamic addressing scheme



Burned MAC

Question NO 5

When an application ----- data, it makes a copy of the data available to all other computers on the network.

Select Correct Option



Broadcasting



550. Multicasting (For More Visit VUStudents.pk)

Unicasting



Anycasting

Question NO 6

Formally name _____ informally known as the thick wire Ethernet or Thick net.

Select Correct Option



10 Base 2



10 Base 5

555. 10 Base T.....(For More Visit VUStudents.pk)



100 Base T

Question NO 7

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

Select Correct Option



Connectionless service paradigm

558. Connection-oriented service paradigm.....(For More Visit VUStudents.pk)



Both Connectionless service paradigm



Connection-oriented service without acknowledgement

Question NO 8

In which type of topology computers are connected to a central device?

Select Correct Option

561. Ring.....(For More Visit VUStudents.pk)



Mech



Bus



Star

Question No 9

----- was especially most concerned about the lack of highly powered computers.

Select Correct Option



ARPA.....(For More Visit VUStudents.pk)

- ✓ IEEE
- ✓ EIA
- ✓ Government

Question NO 10

“Frame type” field of the header consists of _____ bits.

Select Correct Option

    8

    10

    16

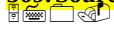
568. 32.....(For More Visit VUStudents.pk)

Question NO 11

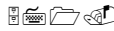
A bridge uses the _____ to determine which computers connect to which segment and use the _____ to determine whether to forward a copy of frame.

Select Correct Option

569. Source address, Destination address.....(For More Visit VUStudents.pk)



Destination address, Source address



Destination address, MAC Address



Destination address, Destination address

Question No 12

Generally a Bridge is used to perform _____.

Select Correct Option



Extend a LAN, Filter and Forward a packet



Routing

575. Filter and Forward a frame.....(For More Visit VUStudents.pk)



Filter and Forward a packet

Question NO 13

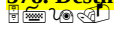
If bridge cannot find _____ in its tables then forwards frames to the other segment.

Select Correct Option



IP address

578. Destination address.....(For More Visit VUStudents.pk)



Source address



Port address

Question NO 14

_____ technology can support both LAN and WAN connections.

Select Correct Option



Frame relay



Ethernet



SMFS

584. ATM.....(For More Visit VUStudents.pk)

Question NO 15

Virtual Circuit connection identifier occupies _____ bits.

Select Correct Option

585. 16.....(For More Visit VUStudents.pk)



24



32



8

Question No 16

_____ is used for ordinary data.

Select Correct Option

589. Available Bit Rate (ABR) service.....(For More Visit VUStudents.pk)



Uniform Bit Rate (UBR) service



Constant Bit Rate (CBR) service



Variable Bit Rate (VBR) service

Question NO 17

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers _____.

Select Correct Option

593. Connectionless service paradigm.....(For More Visit VUStudents.pk)



Connection oriented service paradigm



Both Connectionless and Connection-oriented service paradigm



Stateless Connection oriented service paradigm

Question NO 18

The term _____ refers to the general concept of a small block of data in a network.

Select Correct Option

597. Packet.....(For More Visit VUStudents.pk)



Byte



Cell



Information

Question No 19

LAN that uses ATM technology has a _____.

Select Correct Option

601. Star topology.....(For More Visit VUStudents.pk)



Bus topology



Ring topology



Mesh topology

Question NO 20

The next hop to which a packet is sent depends only on _____ .

Select Correct Option

605. Packet's destination.....(For More Visit VUStudents.pk)



Packet's original source



Path the packet has taken



Routed Protocol

(File 2)

Question No 1

ATM was designed for _ and use a _.

Select Correct Option



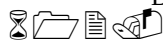
LANs, Connectionless service paradigm



WANs, Connectionless service paradigm



Both LANs and WANs, Connectionless service paradigm



Both LANs and WANs, Connection-oriented service paradigm

Question NO 2

In Point-to-Point type of topologies there are two topologies _.

Select Correct Option



Tree and Ring



Star and Ring

615. Star and Tree.....(For More Visit VUStudents.pk)



Bus and Mesh

Question NO 3

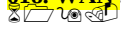
In ----- , network occupies larger areas like cities & countries.

Select Correct Option



LAN

618. WAN..... (For More Visit VUStudents.pk)



MAN



CAN

Question No 4

Hardware that calculates a CRC uses two simple components.

Select Correct Option



AND unit with XOR unit

622. Shift register with XOR unit (For More Visit [VUStudents.pk](https://www.vustudents.pk))



Shift register with AND unit



Shift register with shift XOR unit

Question NO 5

A network that uses a -----usually consist of a single long cable to which computers attach.

Select Correct Option



Star Topology



Ring Topology

627. Bus Topology (For More Visit [VUStudents.pk](https://www.vustudents.pk))



Tree Topology

Question NO 6

A -----provides mechanism that a customer can use to set a physical address.

Select Correct Option



Static addressing scheme



Configurable addressing scheme



Dynamic addressing scheme



Masking

Question No 7

Formally name _____ informally known as the thick wire Ethernet or Thick net.

Select Correct Option



10 Base 2



10 Base 5

635. 10 Base T (For More Visit VUStudents.pk)



100 Base T

Question NO 8

Preamble field in Ethernet frame consist of _____ bits.

Select Correct Option



8

638. 32.....(For More Visit VUStudents.pk)



48



64

Question NO 9

A bridge functions in the _____ layer(s)

Select Correct Option

641. Physical (For More Visit VUStudents.pk)



Data link



Network and Data link



Physical and Data link

Question No 10

Generally a Bridge is used to perform _____ .

Select Correct Option



Extend a LAN, Filter and Forward a packet



Routing

647. Filter and Forward a frame (For More Visit VUStudents.pk)



Filter and Forward a packet

Question NO 11

_____ technology can support both LAN and WAN connections.

Select Correct Option



Frame relay



Ethernet



SMFS

652. ATM.....(For More Visit VUStudents.pk)

Question NO 12

One repeater _____, two repeaters _____ the maximum cable length limitation.

Select Correct Option



Square root, Cube root

654. Double, Triple.....(For More Visit VUStudents.pk)



Triple, Quad



Double, Quad

Question No 13

_____ is used for compressed audio and video where the data rate is related to level of compression.

Select Correct Option



Constant Bit Rate (CBR) service

658. Variable Bit Rate (VBR) service.....(For More Visit VUStudents.pk)



Available Bit Rate (ABR) service



Uniform Bit Rate (UBR) service

Question NO 14

_____ is used for ordinary data.

Select Correct Option

661. Available Bit Rate (ABR) service.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))



Uniform Bit Rate (UBR) service



Constant Bit Rate (CBR) service



Variable Bit Rate (VBR) service

Question NO 15

_____ sends a message to a remove and reports whether the computer responds.

Select Correct Option



Telnet

666. Ping.....(For More Visit

[VUStudents.pk](https://www.vustudents.pk)) Ping



Traceroute



ICMP

Question No 16

A network with throughput T and delay D would have a total no of _____ bits in transit at any time.

Select Correct Option



T+B



T x D/Unit Time



T/D

672. T x D.....(For More Visit VUStudents.pk)

Question NO 17

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____.

Select Correct Option



Phase Division Multiplexing



Time Division Multiplexing



Frequency Division Multiplexing



Ware Division Multiplexing

Question NO 18

A network which is based on packets uses a computer which may have _____, memory and _____ for switching.

Select Correct Option



Administration Rights, Algorithm



Source address, Destination address



Interface, Program



Input, Output

Question No 19

In direct point to point communication adding the N^{th} computer requires _____ new connections.

Select Correct Option

   $N(N-1)$

   N^2

$N-1$

680. $(N^2 - N)/2$ (For More Visit VUStudents.pk)

Question NO 20

Computers attached to an ethernet use _____ in which a computer waits for the Ether to be idle before transmitting a frame.

Select Correct Option

681. CSMA/CD(For More Visit VUStudents.pk)

   CSMA/CA

   Token passing

   Round Robin

(File 3)

Question No 1

In ----- , network occupies the smaller area like a room a floor or a building.

Select Correct Option

685.LAN (For More Visit VUStudents.pk)



WAN



MAN



PAN

Question NO 2

A network that uses a -----usually consist of a single long cable to which computers attach.

Select Correct Option



Star Topology



Ring Topology



Bus Topology



Tree Topology

Question NO 3

A network uses a ----- if all computers attach to a central point

Select Correct Option



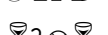
Star Topology



Ring Topology



Bus Topology



Mesh Topology

Question No 4

A ----- provides mechanism that automatically assigns a physical address to a station when the station first boots.

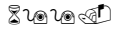
Select Correct Option



Static addressing scheme



Configurable addressing scheme



Dynamic addressing scheme



Dynamic Host Configuration

Question NO 5

----- was especially most concerned about the lack of highly powered computers.

Select Correct Option



ARPA



IEEE



EIA



Government

Question NO 6

A bridge functions in the _____ layer(s)

Select Correct Option

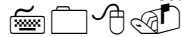
705. Physical.....(For More Visit VUStudents.pk)



Data link



Network and Data link

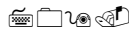


Physical and Data link

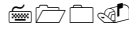
Question No 7

Which of the following is not used to extend LAN?

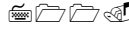
Select Correct Option



Bridge



Repeater



Fiber modem

☐ **Router**

Question NO 8

The working of connection-oriented service for networking is similar to the _____.

Select Correct Option



Ethernet



Telephone



IP

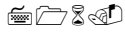


UDP

Question NO 9

_____ senses a signal on a cable and then transmits an amplified copy of the signal.

Select Correct Option



Repeater



Bridge



Hub



Transceiver

Question No 10

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers _____.

Select Correct Option



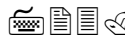
Connectionless service paradigm



Connection oriented service paradigm



Both Connectionless and Connection-oriented service paradigm



Stateless Connection oriented service paradigm

Question NO 11

A Bridge forward or filters a frame by comparing the information in its address table to the frame's _____.

Select Correct Option

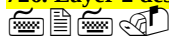


Layer 2 source address



Layer 3 source address

726. Layer 2 destination address.....(For More Visit VUStudents.pk)



Layer 3 destination address

Question NO 12

ATM is designed to work on _____.

Select Correct Option



Twisted Pair



Coaxial



Radio Frequency

731. Fiber.....(For More Visit VUStudents.pk)

Question No 13

_____ identifies intermediate computers along a path to remote destination.

Select Correct Option



Ping

733. Traceroute.....(For More Visit VUStudents.pk)



ICMP



OSPF

Question NO 14

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Select Correct Option

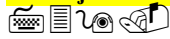


Greedy Algorithm



Distance vector Algorithm

738. Dijkstra's algorithm.....(For More Visit VUStudents.pk)



Binary Search Algorithm

Question NO 15

_____ has no way to determine the cause of the problem.

Select Correct Option

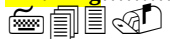


ICMP



ARP

742. Ping.....(For More Visit VUStudents.pk)



Traceroute

Question No 16

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____.

Select Correct Option

-  Phase Division Multiplexing
-  Time Division Multiplexing
-  Frequency Division Multiplexing
-  Wave Division Multiplexing

Question NO 17

No error detection scheme is perfect because transmission errors can effect _____.

Select Correct Option

-  Additional information
-  **749. Additional information & data**.....(For More Visit VUStudents.pk)
-  Transmission medium
-  Data

Question NO 18

Computers attached to an ethernet use _____ in which a computer waits for the Ether to be idle before transmitting a frame.

Select Correct Option

-  **752. CSMA/CD**.....(For More Visit VUStudents.pk)
-  CSMA/CA
-  Token passing
-  Round Robin

Question No 19

Most WAN system include mechanism that can be used to eliminate the common case of

duplication routing is called _____.

Select Correct Option



Dynamic route



Hierarchical address

758. Default route.....(For More Visit VUStudents.pk)



Shortest path

Question NO 20

A typical port on an ATM switch operates at _____ or higher.

Select Correct Option



OC-2 speed (155Mbps)



OC-3 speed (100Mbps)

762. OC-3 speed (155Mbps)(For More Visit VUStudents.pk)



OC-3 speed (155Gbps)

(File 4)

Question No 1

TM was designed for _____ and use a _____.

Select Correct Option



LANs, Connectionless service paradigm



WANs, Connectionless service paradigm



Both LANs and WANs, Connectionless service paradigm



Both LANs and WANs, Connection-oriented service paradigm

Question NO 2

In broadcast topology there are two categories _____.

Select Correct Option

768. Satellite/Radio and Ring.....(For More Visit VUStudents.pk)



Satellite/Radio and Star



Ring and Star



Satellite/Radio

Question NO 3

----- have advantages arisen from the size and ease of computation.

Select Correct Option



CRC



Parity

774. Checksums.....(For More Visit VUStudents.pk)



VRS

Question No 4

When an application ----- data, it makes a copy of the data available to all other computers on the network.

Select Correct Option



Broadcasting

777. Multicasting (For More Visit VUStudents.pk)



Unicasting



Anycasting

Question NO 5

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____ .

Select Correct Option



Connectionless service paradigm

781. Connection-oriented service paradigm (For More Visit VUStudents.pk)



Both Connectionless service paradigm



Connection-oriented service without acknowledgement

Question NO 6

If the sending computer crashes before transmitting a complete frame, the_____ - will not arrive and the receiver will know that the frame is _____.

Select Correct Option



soh, complete



soh, incomplete

786. eot, incomplete.....(For More Visit VUStudents.pk)



eot, complete

Question No 7

----- was especially most concerned about the lack of highly powered computers.

Select Correct Option



ARPA.....(For More Visit VUStudents.pk)



IEEE



EIA



Government

Question NO 8

“Frame type” field of the header consists of _____ bits.

Select Correct Option



8



10



16

791. 32.....(For More Visit VUStudents.pk)

Question NO 9

Virtual Circuit connection identifier occupies _____ bits.

Select Correct Option

792. 16.....(For More Visit VUStudents.pk)

 24

 32

 8

Question No 10

_____ senses a signal on a cable and then transmits an amplified copy of the signal.

Select Correct Option

796. Repeater.....(For More Visit VUStudents.pk)

 Bridge

 Hub

 Transceiver

Question NO 11

One repeater _____, two repeaters _____ the maximum cable length limitation.

Select Correct Option



Square root, Cube root

801. Double, Triple.....(For More Visit VUStudents.pk)



Triple, Quad



Double, Quad

Question NO 12

A Bridge forward or filters a frame by comparing the information in its address table to the frame's _____.

Select Correct Option

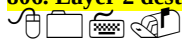


Layer 2 source address



Layer 3 source address

806. Layer 2 destination address.....(For More Visit VUStudents.pk)



Layer 3 destination address

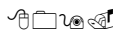
Question No 13

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____.

Select Correct Option



Phase Division Multiplexing



Time Division Multiplexing



Frequency Division Multiplexing



Wave Division Multiplexing

Question NO 14

No error detection scheme is perfect because transmission errors can effect _____.

Select Correct Option

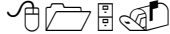


Additional information

813. Additional information & data.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))



Transmission medium



Data

Question NO 15

LAN that uses ATM technology has a _____.

Select Correct Option

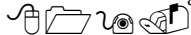
816. Star topology.....(For More Visit [VUStudents.pk](https://www.vustudents.pk))



Bus topology



Ring topology



Mesh topology

Question No 16

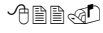
FDDI can transmits data at a rate of _____ .

Select Correct Option

820. 100 million bits per second.....(For More Visit VUStudents.pk)



10 million bits per second



1000 million bits per second



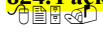
3 million bits per second

Question NO 17

The next hop to which a packet is sent depends only on _____ .

Select Correct Option

824. Packet's destination.....(For More Visit VUStudents.pk)



Packet's original source



Path the packet has taken



Routed Protocol

Question NO 18

Most WAN system include mechanism that can be used to eliminate the common case of duplication routing is called _____ .

Select Correct Option



Dynamic route



Hierarchal address

830. Default route.....(For More Visit VUStudents.pk)



Shortest path

Question No 19

_____ in which delay among packets can vary.

Select Correct Option

- ✓ Virtual Private Network
- ✓ Isochronous Networks
- ✓ Asynchronous Networks
- ✓ Synchronous Networks

Question NO 20

A typical port on an ATM switch operates at _____ or higher.

Select Correct Option

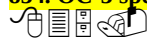


OC-2 speed (155Mbps)



OC-3 speed (100Mbps)

834. OC-3 speed (155Mbps)(For More Visit VUStudents.pk)



OC-3 speed (155Gbps)

(File 5)

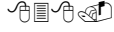
Question No 1

In ----- , network occupies the smaller area like a room a floor or a building.

836. LAN (For More Visit VUStudents.pk)



WAN



MAN



Wireless WAN

Question NO 2

In which type of topology computers are connected to a central device?

Select Correct Option

840. Ring (For More Visit VUStudents.pk)



Mech



Bus



Star

Question NO 3

The term _____ refers to the general concept of a small block of data in a network.

Select Correct Option

844. Packet (For More Visit VUStudents.pk)



Byte



Cell



Information

Question No 4

_____ is the scheme, which is designed to help in detecting transmission errors by sending one extra bit of information with each character.

Select Correct Option

848. Parity checking (For More Visit VUStudents.pk)



Checksums



CRC



LRC

Question NO 5

----- have advantages arisen from the size and ease of computation.

Select Correct Option

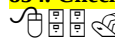


CRC



Parity

854. Checksums (For More Visit VUStudents.pk)



VRS

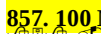
Question NO 6

The Fast Ethernet hardware operates at a rate of _____.

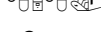
Select Correct Option



10 Mbps



857. 100 Mbps.....(For More Visit VUStudents.pk)



1000 Mbps



1 Gbps

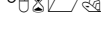
Question No 7

The Global Ethernet hardware operates at a rate of _____.

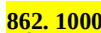
Select Correct Option



10 Mbps



100 Mbps



862. 1000 Mbps.....(For More Visit VUStudents.pk)



10 Gbps

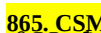
Question NO 8

Wireless LAN uses _____ as a media access method.

Select Correct Option



CSMA/CD



865. CSMA/CA.....(For More Visit VUStudents.pk)



Token passing



Round Robin

Question NO 9

A -----relies on the hardware manufacturer to assign a unique physical to each network interface.

Select Correct Option

868. Static addressing scheme (For More Visit VUStudents.pk)



Configurable addressing scheme



Dynamic addressing scheme



Burned MAC

Question No 10

IEEE LLC/SNAP header is consist of _____ which is used to specify the type of data.

Select Correct Option

872. 8 octets (For More Visit VUStudents.pk)



10 octets



12 bytes



16 bits

Question NO 11

Formally name _____ informally known as the thick wire Ethernet or Thick net.

Select Correct Option

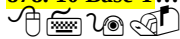


10 Base 2



10 Base 5

878. 10 Base T.....(For More Visit VUStudents.pk)



100 Base T

Question NO 12

_____ senses a signal on a cable and then transmits an amplified copy of the signal.

Select Correct Option

880. Repeater.....(For More Visit VUStudents.pk)



Bridge



Hub



Transceiver

Question No 13

Generally a Bridge is used to perform _____ .

Select Correct Option



Extend a LAN, Filter and Forward a packet



Routing

886. Filter and Forward a frame.....(For More Visit VUStudents.pk)



Filter and Forward a packet

Question NO 14

Which of the following is not used to extend LAN?

Select Correct Option



Bridge



Repeater



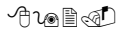
Fiber modem

891. HUB.....(For More Visit VUStudents.pk)

Question NO 15

If bridge cannot find _____ in its tables then forwards frames to the other segment.

Select Correct Option



IP address

893. Destination address.....(For More Visit VUStudents.pk)



Source address



Port address

Question No 16

Most WAN system include mechanism that can be used to eliminate the common case of duplication routing is called _____.

Select Correct Option



Dynamic route



Hierarchal address

898. Default route.....(For More Visit VUStudents.pk)



Shortest path

Question NO 17

The working of connection-oriented service for networking is similar to the _____.

Select Correct Option

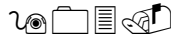


Ethernet

901. Telephone.....(For More Visit VUStudents.pk)



IP



UDP

Question NO 18

ATM assigns each VC a _____ identifier which is divided into two parts to produce a hierarchy.

Select Correct Option



23-bit

905. 24-bit.....(For More Visit VUStudents.pk)



21-bit



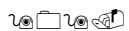
22-bit

Question No 19

_____ is used for delay sensitive applications requiring high data rate.

Select Correct Option

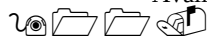
908. Constant Bit Rate (CBR) service.....(For More Visit VUStudents.pk)



Variable Bit Rate (VBR) service



Available Bit Rate (ABR) service



Uniform Bit Rate (UBR) service

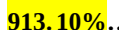
Question NO 20

Header of cell takes _____ of network capacity.

Select Correct Option



5%



913.10%.....(For More Visit VUStudents.pk)



12%



15%



Prouds to Be Virtualian
MC100401285

CS610- Computer Network

Solved MCQS

From Midterm Papers

April 25,2012

Moaaz.pk@gmail.com

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PSMD01

CS610- Computer Network

MIDTERM EXAMINATION

Spring 2010

Question No: 1 (Marks: 1)

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

► True (Computer Networks and Internets, page82)

916. False

Question No: 2 (Marks: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

► [Ping](#) (Computer Networks and Internets, page9)

917. Traceroute

918. ICMP

919. Non of the given

Question No: 3 (Marks: 1)

----- was especially concerned about the lack of high powered computers.

► [ARPA](#) (Computer Networks and Internets, page5)

920. IEEE

921. EIA

922. Non of the given

Question No: 4 (Marks: 1)

The term ----- is used to denote the definition of a packet used with a specific type of network.

923. Packet

► [Frame](#) (Computer Networks and Internets, page 73)

924. Data

925. None of the given

Question No: 5 (M a r k s: 1)

Computer networks are often called ----- because they use packet technology.

- 926. Ethernet
- 927. Switch networks

► [Packet networks](#) (Computer Networks and Internets, page 73)

- 928. None of the given

Question No: 6 (M a r k s: 1)

----- have advantages arisen from the size and ease of computation.

- 929. CRC

► [Parity](#) (Page 19)

- 930. Checksums
- 931. None of given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- 932. CSMA/CD
- 933. CSMA/CA

► [TOKEN PASSING](#) (Computer Networks and Internets, page95)

- 934. None of the given

Question No: 8 (M a r k s: 1)

IEEE LLC/SNAP header is ----- , which is used to specify the type of data.

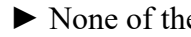
935. 8 octets (Computer Networks and Internets, page112)

936. ~~8 bytes~~

- 937. 8 bits
- 938. None of the given

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- ▶ 10 Base 2
- ▶ 10 Base 5
- ▶ **10 Base T (Page 43)**
- ▶  
- ▶ None of the given

Question No: 10 (M a r k s: 1)

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

939. RJ-45, 10 Base T (Computer Networks and Internets, page 126)

940. RJ-45, 10 Base 5

941. BNC, 10 Base 2

942. BNC, 10 Base T

Question No: 11 (M a r k s: 1)

A bridges function in the _____ layers(s).

► **Physical (MAC)** [Click here for Detail](#)



- 943. Data link
- 944. Network
- 945. Physical (MAC) and Data link

Question No: 12 (M a r k s: 1)

A Bridge can _____

- 946. Filter a frame
- 947. Forward a frame
- 948. Extend a LAN

► **Do all the above** [Click here for Detail](#)

Question No: 13 (M a r k s: 1)

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- 949. Layer 2 source address
- 950. Source node's physical address

► **Layer 2 destination address** [Click here for Detail](#)

- 951. Layer 3 destination address

Question No: 14 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- 952. Greedy algorithm
- 953. Distance vector algorithm

► **Dijkstra's algorithm** (Computer Networks and Internets, page112)

954. Non of the given _____
is used for audio and video, since these have predefined maximum data rates

► [Constant Bit Rate \(CBR\) service](#) (Page 71)

- 955. Variable Bit Rate (VBR) service
- 956. Available Bit Rate (ABR) service
- 957. None of the given

Question No: 16 (M a r k s: 1)

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers _____.

► [Connectionless service paradigm](#) (Computer Networks and Internets, page112)

- 958. Connection oriented service paradigm
- 959. Both Connectionless and Connection-oriented service paradigm
- 960. None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____bit in transit at any time.

961. T / D

962. **T x D**(Computer Networks and Internets, page203)

963. ~~T~~ + D

964. None of the given

Question No: 18 (M a r k s: 1)

ATM is designed to work on _____ .

965. Twisted Pair

966. Coaxial

967. Radio Frequency

► **Fiber** (Computer Networks and Internets, page 72)

Question No: 19 (M a r k s: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

► **CSMA/CD** (Computer Networks and Internets, page 93)

968. CSMA/CA

969. TOKEN PASSING

970. None of the given

Question No: 15 (M a r k s: 1)

Question No: 20 (M a r k s: 1)

FDDI can transmits data at a rate of -----

► **100 million bits per second** (Page 314)

971. 10 million bits per second

972. 1000 million bits per second

973. None of the given

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Spring 2010

Question No: 1 (M a r k s: 1)

In Point-to-Point topology there are two topologies.

- 974. Star and Tree
- 975. Tree and Ring
- 976. Star and Ring

977. [Star and Tree](#) (Page 5)

- 978. None of the given

Question No: 2 (M a r k s: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

► [Ping](#) (Computer Networks and Internets, page9)

- 979. Traceroute
- 980. ICMP
- 981. Non of the given

Question No: 3 (Marks: 1)

----- has no way to determine the cause of the problem.

- 982. ICMP

► [Ping](#) (Computer Networks and Internets, page 11)

- 983. Trace route
- 984. Non of the given

Question No: 4 (Marks: 1)

The term ----- refers to the general concept of a small block of data

985. [Packet](#) (Page 16)

- 986. Frame
- 987. Data
- 988. None of the given

Question No: 5 (Marks: 1)

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

► [Parity](#) (Computer Networks and Internets, page 77)

- 989. Checksums
- 990. CRC
- 991. None of given

Question No: 6 (Marks: 1)

Local Talk is a LAN technology that employs -----

► **Bus topology (Page 30)**

- 992. Ring topology
- 993. Star topology
- 994. None of the given

Question No: 7 (Marks: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- 995. CSMA/CD
- 996. CSMA/CA

► **TOKEN PASSING (Computer Networks and Internets, page95) rep**

- 997. None of the given

Question No: 8 (Marks: 1)

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

998. 64

999. 48 (Computer Networks and Internets, page 109)

~~1000.~~ 32

1001. 8

Question No: 9 (M a r k s : 1)

Formally named _____ informally known as the thick wire Ethernet or Thick net.

1002. 10 Base 2

1003. 10 Base 5 Computer Networks and Internets, page 120)

~~1004.~~ 10 Base T

1005. None of the given

Question No: 10 (M a r k s : 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

1006. 10 Base 2

1007. 10 Base 5

► 10 Base T (Page 43) rep

1008. None of the given

Question No: 11 (M a r k s : 1)

The maximum size of an Ethernet segment is _____

1009. 250 meters

1010. 500 meters (Page 27)

1011. 700 meters

1012. None of the given

Question No: 12 (M a r k s : 1)

A Bridge can _____

- ▶ Filter a frame
- ▶ Forward a frame
- ▶ Extend a LAN

▶ Do all the above rep [Click here For Detail](#)

Question No: 13 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- ▶ Greedy algorithm
- ▶ Distance vector algorithm

▶ Dijkstra's algorithm (Computer Networks and Internets, page112) rep ▶
Non of the given

Question No: 14 (M a r k s: 1)

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

1013. Constant Bit Rate (CBR) service

► [Variable Bit Rate \(VBR\) service](#) (Page 71)

1014. Available Bit Rate (ABR) service

1015. None of the given

Question No: 15 (Marks: 1)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

► [Connectionless service paradigm](#) (Computer Networks and Internets, page112)

1016. Connection-oriented service paradigm

1017. Both Connectionless and Connection-oriented service paradigm

1018. None of the given

Question No: 16 (Marks: 1)

The product of delay and throughput measures the _____ of data that can be present on the network.

► Area

► [Volume](#) (Page 80)

► [Length](#)

► Length

► None of the given

Question No: 17 (Marks: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

1019. T / D

[1020. \$T \times D\$](#) (Computer Networks and Internets, page203) rep

1021. $T + D$

1022. None of the given

1023. square roots, cube roots

1024. and, triple

1025. doubles, cancel

Question No: 18 (Marks: 1)

One repeater _____, two repeaters

► [doubles, triple](#) (Page 49) _____

► [Length](#)

the maximum cable length limitation.

Question No: 19 (M a r k s: 1)

A network uses a ----- if all computers attach to a central point

1026. [Star Topology](#) (Page 25)

1027. Ring Topology

1028. Bus Topology

1029. None of the given

Question No: 20 (M a r k s: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

[▶ CSMA/CD](#) (Computer Networks and Internets, page 93) rep

1030. CSMA/CA

1031. TOKEN PASSING

1032. None of the given

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Fall 2008

Question No: 1 (Marks: 1) - Please choose one

A typical port on an ATM switch operates at _____ or higher.

- ▶ OC-2 speed (155Mbps)
 - ▶ OC-3 speed (100Mbps)
 - ▶ **OC-3 speed (155Mbps) (Page 72)**
-
- ▶ OC-3 speed (155Gbps)

Question No: 2 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

- ▶ Area
- ▶ **Volume (Page 80) rep**

-
- 1033. Length
 - 1034. None of the given

Question No: 3 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

□□□□□□▶ ~~Constant~~ Bit Rate (CBR) service

□□□□□□▶ ~~Variable~~ **riable Bit Rate (VBR) service (Page 71) rep**

_____ailable Bit Rate (ABR) service

_____ of the given

Question No: 4 (Marks: 1) - Please choose one

Which of the following is a connecting device?

Bridge




[Click here for Detail](#)

Question No: 5 (Marks: 1) - Please choose one

A bridges function in the layers(s).

☐ ☐ ☐ ☐ ☐ ☐ ☒ All the given Physical (MAC)rep [Click here for Detail](#)

ata link

□ □ □ □ □ □ □ ► □ network


 Physical (MAC) and Data link

□ □ □ □ □ □ □ ▶ □ N¹

□ □ □ □ □ □ □ ▶ □ P

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

octets (Computer Networks and Internets, page112) rep

bytes

A diagram showing a horizontal rectangle divided into eight equal vertical sections. A black arrow points from the right side of the rectangle to the number '8', which is followed by the word 'bits'.

 of the given

Ne

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

☐ 8
☐ 32
☒ 16

(Computer Networks and Internets, page110)

☐ 8

Question No: 8 (Marks: 1) - Please choose one

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

☐ Explicit frame type (Page 35)

Ideal frame type

☐ Implicit frame type

☐ of the given

☐ None

Question No: 9 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

☐ Star

topology

☒ Bus topology (Page 30) rep

☐ Ring topology

☐ None of the given

Question No: 10 (Marks: 1) - Please choose one

The Fast Ethernet hardware operates at a rate of -----

☐ 10 Mbps

☒ 100 Mbps (Page 27)

☐ 1000 Mbps

☐ None of the given

Question No: 11 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

☐ Parity

(Computer Networks and Internets, page 77) rep

☐Checksums

☐CRC

☐ None of given

Question No: 12 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

ernet

witch networks

Packet networks (Computer Networks and Internets, page 73) rep

of the given

Question No: 13 (Marks: 1) - Please choose one

☐ S

Program sends a message to a remote computer and reports whether the computer responds.

☐ Ping

(Computer Networks and Internets, page

9) rep

☐ traceroute

☐ ICMP

☐ of the given

☐ None

Question No: 14 (Marks: 1) - Please choose one

Question No: 1 (Marks: 1) - Please choose one

_____ has a jitter zero

1036. None of the given

1037. Virtual Private Network

1038. Isochronous Network

1039. Asynchronous Network (Page 66)

Question No: 2 (Marks: 1) - Please choose one

☐ Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers _____.

☐

☐ Connectionless

service paradigm (Computer Networks and Internets, page112)

☐ ☐ Connection oriented service paradigm

☐ ☐ Both Connectionless and Connection

-oriented service paradigm

☐ ☐ None of the given

Question No: 3 (Marks: 1) - Please choose one

☐ ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

☐

☐ 21

-bit

☐ ☒ 22

-bit

☐ ☒ 23bit

☐ ☒ 24

-bit(Page 67) Question No: 4 (Marks: 1) - Please choose one

☐ Most WAN systems include a mechanism that can be used to eliminate the common case of duplication

routing is called

☐

☒ Hierarchal address

☐ ☒ Default

route (Computer Networks and Internets, page172)

☐ ☒ Shortest path

☐ ☒ None of the given

Question No: 5 (Marks: 1) - Please choose one

☐ The next hop to which a packet is sent depends only on

☐

☒ Packet's

destination (Computer Networks and Internets, page169)

☐ ☒ Packet's original source

☐ ☒ Path the packet has taken

☐ ☒ Non of the given

Question No: 6 (Marks: 1) - Please choose one

_____ have an _____ connector , and must generate signals

☐ An interface for twisted pair Ethernet must _____ according to the _____ specification.

☐

☒ RJ

-45, 10 Base T (Computer Networks and Internets, page 126) rep

-45, 10 Base 5

☐ ☒ RJ

☐ ☒ BNC, 10 Base

2

☐ ☒ BNC, 10 Base

T

Question No: 7 (Marks: 1) - Please choose one

☐ When an application ----- data, it makes a copy of the data available to all other computers on the network.

☐

☒ [Broadcasting](#)

[Click here for Detail](#)

- ☐ ☒ Multicasting
- ☐ ☒ Unicasting
- ☐ ☒ None of the given

Question No: 8 (Marks: 1) - Please choose one

☐ ~~A~~

----- provide a mechanism that a customer can use to set a physical address.

☐

☒ ☐ Static addressing scheme

☒ [Configurable](#)

[addressing scheme \(Page 34\)](#)

- ☐ ☒ Dynamic addressing scheme
- ☐ ☒ None of the given

Question No: 9 (Marks: 1) - Please choose one

☐ FDDI can transmits data at a rate of -----

☐

☒ [100 million bits](#)

[per second \(Page 314\) rep](#)

- ☐ ☒ 100 million bits per second
- ☐ ☒ 100 million bits per second
- ☐ ☒ None of the given

Question No: 10 (Marks: 1) - Please choose one

☐ Computers attached to an ether use ---- in which a computer waits for the ether to be idle before transmitting a frame.

☐

☒ [CSMA/CD](#)

(Computer Networks and Internets, page 93) rep

☐ ☒ ~~CSMA/CA~~

☐ ☒ TOKEN PASSING

☐ ☒ None of the given

Question No: 11 (Marks: 1) - Please choose one

☐ ----- have advantages arisen from the size and ease of computation.

☐

☒ ☐ CRC

☒ ☐ [Parity](#)

(Page 19) rep

☐ ☒ Checksums

☐ ☒ None of given

Question No: 12 (Marks: 1) - Please choose one

☐ The term ----- is used to denote the definition of a packet used with a specific type of network.

☐

☒ ☐ Packet

☒ ☐ [Frame](#)

(Computer Networks and Internets, page 73)

☐ ☒ ~~Da~~

ta

☐ ☒ None of the given

Question No: 13 (Marks: 1) - Please choose one

- ☐ --- has no way to determine the cause of the problem.
- ☐

[▶ Ping](#) (Computer Networks and Internets, page 11) rep

- ☐ ▶ ☐ Trace route
- ☐ ▶ ☒ ICMP
- ☐ ▶ ☐ Non of the given

Question No: 14 (Marks: 1) - Please choose one

- ☐ --- Program sends a message to a remote computer and reports whether the computer responds.
- ☐

[▶ Ping](#) (Computer Networks and Internets, page 9) rep

- ☐ ▶ ☐ Traceroute
- ☐ ▶ ☒ ICMP
- ☐ ▶ ☐ Non of the given

Question No: 15 (Marks: 1) - Please choose one

- ☐ In ----- , network occupies the smaller area like a room a floor or a building
- ☐

[▶ LAN](#) (Page 4) rep

- ☐ ▶ ☒ WAN
- ☐ ▶ ☒ MAN
- ☐ ▶ ☐ None of the given

Question No: 16 (Marks: 1) - Please choose one

- ☐ No error detection scheme is perfect because transmission errors can affect the additional information as well

as the data.

► **True** (Computer Networks and Internets, page82)

1040. False

1041. A _____ address-binding table is used for each physical network.

Separate

1042. The Network Layer Protocol ICMP stands for _____

Internet Control Message Protocol

1043. Hashing is the transformation of a string of characters into a usually shorter fixed-length value or a key that represents the original string.

True

1044. The amount of buffer space available at any time is called the window.

True

1045. Each autonomous system used to communicate among autonomous systems by chooses an IGP.

True

1046. TCP is not connection-oriented service.

False

1047. NEXT HEADER field in the base header defines type of header it appears at the end of fixed-size base header.

True

1048. There are two standard implementations to improve computational efficiency:
Hashing and Direct indexing

True

1049. _____ sends ICMP echo messages with increasing TTL.

Tracert

1050. _____ uses window mechanism to control the flow of data.

TCP

1051. _____ is a type of address used for collection of computers with same prefix.

Cluster

1052. Due to revolutionalization of IP-V6 the speed has increased from _____

56Kbps to 1Gbps

- 4 System rebooting is one of the major problems in the reliable delivery

True

- 5 IPV6 addresses are _____ bits.

128 bits

- 6 A popular use of _____ is TCP Splicing. It interconnects two independent TCP connections and performs segment rewriting.

NAPT

- 7 _____ layer Provides reliable delivery of datagram.

Transport

- 8 In TCP when a computer sends a segment, the _____ and _____ fields refer to incoming data.
- 9 _____ protocol uses three way handshake to begin a connection.

TCP

- 10 One of the parameters, which motivated IP for change is address space. The _____ address space allows for over a million networks. But most networks are class C and too small for many organizations.

32 bits

11 The Current version of IP-Version 4 is _____ old

20 years

12 Twice NAT fails if an application uses the IP addresses instead of Domain Name.

True

4. Mapping between a protocol address and a hardware address is called Address Resolution.

true

3 Fragmentation when using ICMP for path MTU should be avoided.

True

24. In 3-way handshake TCP requires to generate a random _____ sequence number.

c bits

K. The UDP stands for_____.

User Datagram Protocol

L. _____ message is sent in response to incoming datagrams with problems.

ICMP

M. _____ contains all information needed to deliver datagram to the destination.

Header

KK. UPD is operating system independent.

Yes

e. NAPT stands for _____

Network Address and Port Translation

f. _____ Serve Same purpose in Internet as frames on LAN

Virtual Packets

g. The amount of buffer space available at any time is called the window.

True

h. A _____ address-binding table is used for each physical network.

separate

i. Class A mask is 255.0.0.0 which is used for _____

Subnetting

j. Dotted decimal represents each octet in _____ and uses a dot to separate octets. **decimal**

k. _____ shows senders preference for low latency, high Reliability.

SERVICE TYPE

l. UDP packet is encapsulated in _____ datagram.

IP

- 4 HEADER LEN field gives size of extension header.

True

- 5 MTU Stands for _____

Maximum transmission unit

7. No time for save.
8. As the Internet grew, the original Classful addressing scheme became a limitation. The IP address space was being exhausted because all networks had to choose one of four possible sizes.

False

Mid term data of 610

Question No: 1 (M a r k s: 1)

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

1053. True (Computer Networks and Internets, page82)

1054. False

Question No: 2 (M a r k s: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

13 [Ping](#) (Computer Networks and Internets, page9)

14 Traceroute

15 ICMP

16 Non of the given

Question No: 3 (M a r k s: 1)

----- was especially concerned about the lack of high powered computers.

► [ARPA](#) (Computer Networks and Internets, page5)

5. IEEE

6. EIA

7. Non of the given

Question No: 4 (M a r k s: 1)

The term ----- is used to denote the definition of a packet used with a specific type of network.

► Packet

► [Frame](#) (Computer Networks and Internets, page 73)

4 Data

5 None of the given

Question No: 5 (M a r k s: 1)

Computer networks are often called ----- because they use packet technology.

- 5. Ethernet
- 6. Switch networks

► [Packet networks](#) (Computer Networks and Internets, page 73)

- . None of the given

Question No: 6 (M a r k s: 1)

----- have advantages arisen from the size and ease of computation.

LLLLLLLLLLLLLLLLLL. CRC

MMMMMMMMMMMMMMMMMM. [Parity](#) (Page 19)

NNNNNNNNNNNNNNNNNN. Checksums
OOOOOOOOOOOOOOOOOO. None of given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- w. CSMA/CD
- x. CSMA/CA

y. [TOKEN PASSING](#) (Computer Networks and Internets, page95)

- z. None of the given

Question No: 8 (M a r k s: 1)

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

5888 [8 octets](#) (Computer Networks and Internets, page112)

5889 8 bytes

5890 8 bits

5891 None of the given

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

23. 10 Base 2

24. 10 Base 5

25. **10 Base T (Page 43)**

26. None of the given

Question No: 10 (M a r k s: 1)

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

5888. [RJ-45, 10 Base T \(Computer Networks and Internets, page 126\)](#)

5889. RJ-45, 10 Base 5

5890. BNC, 10 Base 2

5891. BNC, 10 Base T

Question No: 11 (M a r k s: 1)

A bridges function in the _____ layers(s).

► [Physical \(MAC\)](#) [Click here for Detail](#)

23 Data link

24 Network

25 Physical (MAC) and Data link

Question No: 12 (M a r k s: 1)

A Bridge can _____

5888. Filter a frame

5889. Forward a frame

5890. Extend a LAN

► [Do all the above](#) [Click here for Detail](#)

Question No: 13 (M a r k s: 1)

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- 23 Layer 2 source address
- 24 Source node's physical address

► [Layer 2 destination address](#) [Click here for Detail](#)

5888. Layer 3 destination address

Question No: 14 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- 23 Greedy algorithm
- 24 Distance vector algorithm

► [Dijkstra's algorithm](#) (Computer Networks and Internets, page112)

5888. Non of the given

Question No: 15 (M a r k s: 1)

_____ is used for audio and video, since these have predefined maximum data rates

► [Constant Bit Rate \(CBR\) service](#) (Page 71)

- w. Variable Bit Rate (VBR) service
- x. Available Bit Rate (ABR) service
- y. None of the given

Question No: 16 (M a r k s: 1)

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

► [Connectionless service paradigm](#) (Computer Networks and Internets, page112)

- lllllllllll. Connection oriented service paradigm
- mmmmmmmmmmmmmmmmmmmm. Both Connectionless and Connection-oriented service paradigm
- nnnnnnnnnnnnnnnnnn. None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

- w. T / D
- x. [T x D](#) (Computer Networks and Internets, page203)
- y. $T + D$
- z. None of the given

Question No: 18 (M a r k s: 1)

ATM is designed to work on_____.

- lllllllllll. Twisted Pair
- mmmmmmmmmmmmmmmmmmmm. Coaxial
- nnnnnnnnnnnnnnnnnn. Radio Frequency

► [Fiber](#) (Computer Networks and Internets, page 72)

Question No: 19 (Marks: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

► [CSMA/CD](#) (Computer Networks and Internets, page 93)

- w. CSMA/CA
- x. TOKEN PASSING
- y. None of the given

Question No: 20 (M a r k s: 1)

FDDI can transmits data at a rate of -----

► [100 million bits per second](#) (Page 314)

llllllllllll. 10 million bits per second
mmmmmmmmmmmmmmmmmmmm. 1000 million bits per second
nnnnnnnnnnnnnnnnnn. None of the given

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Spring 2010

Question No: 1 (M a r k s: 1)

In Point-to-Point topology there are two topologies.

- w. Star and Tree
- x. Tree and Ring
- y. Star and Ring

z. [Star and Tree](#) (Page 5)

aa. None of the given

Question No: 2 (M a r k s: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

► [Ping](#) (Computer Networks and Internets, page9)

llllllllllll. Traceroute

mmmmmmmmmmmmmmmmmmmmmm. ICMP
nnnnnnnnnnnnnnnnnnnn. Non of the given

Question No: 3 (M a r k s: 1)

----- has no way to determine the cause of the problem.

w. ICMP

x. Ping (Computer Networks and Internets, page 11)

y. Trace route

z. Non of the given

Question No: 4 (M a r k s: 1)

The term----- refers to the general concept of a small block of data

5888. [Packet \(Page 16\)](#)

5889. Frame

5890. Data

5891. None of the given

Question No: 5 (M a r k s: 1)

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

► [Parity \(Computer Networks and Internets, page 77\)](#)

w. Checksums

x. CRC

y. None of given

Question No: 6 (M a r k s: 1)

Local Talk is a LAN technology that employs -----

||||||| [Bus topology \(Page 30\)](#)

mmmmmmmmmmmmmmmmmmmmmm. Ring topology

nnnnnnnnnnnnnnnnnnnnnn. Star topology

oooooooooooooooooooo. None of the given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

w. CSMA/CD

x. CSMA/CA

y. [TOKEN PASSING \(Computer Networks and Internets, page95\)](#) rep

z. None of the given

Question No: 8 (M a r k s: 1)

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

llllllllllllll. 64

[mmmmmmmmmmmmmmmmmmmm. 48 \(Computer Networks and Internets, page 109\)](#)

nnnnnnnnnnnnnnnnnn. 32

oooooooooooooooooooo. 8

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the thick wire Ethernet or Thick net.

w. 10 Base 2

x. [10 Base 5 Computer Networks and Internets, page 120](#))

y. 10 Base T

z. None of the given

Question No: 10 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

llllllllllll. 10 Base 2

mmmmmmmmmmmmmmmmmmmm. 10 Base 5

nnnnnnnnnnnnnnnnnn. [10 Base T \(Page 43\) rep](#)

oooooooooooooooooooo. None of the given

Question No: 11 (M a r k s: 1)

The maximum size of an Ethernet segment is _____

w. 250 meters

x. [500 meters \(Page 27\)](#)

y. 700 meters

z. None of the given

Question No: 12 (M a r k s: 1)

A Bridge can _____

llllllllllll. Filter a frame

mmmmmmmmmmmmmmmmmmmm. Forward a frame

nnnnnnnnnnnnnnnnnn. Extend a LAN

► [Do all the above](#) rep [Click here For Detail](#)

Question No: 13 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

w. Greedy algorithm

x. Distance vector algorithm

► [Dijkstra's algorithm](#) (Computer Networks and Internets, page112) rep

llllllllllll. Non of the given

Question No: 14 (M a r k s: 1)

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

w. Constant Bit Rate (CBR) service

► [Variable Bit Rate \(VBR\) service \(Page 71\)](#)

llllllllllll. Available Bit Rate (ABR) service
mmmmmmmmmmmmmmmmmmmm. None of the given

Question No: 15 (M a r k s: 1)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

w. [Connectionless service paradigm \(Computer Networks and Internets, page112\)](#)

- x. Connection-oriented service paradigm
- y. Both Connectionless and Connection-oriented service paradigm
- z. None of the given

Question No: 16 (M a r k s: 1)

The product of delay and throughput measures the _____ of data that can be present on the network.

llllllllllll. Area

mmmmmmmmmmmmmmmmmmmm. [Volume \(Page 80\)](#)

nnnnnnnnnnnnnnnnnnnn. Length
oooooooooooooooooooo. None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

w. T / D

x. [T x D \(Computer Networks and Internets, page203\) rep](#)

- y. $T + D$
- z. None of the given

Question No: 18 (M a r k s: 1)

One repeater _____, two repeaters _____ the maximum cable length limitation.

IIIIIIIIIIII. doubles, triple (Page 49)

mmmmmmmmmmmmmmmmmmmmmm. square roots, cube roots

nnnnnnnnnnnnnnnnnnnn. and, triple

oooooooooooooooooooo. doubles, cancel

Question No: 19 (Marks: 1)

A network uses a ----- if all computers attach to a central point

w. [Star Topology \(Page 25\)](#)

x. Ring Topology

y. Bus Topology

z. None of the given

Question No: 20 (Marks: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

► [CSMA/CD \(Computer Networks and Internets, page 93\)](#) rep

IIIIIIIIIIII. CSMA/CA

mmmmmmmmmmmmmmmmmmmmmm. TOKEN PASSING

nnnnnnnnnnnnnnnnnnnn. None of the given

**CS610- Computer Network
MIDTERM EXAMINATION**

Fall 2008

Question No: 1 (Marks: 1) - Please choose one

A typical port on an ATM switch operates at _____ or higher.

w. OC-2 speed (155Mbps)

x. OC-3 speed (100Mbps)

► [OC-3 speed \(155Mbps\) \(Page 72\)](#)

IIIIIIIIIIII. OC-3 speed (155Gbps)

Question No: 2 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

w. Area

x. [Volume \(Page 80\) rep](#)

y. Length

z. None of the given

Question No: 3 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

lllllllllllllll. Constant Bit Rate (CBR) service

[mmmmmmmmmmmmmmmmmmmmmmmm. Variable Bit Rate \(VBR\) service\(Page 71\) rep](#)

nnnnnnnnnnnnnnnnnn. Available Bit Rate (ABR) service

oooooooooooooooooooo. None of the given

Question No: 4 (Marks: 1) - Please choose one

Which of the following is a connecting device?

- ▶ Bridge
- ▶ Repeater
- ▶ Hub

▶ [All the given](#) [Click here for Detail](#)

Question No: 5 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

▶ [Physical \(MAC\)](#) [rep](#) [Click here for Detail](#)

w. Data link

x. Network

y. Physical (MAC) and Data link

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

▶ [8 \(Computeroctets Networks and Internets, page112\)](#) [rep](#)

llllllllllllllll. 8 bytes

mmmmmmmmmmmmmmmmmmmm. 8 bits

nnnnnnnnnnnnnnnnnnnn. None of the given

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

w. 48

2 ▶ 3

[\(Computer▶16 Networks and Internets, page110\)](#)

x. 8

Question No: 8 (Marks: 1) - Please choose one

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

|||||||**Explicit frame(Pagtype 35)**

mmmmmmmmmmmmmmmmmmmmmmmm. Ideal frame type

nnnnnnnnnnnnnnnnnnnnnnnn. Implicit frame type

oooooooooooooooooooo. None of the given

Question No: 9 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

w. Star topology

x. Bus topology(Page 30) rep

y. Ring topology

z. None of the given

Question No: 10 (Marks: 1) - Please choose one

The Fast Ethernet hardware operates at a rate of -----

lllllllllllll. 10 Mbps

mmmmmmmmmmmmmmmmmmmm. 100 Mbps(Page 27)

nnnnnnnnnnnnnnnnnnnn. 1000 Mbps

oooooooooooooooooooo. None of the given

Question No: 11 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with

each character

w. Parity(Computer Networks and Internets, page 77) rep

x. Checksums

y. CRC

z. None of given

Question No: 12 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

lllllllllllll. Ethernet

mmmmmmmmmmmmmmmmmmmm. Switch networks

nnnnnnnnnnnnnnnnnnnn. Packet networks(Computer Networks and Internets, page 73) rep

oooooooooooooooooooo. None of the given

Question No: 13 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

► Ping (Computer Networks and Internets, page 9) rep

w. Traceroute

x. ICMP

y. Non of the given

Question No: 14 (Marks: 1) - Please choose one

In -----, network occupies larger areas like cities & countries.

w. LAN

x. [WAN](#) (Page 4)

y. MAN

z. None of the given

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

w. [LAN](#)(Page 4)

x. WAN

y. MAN

z. None of the given

Question No: 16 (Marks: 1) - Please choose one

In Point-to-Point topology there are two topologies.

1. Tree and Ring

2. Star and Ring

3. [Star and Tree](#)(Page 5)

4. None of the given

CS610- Computer Network
MIDTERM EXAMINATION
Fall 2008

Question No: 1 (Marks: 1) - Please choose one

_____ has a jitter zero

- 5. None of the given
- 6. Virtual Private Network
- 7. Isochronous Network

8. [Asynchronous Network \(Page 66\)](#)

Question No: 2 (Marks: 1) - Please choose one

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

► [Connectionless service paradigm \(Computer Networks and Internets, page112\)](#)

- 9. Connection oriented service paradigm
 - a. Both Connectionless and Connection-oriented service paradigm ► None of the given

Question No: 3 (Marks: 1) - Please choose one

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

- 10. 21-bit
 - a. 22-bit
 - b. 23-bit

11. [24-bit \(Page 67\)](#)

Question No: 4 (Marks: 1) - Please choose one

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

12. Hierarchical address

► [Default route](#) (Computer Networks and Internets, page 172)

► Shortest path ►

None of the given

Question No: 5 (Marks: 1) - Please choose one

The next hop to which a packet is sent depends only on

13. [Packet's destination](#) (Computer Networks and Internets, page 169)

- a. Packet's original source
- b. Path the packet has taken
- c. None of the given

Question No: 6 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector, and must generate signals according to the _____ specification.

14. [RJ-45, 10 Base T](#) (Computer Networks and Internets, page 126) rep

- 15. RJ-45, 10 Base 5
- 16. BNC, 10 Base 2
- 17. BNC, 10 Base T

Question No: 7 (Marks: 1) - Please choose one

When an application----- data, it makes a copy of the data available to all other computers on the network.

18. [Broadcasting](#) [Click here for Detail](#)

- a. Multicasting ►
- Unicasting ► None

of the given

Question No: 8 (Marks: 1) - Please choose one

A----- provide a mechanism that a customer can use to set a physical address.

19. Static addressing scheme

20. [Configurable addressing scheme](#) (Page 34)

- a. Dynamic addressing scheme
- b. None of the given

Question No: 9 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of-----

► [100 million bits per second](#) (Page 314) rep

- 21. 100 million bits per second
- 22. 100 million bits per second
- 23. None of the given

Question No: 10 (Marks: 1) - Please choose one

Computers attached to an ether use----- in which a computer waits for the ether to be idle before transmitting a frame.

24. [CSMA/CD](#) (Computer Networks and Internets, page 93) rep

- 25. CSMA/CA
- 26. TOKEN PASSING
- 27. None of the given

Question No: 11 (Marks: 1) - Please choose one

----- have advantages arisen from the size and ease of computation.

- 28. CRC

► [Parity](#) (Page 19) rep

- 29. Checksums
- 30. None of given

Question No: 12 (Marks: 1) - Please choose one

The term----- is used to denote the definition of a packet used with a specific type of network.

- 31. Packet

32. [rnf](#) (Computer Networks and Internets, page 73)

33. Data

34. None of the given

Question No: 13 (Marks: 1) - Please choose one

----- has no way to determine the cause of the problem.

35. [Ping](#)(Computer Networks and Internets, page 11) rep

36. Trace route

37. ICMP

38. Non of the given

Question No: 14 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

► Ping (Computer Networks and Internets, page9) rep

- a. Traceroute
- b. ICMP
- 39. Non of the given

Question No: 15 (Marks: 1) - Please choose one

In-----, network occupies the smaller area like a room a floor or a building

40. LAN (Page 4) rep

- 41. WAN
- 42. MAN
- 43. None of the given

Question No: 16 (Marks: 1) - Please choose one

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

44. True (Computer Networks and Internets, page82)

- 45. False

Cs610-computer Networks

Quiz (1 & 2) spring 2012



CS610: Quiz 1

Question # 2 of 10 (Start time: 10:29:56 PM) Total Marks: 1

A network that uses _____ usually consists of a single long cable to which computer attach.

Select correct option:

Star Topology

Ring Topology

Bus Topology (Page 26)

Bus and Ring Topology

Question # 3 of 10 (Start time: 10:30:16 PM) Total Marks: 1

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

Select correct option:

64

32

8

Question # 4 of 10 (Start time: 10:30:36 PM) Total Marks: 1

No error detection scheme is perfect because transmission errors can affect _____.

Select correct option:

Data

Additional Information

Additional information & data **(Computer Networks and Internets, page82)**
Transmission medium

Question # 5 of 10 (Start time: 10:30:58 PM) Total Marks: 1

Star topolgy is the kind of

Select correct option:

Tree topology

Point-to-point topology **(Page 5)**

Broadcast topolgy

Ring topology

Question # 6 of 10 (Start time: 10:31:12 PM) Total Marks: 1

The Gigabit Ethernet hardware operates at a rate of -----

Select correct option:

10 Mbps

100 Mbps

1000 Mbps [Click here for Detail](#)

1000 Gbps

Question # 7 of 10 (Start time: 10:31:34 PM) Total Marks: 1

Fixed network is a type of networks which is classified with respect to the.....factor

Select correct option:

Size

Connectivity

Medium

Mobility (Page 6)

Question # 8 of 10 (Start time: 10:31:44 PM) Total Marks: 1

A network uses _____ to arrange computers to be connected in a single closed loop.

Select correct option:

Star Topology

Dual Ring Topology

Ring Topology (Page 25)

Bus Topology

Question # 9 of 10 (Start time: 10:32:02 PM) Total Marks: 1

Hardware that calculates a CRC uses two simple components.

Select correct option:

AND unit and XOR unit

Shift register and XOR unit (Page 20)

Shift register & AND unit

Shift register and shift XOR unit

Question # 10 of 10 (Start time: 10:32:53 PM) Total Marks: 1

When an application ----- data, it makes a copy of the data available to all other computers on the network.

Select correct option:

Broadcasting **rep**

Multicasting

Unicasting

None of the given



Question # 1 of 10 (Start time: 09:55:01 PM) Total Marks: 1

Fixed network is a type of networks which is classified with respect to the.....factor

Select correct option:

Size

Connectivity

Medium

Mobility **rep**

Question # 2 of 10 (Start time: 09:55:38 PM) Total Marks: 1

----- is also called self healing network

Select correct option:

ATM

FDDI [Click here for detail](#)

Token Ring

None of the given

Question # 3 of 10 (Start time: 09:56:01 PM) Total Marks: 1

Local Talk is a LAN technology that employs -----

Select correct option:

Star topology

Bus topology **rep**

Ring topology

Ethernet

Question # 4 of 10 (Start time: 09:56:26 PM) Total Marks: 1

Some applications have large data transfer. In this way they hold the network for long time.this problem can be eliminated by

Select correct option:

Divide the application into small block or chunks called 'Packets'.

Divide the application into small block or chunks called 'Segments'.

Divide the data into small block or chunks called 'Segments'.

Divide the data into small block or chunks called 'Packets'. (Page 14)

Question # 5 of 10 (Start time: 09:56:54 PM) Total Marks: 1

For how much time would other computers be in wait while one computer was transferring 56MB file with packet size of 1000 bytes and transmission speed of 56Kbps

Select correct option:

0.143 sec (Page 15)

0.243 0.343

0.343 0.343

0.443 0.343

Question # 6 of 10 (Start time: 09:57:20 PM) Total Marks: 1

The Gigabit Ethernet hardware operates at a rate of -----

Select correct option:

10 Mbps

100 Mbps

1000 Mbps **rep**

1000 Gbps

Question # 7 of 10 (Start time: 09:57:37 PM) Total Marks: 1

The Fast Ethernet hardware operates at a rate of -----

Select correct option:

1000 bps

100 Mbps [Click here for detail](#)

10 Mbps

1000 Mbps

Question # 8 of 10 (Start time: 09:57:52 PM) Total Marks: 1

----- which is designed to help detect transmissions errors, send one extra bit of information with each character.

Select correct option:

Checksum

CRC

Parity **rep**

VRC

Question # 9 of 10 (Start time: 09:58:30 PM) Total Marks: 1

LAN that use ATM technology have a -----

Select correct option:

Bus topology

Star topology (Page 31)

Ring topology

Mesh Topology

Question # 10 of 10 (Start time: 09:58:58 PM) Total Marks: 1

FDDI can transmits data at a rate of -----

Select correct option:

3 million bits per second

1000 million bits per second

10 million bits per second

100 million bits per second rep



Question # 1 of 10 (Start time: 11:14:05 PM) Total Marks: 1

----- is also called self healing network

Select correct option:

ATM

FDDI **rep**

Token Ring

None of the given

Question # 2 of 10 (Start time: 11:15:34 PM) Total Marks: 1

Most LANs that employ ring topology use an access mechanism known as-----

Select correct option:

CSMA/CD

CSMA/CA

CSMA

TOKEN PASSING **rep**

Question # 3 of 10 (Start time: 11:16:55 PM) Total Marks: 1

An ----- method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Select correct option:

Explicit frame type **rep**

Ideal frame type

Implicit frame type

None of the given

Question # 4 of 10 (Start time: 11:17:47 PM) Total Marks: 1

Hardware that calculates a CRC uses two simple components.

Select correct option:

AND unit and XOR unit

Shift register and XOR unit rep

Shift register & AND unit

Shift register and shift XOR unit

Question # 5 of 10 (Start time: 11:18:59 PM) Total Marks: 1

The Ethernet standard specifies that frames are sent using the -----

Select correct option:

Differential Manchester

Not Return to Zero (NRZ)

Manchester Encoding (Page 27)

Return to Zero (RZ)

Question # 6 of 10 (Start time: 11:20:08 PM) Total Marks: 1

When an application ----- data, it makes a copy of the data available to all other computers on the network.

Select correct option:

Broadcasting rep

Multicasting

Unicasting

None of the given

Question # 7 of 10 (Start time: 11:21:26 PM) Total Marks: 1

LAN interface may use ----- to copy frame data directly from main memory.

Select correct option:

DMA (Page 32)

FDDI

CSMA/CD

None of the given

Question # 8 of 10 (Start time: 11:22:19 PM) Total Marks: 1

LAN and WAN are classified according to their.....

Select correct option:

Size (Page 4)

Connectivity

Medium

Mobility

Question # 9 of 10 (Start time: 11:23:43 PM) Total Marks: 1

For how much time would other computers be in wait while one computer was transferring 56MB file with transmission speed of 56Kbps

Select correct option:

10 Sec [Click here for detail](#)

46. Sec

47. Sec

48. Sec

Question # 10 of 10 (Start time: 11:25:05 PM) Total Marks: 1

A network uses _____ to arrange computers to be connected in a single closed loop.

Select correct option:

Star Topology

Dual Ring Topology

Ring Topology **rep**

Bus Topology



CS610: Quiz 2

Question # 1 of 10 (Start time: 10:52:55 PM) Total Marks: 1

_____network topology is a bus but wiring topology is a star

Select correct option:

10Base-T **(Page 44)**

Ring

10Base-W

Star

Question # 2 of 10 (Start time: 10:53:05 PM) Total Marks: 1

ATM is also called-----

Select correct option:

Label rewriting

Label switching system

Label rewriting and label switching system **(Page 68)**

All of above

Question # 3 of 10 (Start time: 10:53:26 PM) Total Marks: 1

The most important task a bridge performs is _____

Select correct option:

Packet switching

Line controlling

Noise controlling

Frame filtering (Page 50)

Question # 4 of 10 (Start time: 10:53:41 PM) Total Marks: 1
ABR stands for.....

Select correct option:

Asynchronous Bit Rate

Asynchronous Byte Recovery

Asynchronous Bit Redundancy

Available Bit Rate (Page 71)

Question # 5 of 10 (Start time: 10:53:57 PM) Total Marks: 1

What will be the bandwidth rate for short distance 10Base-T?

Select correct option:

16 MHz

20 MHz (Computer Networks and Internets, page 128)

49. MHz

50. MHz

Question # 6 of 10 (Start time: 10:54:14 PM) Total Marks: 1

The maximum cable length of a LAN is _____

Select correct option:

Fixed (Page 48)

Unlimited

Fixed and Unlimited in different conditions

None of these

Question # 7 of 10 (Start time: 10:54:32 PM) Total Marks: 1

SMDS is.....

Select correct option:

Connectionless service (Page 65)

Connection oriented service

Used for delivering blocks of data

None of the above

Question # 8 of 10 (Start time: 10:54:46 PM) Total Marks: 1

All destinations on same switch have same

Select correct option:

Router

Information

Next hop (Page 58)

None of the above

Question # 9 of 10 (Start time: 10:55:04 PM) Total Marks: 1

If a link or a packet switch fails then which one of the following would possibly be happened?

Select correct option:

Link would be reestablished

Request to change failed hardware would appear

The packet switch will modify the tables (Page 63)

None of the above

Question # 10 of 10 (Start time: 10:55:34 PM) Total Marks: 1

..... is effectively a very short Ethernet with very long AUI cables. It can be connected into larger Ethernet Select correct option:

RJ-45

HUBS (Page 43)

10Base-T

None of these

Question No: 1(Marks: 1)

A ----- Relies on the hardware manufacturer to assign a unique physical address to each network interface.

► **Static addressing scheme (Page 34)**

► Configurable addressing scheme

► Dynamic addressing scheme

► None of the given

Question No: 2 (Marks: 1)

An interface for thin Ethernet must have an
connector, and must generate signals according to
the
specification.

► RJ-45, 10 Base T

► RJ-45, 10 Base 5

► BNC, 10 Base 2

(CS610 Reference Book , Page 21)

► BNC, 10 Base T

Question No: 3(Marks: 1)

A system with redundant bridges might have a problem with
in the system.

► Loop

[Click here for detail](#)

► Filters

► Spanning Trees

► All given choices

Question No: 4(Marks: 1)

A Bridge can

► Filter a frame

► Forward a frame

► Extend a LAN

Do the all

► **Do all the** [Click here for detail](#)

Question No: 5(M a r k s: 1)

is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

► Constant Bit Rate (CBR) service

► Variable Bit Rate (VBR) service

► **Available Bit Rate (ABR) service (Page 71)**

► None of the given

Question No: 6(M a r k s: 1)

ATM assigns each VC a

identifier that is divided two parts to produce a hierarchy.

► 21-bit

► 22-bit

► 23-bit

► **24-bit (Page 67)**

Question No: 7(M a r k s: 1)

of TCP/IP layering model, corresponds to basic network hardware.

► **Physical Layer (Page 84)**

► Network Interface Layer

► Internet Layer

► Transport Layer

Question No: 8(Marks: 1)

places the boundary between the second and third octets

► Class A

► **Class B (Computer Networks and Internets, page235)**

► Class C

► Class D

Question No: 9(Marks: 1)

UDP and TCP are both_

layer protocols

► Physical

► Data link

► Network

► **Transport (Page 101) Question No: 10(Marks: 1)**

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by

► IP

► None of the given

► **TCP (Page 123)**

► UDP

Question No: 11(M a r k s: 1)

protocols of TCP/IP layering model specify how to ensure reliable transfer.

- ▶ Physical Layer
- ▶ Network Interface Layer
- ▶ Internet Layer
- ▶ Transport Layer (Page 84)

Question No: 12(M a r k s: 1)

identifies which application program on receiving computer should receive the data

- ▶ Logical address
- ▶ Source port

► Destination Port (Computer Networks and Internets, page313)

► None of the given

Question No: 13(M a r k s: 1)

identifies the application program that sent the data.

► Destination Port

► Source port

(Computer Networks and Internets, page313)

► Logical address

► None of the given

Question No: 14(M a r k s: 1)

Which of the following are interior routing protocols?

► RIP

► OSPF

► BGP

► **RIP and OSPF**

[Click here for detail](#)

Question No: 15(M a r k s: 1)

The Border Gateway Protocol (BGP) uses

for all communication

► UDP

► TCP

[Click here for detail](#)

► Both UDP and TCP

► None of the given

Question No: 16(M a r k s: 1)

measures distance in network hops, where each network between the source and destination counts as single hop.

► BGP

► OSPF

► RIP

(Page 138)

► None of the given

Question No: 17(M a r k s: 1)

OSPF is based on

► Distance vector routing

► Link state routing

(Page 140)

► Path vector routing

► Distance vector routing and Link state routing

Question No: 18(M a r k s: 1)

performs local multicast and uses IP-in-IP encapsulation to send multicast datagrams from one site on the Internet to another.

► **Distance Vector Multicast Routing Protocol (DVMRP) (Page 144)**

► Core Based Trees (CBT)

► Protocol Independent Multicast _ Sparse Mode (PIM-SM)

► Protocol Independent Multicast _ Dense Mode (PIM-DM)

Question No: 19(M a r k s: 1)

The length of time required to send a variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.

► True

► False

(Page 72)

Question No: 20(M a r k s: 1)

NEXT HEADER field in the base header defines type of header and it appears at end of fixed-size base header.

► True

(Page 112)

► False

Question No: 21(M a r k s: 1)

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

► True (Page 99)

► False

Question No: 22(M a r k s: 1)

Address mask defines how many bits of address are in suffix.

► True

► False (Page 103)

Question No: 23(M a r k s: 1)

A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?

► True (Page 81)

► False

Question No: 24(M a r k s: 1)

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False

► True (Page 81)

Question No: 25(M a r k s: 1)

The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address

.

► itself

(Page 87)

► prefix

► suffix

► mask

Question No: 26(M a r k s: 1)

In which method of Address Resolution Protocol the protocol address independent of hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

▶ T, C

▶ D

▶ C

▶ T, D (Page 97)

Question No: 27(M a r k s: 1)

In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

▶ T

▶ D

▶ C (Page 97)

► T, C

Question No: 28(M a r k s: 1)

Reconstruction of original datagram is called reassembly.

► True

(Page 28)

► False

Question No: 29(M a r k s: 1)

A computer needs a complete stack of protocols to run either a client or a server.

▶ **True (Computer Networks and Internets, page 344)**

▶ False

Question No: 30(M a r k s: 1)

TCP uses

mechanism to control the flow of data.

▶ door

▶ **window (Page 126)**

▶ acknowledgment

▶ retransmission

FINALTERM EXAMINATION

Spring 2010

CS610- Data Communication

Question No: 1 (Marks: 1) - Please choose one

In Direct point to point communication adding the Nth computer requires-----
new connections.

► None of the given

2

► N

► N-1

(Page 23)

► $(N^2 - N)/2$

Question No: 2 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

► LAN

(Page 4)

► WAN

► MAN

► None of the given

Question No: 3 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

► 48

► 32

▶ 16

[Click here for detail](#)

▶ 8

Question No: 4 (Marks: 1) - Please choose one

The maximum size of an Ethernet segment is

—

▶ 250 meters

▶ **500 meters**

[Click here for detail](#)

▶ 700 meters

▶ None of the given

Question No: 5 (Marks: 1) - Please choose one

The network with Throughput T and Delay D has a total ----- bits in transit at a time.

► D+T

► D-T

► DXT

(Computer Networks and Internets, page203)

► D/T

Question No: 6 (Marks: 1) - Please choose one

places the boundary between the first and second octets

► Class A

(Computer Networks and Internets, page235)

► Class B

► Class C

► Class D

Question No: 7 (Marks: 1) - Please choose one

Router detects datagram ----- than network MTU and then it splits into pieces and each piece is -----than outbound network MTU.

► Larger, smaller

(Page 108)

► Larger, larger

► Smaller, larger

► Smaller, smaller

Question No: 9 (Marks: 1) - Please choose one

Connectionless service, Message-Oriented protocol, best-effort delivery service, arbitrary interaction & operating system independent are the characteristics of

▶ TCP

▶ UDP (Page 120)

▶ IP

▶ None of the given

Question No: 10 (Marks: 1) - Please choose one

----- provide Application to application communication it also called end to end

communication

▶ IP

▶ TP (Page 119)

▶ RIP

▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

A routing table contains_

- ▶ The destination network ID
- ▶ The hop count to reach the network
- ▶ **The router ID of the next hop (Page 102)**
- ▶ All of the given

Question No: 12 (Marks: 1) - Please choose one

Which of the following protocols allows the sender and receiver to enforce policies.

► RIP

► OSPF

► BGP

(Reference Book 347)

► RIP and OSPF

Question No: 13 (Marks: 1) - Please choose one

measures distance in network hops, where each network between the source and destination counts as single hop.

► BGP

► OSPF

► RIP

(Page 138)

► Non of these

Question No: 14 (Marks: 1) - Please choose one

includes a 32-bits address mask with each address, which allows the address to be classful, classless, or subnetted.

► RIP

► OSPF (Page 140)

► BGP

► None of the given

Question No: 15 (Marks: 1) - Please choose one

One repeater -----, two repeaters ----- the maximum cable length limitation.

► Double, triple

(Page 49)

► Double, 4 time

► half, triple

► Double, half

Question No: 16 (Marks: 1) - Please choose one

ICMP message transport is acted upon by getting ICMP encrypted in IP.

► True (Page 117)

► False

Question No: 17 (Marks: 1) - Please choose one

Like most application programs, a client and server use a transport protocol to communicate.

► True

(Page 146)

► False

Question No: 18 (Marks: 1) - Please choose one

Mapping between a protocol address and a hardware address is called Address Resolution.

► True (Page 93)

► False

Question No: 19 (Marks: 1) - Please choose one

Address mask defines how many bits of address are in suffix?

► True

► False (Page 103)

Question No: 20 (Marks: 1) - Please choose one

A single networking technology is best for all needs.

► True

► False (Page 81)

Question No: 21 (Marks: 1) - Please choose one

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False

► True (Page 81) rep

Question No: 22 (Marks: 1) - Please choose one

Router detects datagram-----than network MTU

► Larger

(Page 108) rep

► Smaller

► None of given

► Equal

Question No: 23 (Marks: 1) - Please choose one

Information can flow in either or both direction between

► Clients

► Clients and servers

(Computer Networks and Internets, page 344)

► Servers

► None of given

Question No: 24 (Marks: 1) - Please choose one

One of the design goals for unicast route propagation is

.

► consistency

► inconsistency

► stability

(Computer Networks and Internets, page 344)

► dynamic addressing

Question No: 25 (Marks: 1) - Please choose one

IPV6 address consists of

▶ 32 Bits

▶ 64 Bits

▶ **128 Bits (Page 128)**

▶ none of the given

Question No: 26 (Marks: 1) - Please choose one

UDP offers application programs a Message-Oriented Interface, applications can depend on protocol to preserve data boundaries.

▶ **True (Page 120)**

► False

Question No: 27 (Marks: 1) - Please choose one

In case TCP, retransmission, acknowledgment from a computer on LAN are expected to arrive within

► Seconds

► Micro seconds

► **Milliseconds**

[Click here for detail](#)

► Nanoseconds

Question No: 28 (Marks: 1) - Please choose one

Twice NAT is another variant of NAT. it is used with site that runs server. In this process NAT box is connected to Domain Name.

► True

(Page 131)

► False

Question No: 29 (Marks: 1) - Please choose one

A network uses a -----arranges for computers to be connected in a closed loop.

► Star Topology

► Ring Topology (Page 25)

► Bus Topology

► None of the given

Question No: 30 (Marks: 1) - Please choose one

Protocol addresses are abstractions provided by

.

► hardware

► software (Page 93)

► operating system

► internet

FINALTERM EXAMINATION

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Question No: 1 (Marks: 1) - Please choose one

In Direct point to point communication adding the Nth computer requires----- new connections.

► None of the given

► N^2

► **N-1 (Page 23)**

► $(N^2 - N)/2$

Question No: 2 (Marks: 1) - Please choose one

In Point-to-Point topology there are two topologies.

- ▶ Tree and Ring
- ▶ Star and Ring
- ▶ Star and Tree (Page 5)
- ▶ None of the given

Question No: 3 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

► LAN (Page 4)

► WAN

► MAN

► None of the given

Question No: 4 (Marks: 1) - Please choose one

Hardware that calculates a CRC uses two simple components.

► AND unit and XOR unit

► Shift register and XOR unit (Page 20)

► Shift register and AND unit

► None of the given

Question No: 5 (Marks: 1) - Please choose one

CRC can detect more errors than a simple checksum.

► true

(Computer Networks and Internets, page 80)

► false

Question No: 6 (Marks: 1) - Please choose one

The Gigabit Ethernet hardware operates at a rate of -----

▶ 10 Mbps

▶ 100 Mbps

▶ **1000 Mbps**

[Click here for detail](#)

▶ None of the given

Question No: 7 (Marks: 1) - Please choose one

Formally named

informally known as the twisted pair Ethernet or TP Ethernet.

► 10 Base 2

► 10 Base 5

► **10 Base T (Page 43)**

► None of the given

Question No: 8 (Marks: 1) - Please choose one

An interface for thin Ethernet must have an

connector , and must generate signals according to

the

specification.

► **RJ-45, 10 Base T**

► **RJ-45, 10 Base 5**

► **BNC, 10 Base 2 (cs610 reference book Page 201)**

► **BNC, 10 Base T**

Question No: 9 (Marks: 1) - Please choose one

A system with redundant bridges might have a problem with

in the system.

► Loop

[Click here for detail](#)

rep

► Filters

► Spanning Trees

► All given choices

Question No: 10 (Marks: 1) - Please choose one

computes shortest paths in a graph by using weights on edges as a measure of distance.

► Greedy algorithm

► Distance vector algorithm

► Dijkstra's algorithm

(Computer Networks and Internets, page 112)

► Non of the given

Question No: 11 (Marks: 1) - Please choose one

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a

.

► Connectionless service paradigm

(Computer Networks and Internets, page 112)

► Connection-oriented service paradigm

► Both Connectionless and Connection-oriented service paradigm

► None of the given

Question No: 12 (Marks: 1) - Please choose one

protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer

► Network Interface Layer

► Internet Layer

► **Transport Layer (Page 84) rep**

Question No: 13 (Marks: 1) - Please choose one

An Internet Address (IP address) is a unique
binary number assigned to a host and used for all
communication with host

▶ 48-bit

▶ **32-bit (Page 85)**

▶ 24-bit

▶ None of the given

Question No: 14 (Marks: 1) - Please choose one

The address

identifies the physical network to which the computer is attached, while the identifies an individual computer on that network.

▶ **prefix , suffix (Page 85)**

▶ suffix , prefix

▶ suffix , suffix

▶ None of the given

Question No: 15 (Marks: 1) - Please choose one

places the boundary between the first and second octets

► **Class A (Computer Networks and Internets, page 235)**

► Class B

► Class C

► Class D

Question No: 16 (Marks: 1) - Please choose one

places the boundary between the third and fourth octets.

► Class A

► Class B

► **Class C**

(Computer Networks and Internets, page 235)

► **Class D**

Question No: 17 (Marks: 1) - Please choose one

field of header indicates whether a datagram is a fragment or a complete datagram.

► **FLAGS**

[Click here for detail](#)

► FLAGMENT OFFSET

► IDENTIFICATION

► None of the given

Question No: 18 (Marks: 1) - Please choose one

provides connectionless service.

► TCP

► UDP (Page 120)

► IP

► None of the given

Question No: 19 (Marks: 1) - Please choose one

UDP and TCP are both_

layer protocols

► Physical

► Data link

► Network

► Transport (Page 101) rep

Question No: 20 (Marks: 1) - Please choose one

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by

► IP

► None of the given

► TCP (Page 123) rep

► UDP

Question No: 21 (Marks: 1) - Please choose one

protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer

► Network Interface Layer

► Internet Layer

► **Transport Layer (Page 84) rep**

Question No: 22 (Marks: 1) - Please choose one

identifies which application program on receiving computer should receive the data

► Logical address

► Source port

► **Destination Port (Computer Networks and Internets, page313) rep**

► None of the given

Question No: 23 (Marks: 1) - Please choose one

identifies the application program that sent the data.

► DestinationPort

► Source port

(Computer Networks and Internets, page313) rep

► Logical address

► None of the given

Question No: 24 (Marks: 1) - Please choose one

The Border Gateway Protocol (BGP) uses

for all communication

► UDP

► TCP [Click here for detail](#)

rep

► Both UDP and TCP

► None of the given

Question No: 25 (Marks: 1) - Please choose one

Which of the following protocols allows the sender and receiver to enforce policies.

► RIP

► OSPF

► BGP

(Reference Book 347) rep

► RIP and OSPF

Question No: 26 (Marks: 1) - Please choose one

uses distance vector approach to define routing

► BGP

► OSPF

► RIP

(Computer Networks and Internets, page332)

► None of the given

Question No: 27 (Marks: 1) - Please choose one

ICMP message transport is acted upon by getting ICMP encrypted in IP.

► True (Page 117)

► False

Question No: 28 (Marks: 1) - Please choose one

Protocol addresses are abstractions provided by

.

► hardware

► software (Page 93) rep

► operating system

► internet

Question No: 29 (Marks: 1) - Please choose one

These packets serve same purpose on

as frames on

► Intranet, LAN

► Internet, WAN

► Intranet, WAN

► **Internet, LAN (Page 101)**

Question No: 30 (Marks: 1) - Please choose one

Address mask defines how many bits of address are in suffix?

► True

► False (Page 103) rep

Question No: 31 (Marks: 1) - Please choose one

A single networking technology is best for all needs.

► True

► False (Page 81) rep

Question No: 32 (Marks: 1) - Please choose one

A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?

► True (Page 81) rep

► False

Question No: 33 (Marks: 1) - Please choose one

The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address

.

► itself (Page 87)

► prefix

► suffix

► mask

Question No: 34 (Marks: 1) - Please choose one

Find the class of the address.

10100111 11011011 10001011 01101111

▶ A

▶ B (Computer Networks and Internets, page 122)

▶ E

▶ C

Question No: 35 (Marks: 1) - Please choose one

Find the class of the address:

11110011 10011011 11111011 00001111

► A

► C

► E

(Computer Networks and Internets, page 122)

► B

Question No: 36 (Marks: 1) - Please choose one

In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

▶ T

▶ D

▶ C

(Page 97) rep

▶ T, C

Question No: 37 (Marks: 1) - Please choose one

Which method of Address Resolution Protocol requires hardware broadcast?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► D (Page 97)

► T

► C

► T, D

Question No: 38 (Marks: 1) - Please choose one

Which method of Address Resolution Protocol resolution with minimum delay?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

▶ T, D

▶ c

▶ T

▶ T, C

(Page 97)

Question No: 39 (Marks: 1) - Please choose one

In which method of Address Resolution Protocol the implimentation is more difficult?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

▶ T, C

▶ T

▶ C

▶ D (Page 97)

Question No: 40 (Marks: 1) - Please choose one

One of the design goals for unicast route propagation is

.

► Consistency

► inconsistency

► stability

(Computer Networks and Internets, page 344) rep

► dynamic addressing

Question No: 41 (Marks: 1) - Please choose one

Propagation of multicast routing information differs dramatically from unicast route propagation?

► True

(Computer Networks and Internets, page 335)

► False

Question No: 42 (Marks: 1) - Please choose one

The IP multicast abstraction allows an application running on an arbitrary computer to leave a multicast group at any time. While

application on a computer remain a member of a group.

► One or more

► only one

► no

► many

Question No: 43 (Marks: 1) - Please choose one

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system.

► True

► False (Computer Networks and Internets, page 329)

Question No: 44 (Marks: 1) - Please choose one

In IPv6 the type of address used for collection of computers with same prefix. Are known as

.

► Anycast

► Unicast

► Multicast

► Non of the given (Page 114)

Question No: 45 (Marks: 1) - Please choose one

Special types of addresses in IPv6 used for multiple destinations; possibly not at same site. Are known as

.

► Unicast

► Anycast

► **Multicast (Page 114)**

► Non of the given

Question No: 46 (Marks: 1) - Please choose one

UDP offers application programs a Message-Oriented Interface, applications can depend on protocol to preserve data boundaries.

► True (Page 120) rep

► False

Question No: 47 (Marks: 1) - Please choose one

Reliability is the responsibility of the

layer

► Network

► Datalink

► Transport (Page 123)

► Application

Question No: 48 (Marks: 1) - Please choose one

TCP uses

mechanism to control the flow of data.

▶ door

▶ window (Page 126) rep

▶ acknowledgment

▶ retransmission

TCP uses window mechanism to control the flow of data.

Question No: 49 (Marks: 1) - Please choose one

The time for acknowledgement to arrival of packet depends on.

▶ Distance to destination and Current traffic conditions (Page 125)

► Current traffic conditions

► Distance to destination

► non of these

Question No: 50 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of -----

► 100 million bits per second

(Page 31)

► 10 million bits per second

► 1000 million bits per second

► None of the given

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Question No: 1

(M a r k s : 1)

Computer networks are often called ----- because they use packet technology.

► Ethernet

► Switch networks

► **Packet networks**

(Computer Networks and Internets, page 73)

► None of the given

Question No: 2

(Marks: 1)

A network uses a -----arranges for computers to be connected in a closed loop.

► Star Topology

► Ring Topology (Page 25) rep

► Bus Topology

► None of the given

Question No: 3

(Marks: 1)

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

► **Explicit frame type (Computer Networks and Internets, page 108)**

► Ideal frame type

► Implicit frame type

► None of the given

Question No: 4

(Marks: 1)

An interface for thin Ethernet must have an

connector , and must generate signals according to

the

specification.

► RJ-45, 10 Base T

► RJ-45, 10 Base 5

► BNC, 10 Base 2

(cs610 reference book Page 201) rep

► BNC, 10 Base T

Question No: 5

(Marks: 1)

A Bridge forwards or filters a frame by comparing the information in its address table to the

frame's

► Layer 2 source address

► Source node's physical address

► **Layer 2 destination address**

[Click here for detail](#)

► Layer 3 destination address

Question No: 6

(M a r k s : 1)

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called

► Hierarchal address

► **Default route**

(Computer Networks and Internets, page 172)

► Shortest path

► None of the given

Question No: 7

(Marks: 1)

of TCP/IP layering model, corresponds to basic network hardware.

► Physical Layer

(Page 84) rep

► Network Interface Layer

► Internet Layer

► Transport Layer

Question No: 8

(Marks: 1)

protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer

► Network Interface Layer

► Internet Layer

► Transport Layer

(Page 84) rep

Question No: 9

(Marks: 1)

is called an end-to-end protocol because it provide a connection directly from an application on one computer to an application on a remote computer.

► IP

► UDP

► TCP

(Computer Networks and Internets, page 306)

► None of the given

Question No: 10

(Marks: 1)

uses distance vector approach to define routing

► BGP

► OSPF

► RIP

(Computer Networks and Internets, page332) rep

► None of the given

Question No: 11

(Marks: 1)

is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

► Flood-and –Prune

(Page 143)

► Configuration-and -Tunneling

► Core-Based Discovery

► None of the given

Question No: 12

(Marks: 1)

Router that decrements TTL to sends ICMP time exceeded message, with router's address as source address

► 3

▶ 2

▶ 1

▶ 0 (Page 118)

Question No: 13

(Marks: 1)

Protocol addresses are abstractions provided by

.

▶ hardware

▶ software (Page 93) rep

▶ operating system

▶ internet

Question No: 14

(Marks: 1)

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

► True

(Page 99) rep

► False

Question No: 15

(Marks: 1)

ARP is almost always used to bind a

-bit IP address to a

-bit Ethernet address.

► **32, 48 (Page 98)**

► 24, 32

► 32, 64

► 32, 128

Question No: 16

(Marks: 1)

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False

► True

(Page 81) rep

Question No: 17

(Marks: 1)

In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

Types of

Features

Resolution

Use full with any hardware

T

Address change affects all hosts T

Protocol address is determined

C

by hardware address

► T

► D

► C (Page 97) rep

► T, C

Question No: 18

(Marks: 1)

The general form of an IP datagram is with a header followed by data. The header contains information that controls where and how the datagram is to be sent.

► True

(Computer Networks and Internets, page 332)

► False

Question No: 19

(Marks: 1)

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system.

► True

► False

(Computer Networks and Internets, page 329)

Question No: 20

(Marks: 1)

Which of the following is a correct representation of the IPv6?

► 105.220.136.100.255.255.255.0.0.18.128.140.10.255.255

(Page 114)

► 105.220.136.100.255.255.255.256.0.0.18.128.140.10.255.255

► 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255.256

► 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255

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